خدمتكم طريق خضناه لرضى الله

2021

ملخص

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الفزياء العامة 102



لجنة السنافر

سنافر البوليتكنك | 0790640139

CH:23 Electric Field

charge (q): الشحنة الكهربائية (C)

$q = ne$ (الشحنة)

- $1.6 * 10^{-19}$ C ⇐ الجسم يفقد (lose)
- $-1.6 * 10^{-19}$ C ⇐ الجسم يكتسب (accept)

n: عدد الإلكترونات المفقودة أو المكتسبة وهي عدد صحيح موجب

Ex: If an object accepts $2 * 10^7$ e, what is its charge?

$$q = ne$$
$$= 2 * 10^7 * -1.6 * 10^{-19}$$
$$= -3.2 * 10^{-12} C.$$

Ex: If an object contains $4 * 10^3$ e^- and $2 * 10^3$ p^+ what is its net charge?

$q = ne + np$ (الشحنة)

$$= 4 * 10^3 * -1.6 * 10^{-19} + 2 * 10^3 * 1.6 * 10^{-19}$$
$$= -6.4 * 10^{-16} + 3.2 * 10^{-16}$$
$$= -3.2 * 10^{-16} C.$$

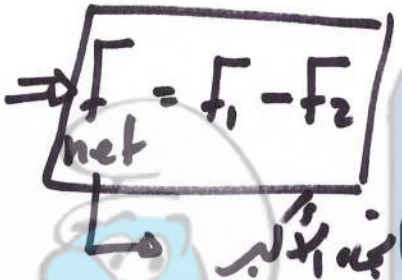
٢) حساب محصلة المتجهات :-

١) إذا كان المتجهين بنفس الاتجاه :

$$F_{net} = F_1 + F_2$$



٢) إذا كانت المتجهات المتجهان متعاكستان :

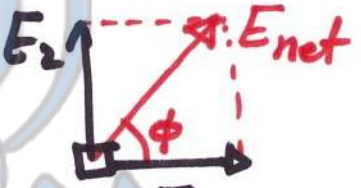




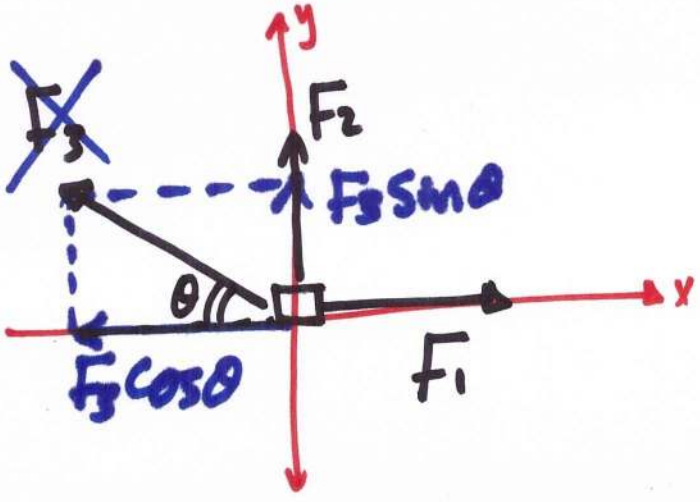
٣) إذا كان المتجهان متعامدان :

$$E_{net} = \sqrt{E_1^2 + E_2^2}$$

$$\tan\phi = \frac{E_2}{E_1} \Rightarrow \phi = \tan^{-1}\frac{E_2}{E_1}$$



٤) إذا أثر على الجسم أكثر من متجهين و أو إذا كانت $\theta \neq 90°$



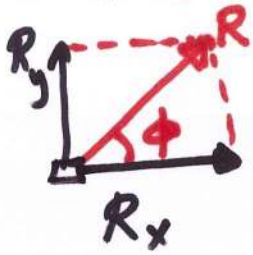
١) تحديد المحاور :

٢) تحلل أي قوة لا تنطبق على المحاور

٣) أجد المحصلة السينية والصادية :

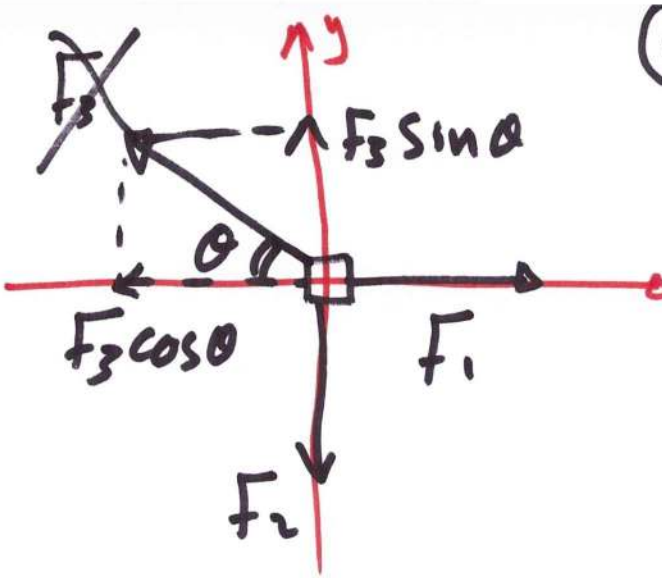
$$R_x = F_1 - F_3\cos\theta$$

$$R_y = F_2 + F_3\sin\theta$$



٤) أجد R و $\tan\phi = \frac{R_y}{R_x}$ ، $R = \sqrt{R_x^2 + R_y^2}$

3)



ج) الجسم متزن (equilibrium)

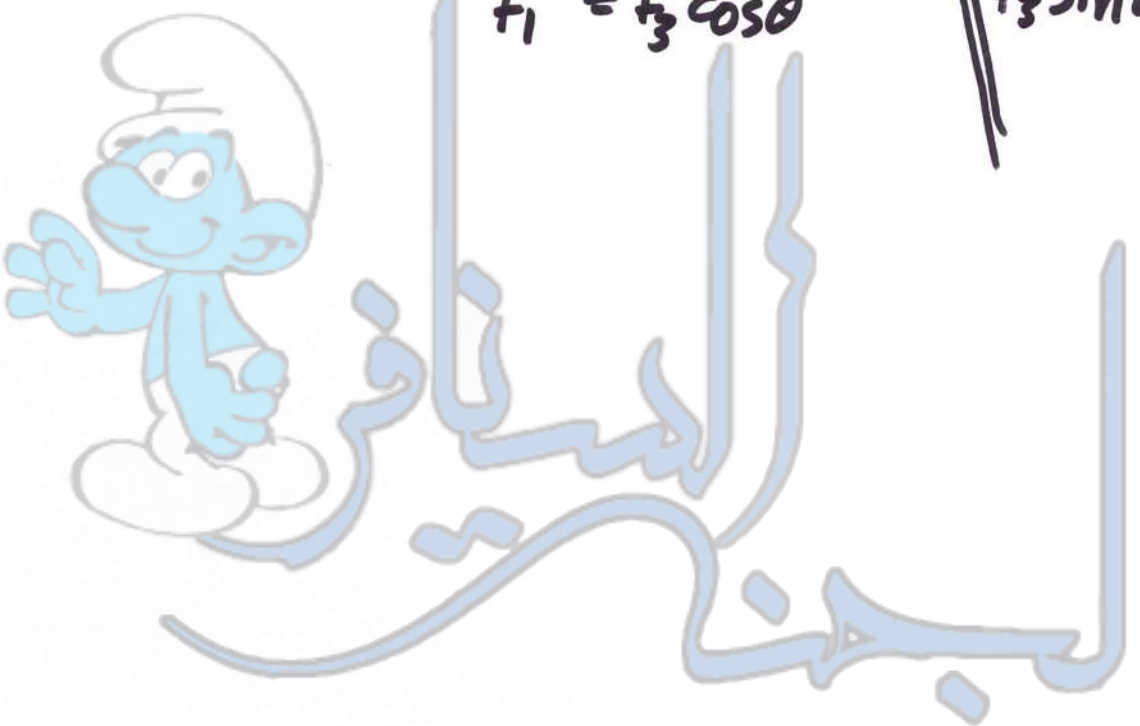
١) تحديد المحاور

٢) تحليل القوة: الغير منطبقة على المحاور.

٣) نطبق قانوني الاتزان :-

$$\Sigma F_{\rightarrow} = \Sigma F_{\leftarrow} \quad , \quad \Sigma F_{\uparrow} = \Sigma F_{\downarrow}$$

$$F_1 = F_3 \cos\theta \qquad F_3 \sin\theta = F_2$$



Coulomb's law



$$F_e = \frac{k q_1 q_2}{r^2}$$

$$k = \frac{1}{4\pi\epsilon_0} = 9*10^9$$

$$\epsilon_0 = 8.85*10^{-12}$$

* استثنت سالبة ؛ فاننا لا نعوض الاشارة السالبة في قانون كولوم

$K = 10^3$, $M = 10^6$, $G = 10^9$

$c = 10^{-2}$, $m = 10^{-3}$, $\mu = 10^{-6}$, $n = 10^{-9}$, $P = 10^{-12}$, $f = 10^{-15}$

Ex :





$$F = \frac{9*10^9 * 8*10^{-6} * 6*10^{-6}}{(3*10^{-2})^2} = 48*10 = 480\ N\ (\text{attraction})$$

Ex :



What is the net force that acting on q_2.

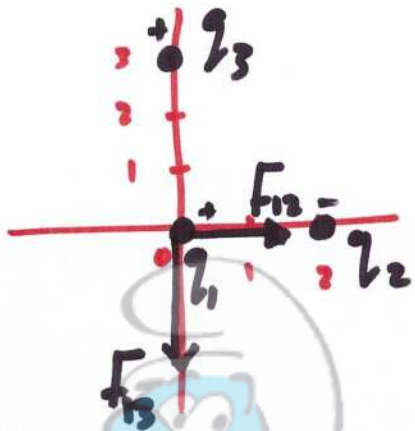
$$F_{12} = \frac{9*10^9 * 9*10^{-6} * 4*10^{-6}}{(2*10^{-3})^2} = 81*10^3\ N$$

$$F_{23} = \frac{9*10^9 * 4*10^{-6} * 1*10^{-6}}{(1*10^{-3})^2} = 36*10^3\ N$$

$$F_{net} = F_{12} + F_{23} = 117*10^3\ N(-\hat{i})$$

* اتجاهات
* حسابات
* محصلة.

Ex: If we have 3-charges, $q_1 = 1nc$ at origin and $q_2 = -4nc$ at $x = 2cm$, while $q_3 = 12nc$ at $y = 3cm$, Find the force acting on q_1.



$$F_{12} = \frac{9\times10^9 \times 1*10^{-9} * 4\times10^{-9}}{(2\times10^{-2})^2}$$

$$= 9 * 10^{-5} N$$

$$F_{13} = \frac{9\times10^9 * 1*10^{-9} * 12*10^{-9}}{(3\times10^{-2})^2}$$

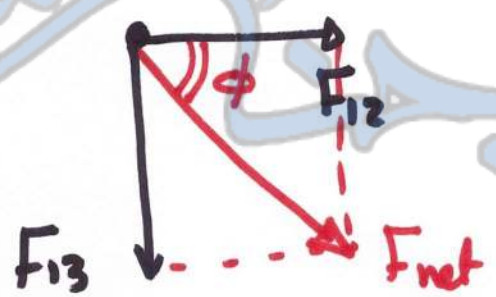
$$= 12 * 10^{-5} N .$$

$$F_{net} = \sqrt{F_{12}^2 + F_{13}^2}$$

$$= 10^{-5}\sqrt{9^2 + 12^2}$$

$$= 10^{-5}\sqrt{81+144}$$

$$= 15 * 10^{-5} N$$

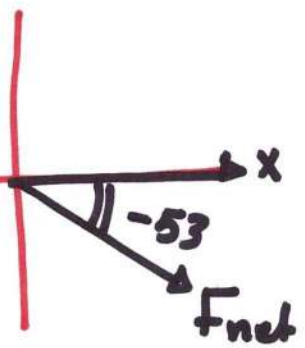


$$\tan\phi = \frac{12*10^{-5}}{9*10^{-5}}$$

$$= \frac{4}{3}$$

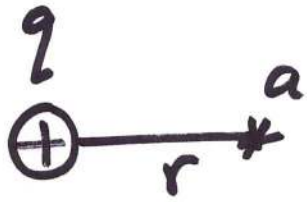
$$\Rightarrow \boxed{\phi = 53.1^\circ}$$

direction of F_{net} is $\boxed{-53^\circ}$ or 307°.

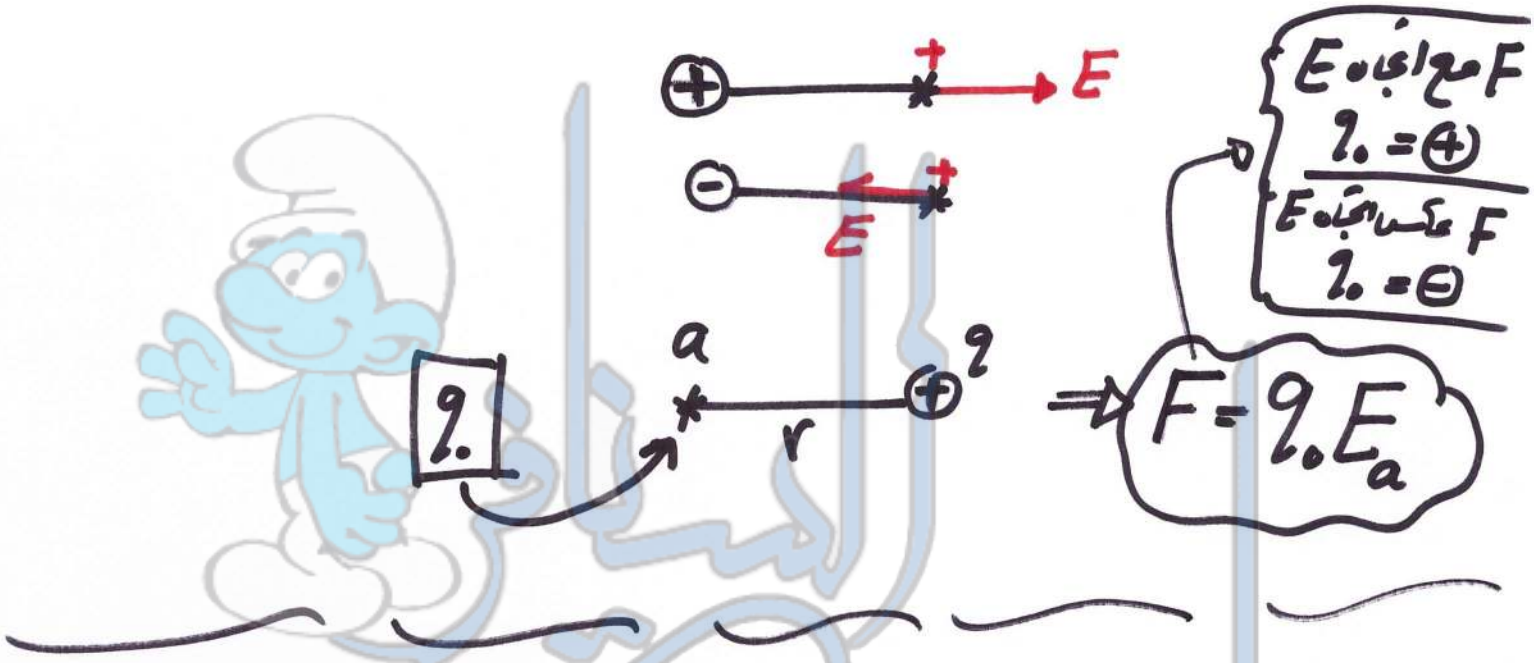


Electric Field

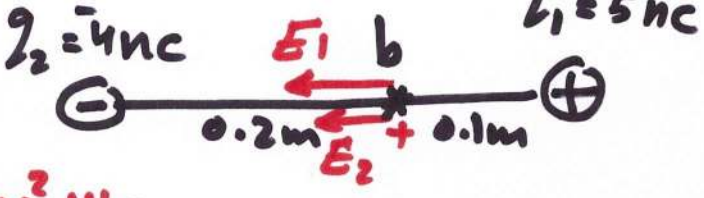
$$E = \frac{k\,q}{r^2}$$



* الإشارة السالبة لا تعوض .



Ex: calculate the E at b .



$$E_1 = \frac{9\times10^9 * 5*10^{-9}}{(1*10^{-1})^2} = 45*10^3\ N/C$$

$$E_2 = \frac{9\times10^9 * 4*10^{-9}}{(2*10^{-1})^2} = 9*10^3\ N/C$$

* اتجاهات .
* حسابات .
* محصلة .

$$E_{net} \equiv E_b = E_1 + E_2 = 54*10^3\ N/C\ (-i)$$

$$F = q_o E_b = 2*10^{-3} * 45*10^2 = 90*10^{-1} = 9N$$

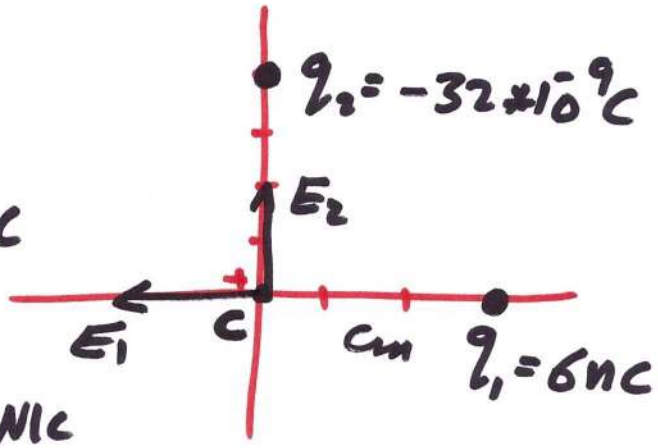
b ← q_o *

Ex: Calculate the Electric Field at C.

Sol:

$$E_1 = \frac{9*10^9 * 6*10^{-9}}{(3*10^{-2})^2} = 6*10^4 \text{ N/C}$$

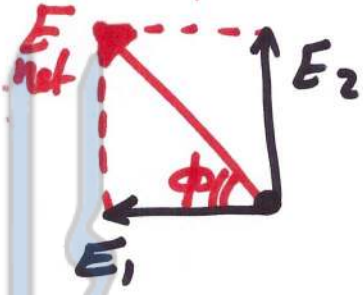
$$E_2 = \frac{9\times10^9 * 32*10^{-9}}{(4\times10^{-2})^2} = 18*10^4 \text{ N/C}$$



$$E_{net} = \sqrt{E_1^2 + E_2^2}$$

$$= \sqrt{6^2 + 18^2} * 10^4$$

$$= 19*10^4 \text{ N/C}$$

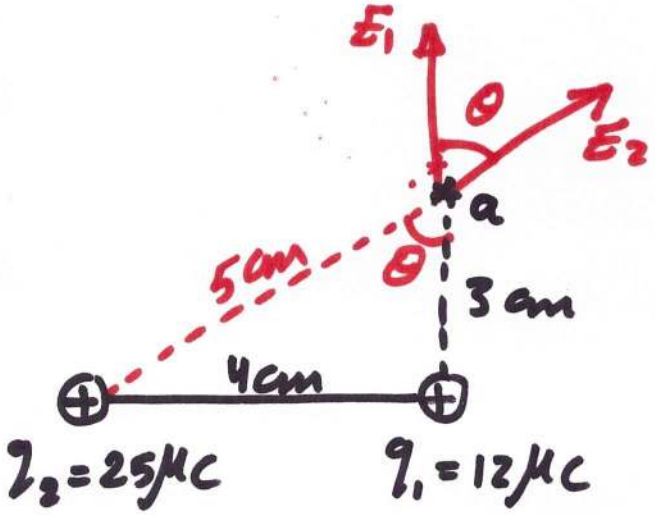


$$\tan\phi = \frac{18*10^4}{6*10^4} = 3 \Rightarrow \boxed{\phi = 71.5^\circ}$$

Ex: What is the Electric field at Point a:

$$E_1 = \frac{9\times10^9 \times 12\times10^{-6}}{9*10^{-4}} = 12*10^7 \text{ N/C}$$

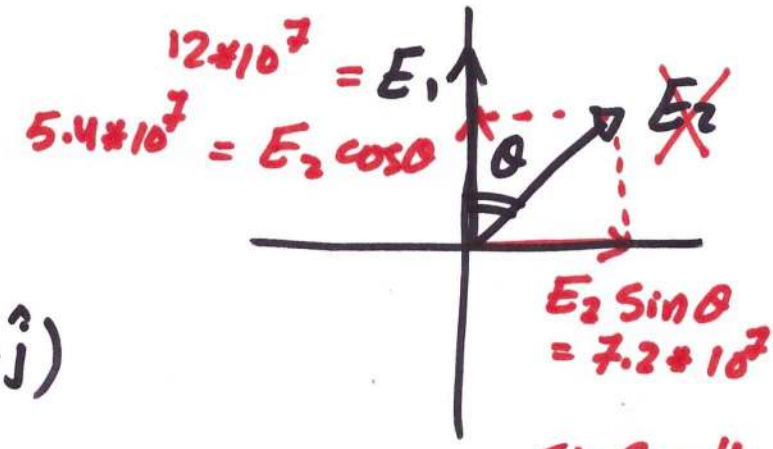
$$E_2 = \frac{9\times10^9 \times 25\times10^{-6}}{(5\times10^{-2})^2} = 9*10^7 \text{ N/C}$$



$R_x = 7.2 * 10^7 \ (+\hat{i})$

$R_y = 12 * 10^7 + 5.4 * 10^7$

$= 17.4 * 10^7 \ (+\hat{j})$

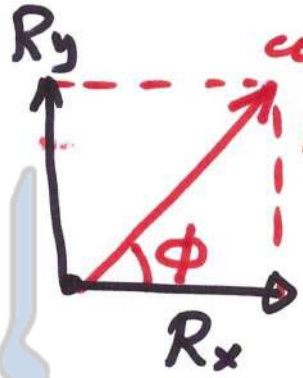


$\sin\theta = \frac{4}{5} = 0.8$

$\cos\theta = \frac{3}{5} = 0.6$

$R = \sqrt{7.2^2 + 17.4^2} * 10^7$

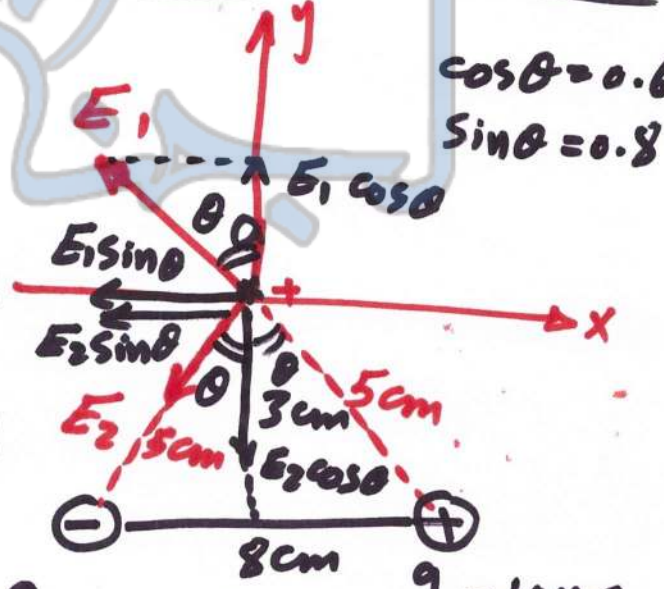
$= 18.8 * 10^7 \ N/C$



$\tan\phi = \frac{17.4*10^7}{7.2*10^7} = 2.4$

$\Rightarrow \phi \simeq 67.4$

Ex: Find E_{net} at a:



$\cos\theta = 0.6$

$\sin\theta = 0.8$

$E_1 = \frac{9\times10^9 * 10\times10^{-9}}{25*10^{-4}} = 3.6*10^4 \ N/C$

$E_2 = \frac{9\times10^9 \times 10\times10^{-9}}{25\times10^{-4}} = 3.6*10^4 \ N/C$

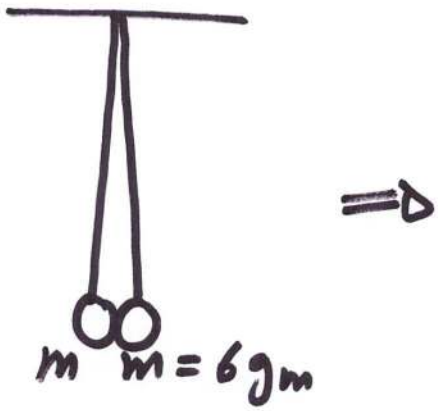
$R_x = E_1 \sin\theta + E_2 \sin\theta$

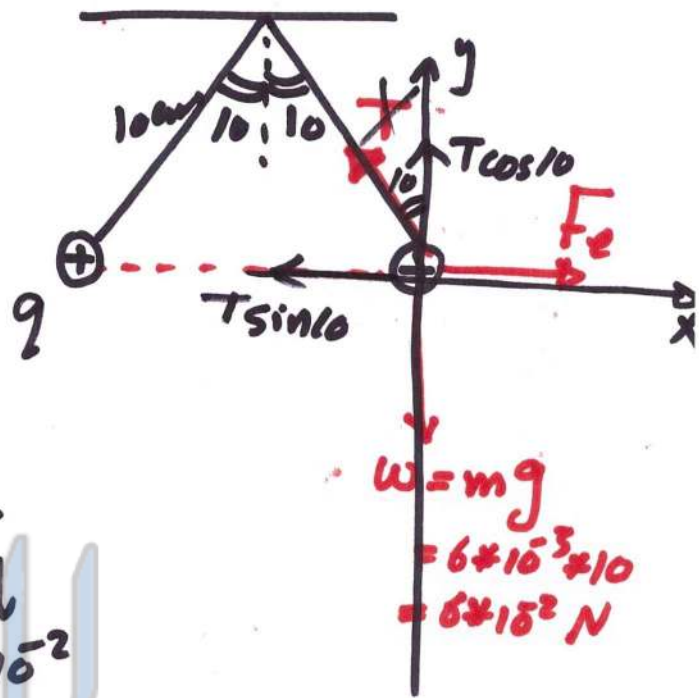
$= 3.6\times10^4 * 0.8 + 3.6\times10^4 * 0.8 = 2.88*10^4 \ N/C \ (-\hat{i})$

$R_y = E_1 \cos\theta - E_2 \cos\theta = 0$

$\Rightarrow R = R_x = 2.88*10^4 \ N/C \ \boxed{-\hat{i}}$.

Ex Find q if the system is at equilibrium.





$$\Sigma F_{\uparrow} = \Sigma F_{\downarrow}$$

$$T\cos 10 = 6*10^{-2}$$

$$\boxed{T = 0.0609\ N}$$

$$\Sigma F_{\rightarrow} = \Sigma F_{\leftarrow}$$

$$Fe = T\sin 10$$

$$= 0.0609 * \sin 10$$

$$\boxed{Fe = 1.06*10^{-2}\ N}$$

$$F_e = \frac{9*10^9\, q_1 q_2}{r^2}$$

$$1.06*10^{-2} = \frac{9*10^9 * q^2}{(3.4*10^{-2})^2}$$

$$q^2 = 1.36*10^{-7}$$

$$\boxed{q = 3.7*10^{-4}\ C}$$



$r = 2x$

$$\sin 10 = \frac{x}{10*10^{-2}}$$

$$\Rightarrow x = 1.7*10^{-2}\ m$$

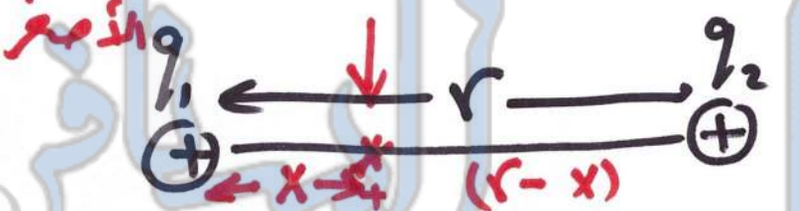
$$\Rightarrow r = 2x = 3.4*10^{-2}\ m$$

equilibrium Point

* هي النقطة التي ينعدم عندها المجال الكهربائي والقوة الكهربائية. ($E=0$).

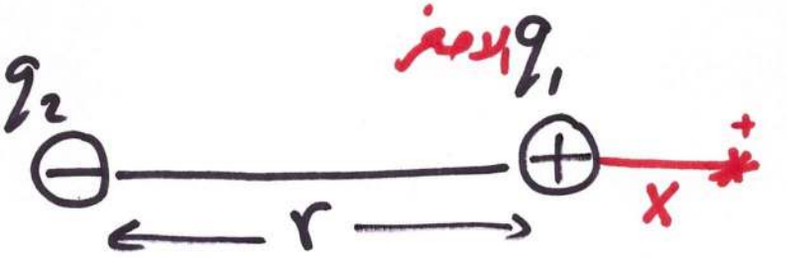
- شحنتان متشابهتان: نقطة التعادل تقع بينهما وأقرب للشحنة الأصغر.

- شحنتان مختلفتان: نقطة التعادل تقع خارجهما أقرب للأصغر



$$E_1 = E_2$$ ← تعادل نقطة

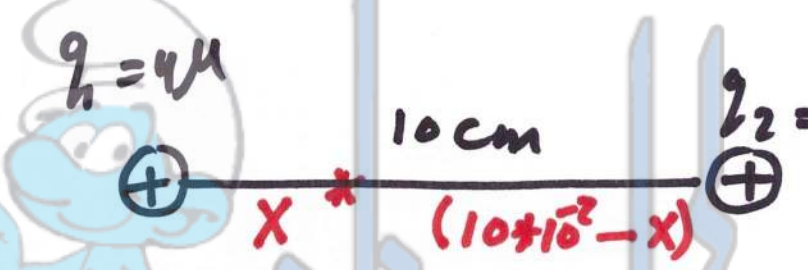
$$\frac{k q_1}{r_1^2} = \frac{k q_2}{r_2^2}$$



Ex: If we two Point charges:

$q_1 = 4\mu c$, $q_2 = 9\mu c$, $r = 10\,cm$

where is the equilibrium Point.
(where is the Point that has no net electric field)



$$E_1 = E_2$$

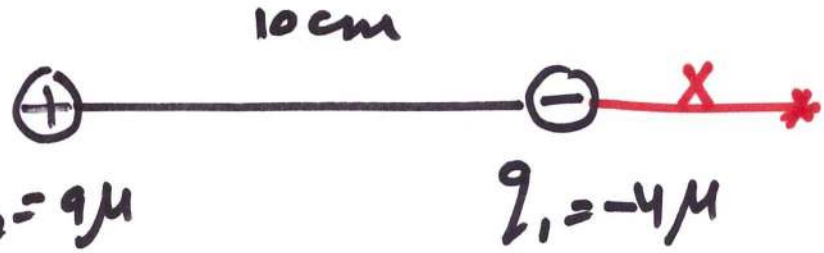
$$\frac{k\ 4*10^{-6}}{X^2} = \frac{k\ 9*10^{-6}}{(10*10^{-2} - X)^2}$$

$$\frac{2}{X} = \frac{3}{10\times10^{-2} - X} \Rightarrow 3X = 20*10^{-2} - 2X$$

$$5X = 20*10^{-2}$$

$$\boxed{X = 4*10^{-2}\ m = 4\,cm}$$

Ex: $q_1 = -4\mu C$, $q_2 = 9\mu C$, $r = 10\,cm$



$$E_1 = E_2$$

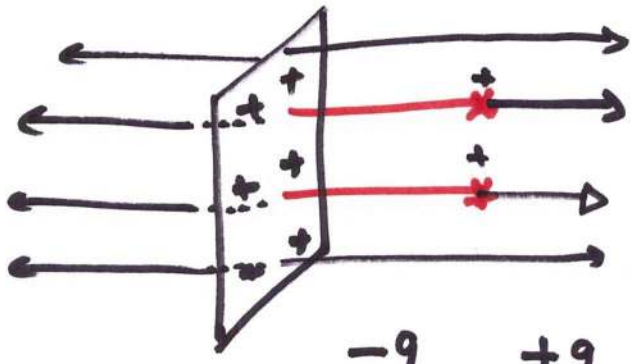
$$\frac{9\times10^{9}\times 4\times10^{-6}}{X^2} = \frac{9\times10^{9}\times 9\times10^{-6}}{(10*10^{-2}+X)^2}$$

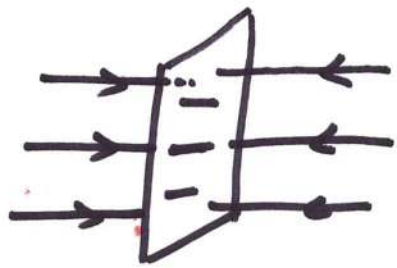
الجذر

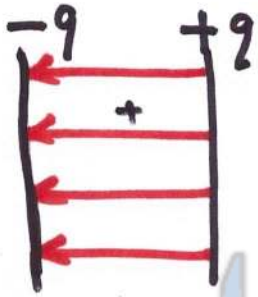
$$\frac{2}{X} = \frac{3}{10\times10^{-2}+X} \Longrightarrow 3X = 20\times10^{-2} + 2X$$

$$X = 20*10^{-2}\,m = 20\,Cm$$

Uniform Electric Field







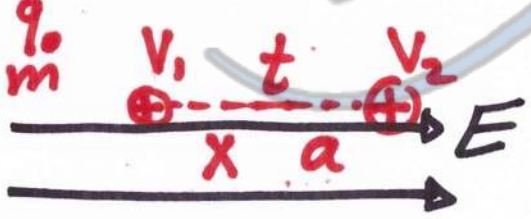
E = Constant

$E_{uni} = \frac{9 * 10^9 * q}{r^2}$

تطبيق 1

حركة جسيم مشحون داخل هذا المجال:

Motion of a small charged particle inside the uniform electric field.



$F_E = q_o E = ma$

$W = F X \cos\theta_{Fx}$

$\Delta K = K_f - K_i$

$= \frac{1}{2} m V_f^2 - \frac{1}{2} m V_i^2$

$W_{total} = \Delta K$

$V_2 = V_1 + at$

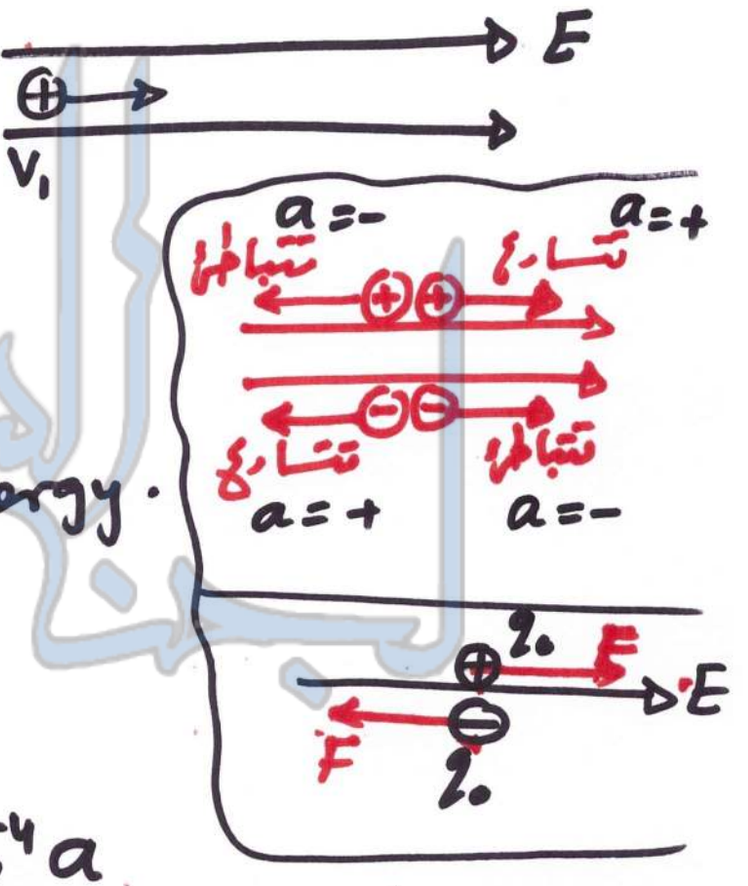
$V_2^2 = V_1^2 + 2aX$

$X = V_1 t + \frac{1}{2} a t^2$

$X = \left(\frac{V_1 + V_2}{2}\right) t$

Ex: An object of mass $2*10^{-4}$ kg (m) and charge of $6*10^{-8}$ C (q_o), enters a Uniform electric field with speed of $2*10^4$ m/s (V_1) for 10^3 sec. (t) if $E = 4000$ N/C (E) as in figure, Find:

1) acceleration
2) Final speed
3) traveled displacement.
4) Force done exerted.
5) Work done.
6) change in Kinetic energy.



Sol:

1) $q_o E = ma$

$6*10^{-8} * 4*10^3 = 2*10^{-4} a$

$a = 12 * 10^{-1} = +1.2\ m/s^2$.

---

2) $V_2 = V_1 + at$

$= 2*10^4 + 1.2*10^3$

$= 20000 + 1200$

$\boxed{V_2 = 21200\ m/s.}$

3) $X = V_1 t + \frac{1}{2} a t^2$

$= 2*10^4 * 10^3 + \frac{1}{2} * 1.2 * 10^6$

$= 2*10^7 + 0.6*10^6$

$\boxed{X = 20.6*10^6\ m}$

4) $F = q_o E = ma$

$= 6 * 10^{-8} * 4 * 10^{3}$

$= 24 * 10^{-5}\ N$

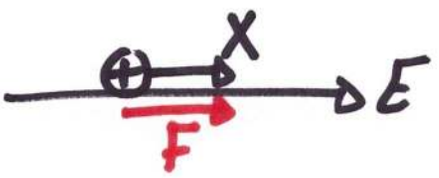
$= 2 * 10^{-4} * 1.2$

$= 2.4 * 10^{-4}$

$= 24 * 10^{-5}\ N$

---

5) $W = F \times \cos\theta_{Fx}$



$= 24 * 10^{-5} * 20.6 * 10^{6} \cos 0$

$= 4944\ J$

---

6) $\Delta K = \frac{1}{2} m V_2^2 - \frac{1}{2} m V_1^2$

$= \frac{1}{2} 2 * 10^{-4} * (21200)^2 - \frac{1}{2} * 2 * 10^{-4} * (2 * 10^{4})^2$

$= 44944 - 40000$

$= 4944\ J$

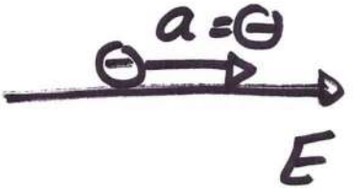
Ex: If an object of mass 0.1 gm (m) and charge $-2 * 10^{-6}$ C (q_o), start moving with speed of 20 m/s (V_1) inside a uniform electric field and with the field direction, the distance traveled is 200 m (x) Untill the particle was stopped ($V_2 = 0$), what is the mag. of Electric field (E).

Sol:

$$V_2^2 = V_1^2 + 2ax$$

$$0 = 400 + 2 * a * 200$$

$$a = -\frac{400}{400} = -1\ m/s^2$$



$$q_o E = ma$$

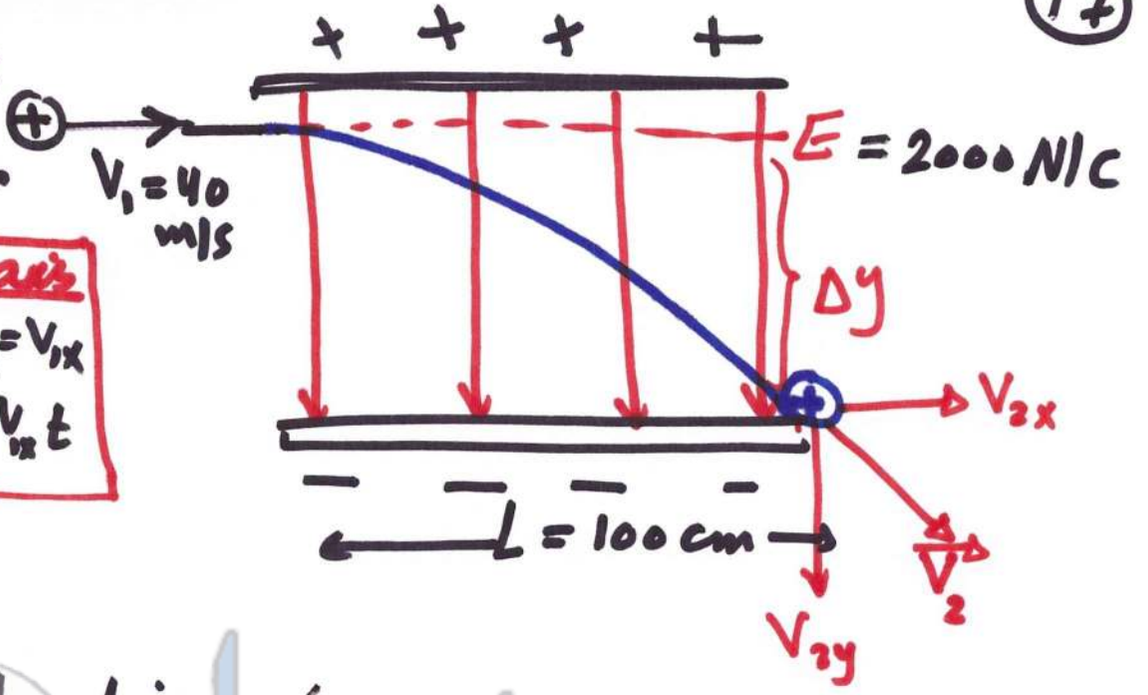
$$E = \frac{ma}{q_o} = \frac{0.1 * 10^{-3} * 1}{2 * 10^{-6}} = 0.5 * 10^2 = 50\ N/C$$

* $V_2 = V_1 + at$

$$0 = 20 - 1t$$

$$\boxed{t = 20\ sec}$$

Ex: $q_o = 10\mu C$, $m = 2gm$, $V_1 = 40$ m/s



y-axis	x-axis
$V_{2y} = V_{1y} + a_y t$	$V_{2x} = V_{1x}$
$V_{2y}^2 = V_{1y}^2 + 2a_y y$	$x = V_{1x} t$
$y = V_{1y} t + \frac{1}{2} a_y t^2$	

Find:

1) acceleration (a_y)
2) final velocity (speed).
3) Vertical displacement (Δy)
4) time.

$V_{1x} = V_1 \cos\theta = 40 \cos 0 = 40$ m/s

$V_{1y} = V_1 \sin\theta = 40 \sin 0 = 0$ m/s.

1) $q_o E_y = m a_y$

$10 * 10^{-6} * 2000 = 2 * 10^{-3} a_y$

$\boxed{a_y = +10 \text{ m/s}^2}$

2) $V_{2x} = V_{1x} = 40$

$V_{2y} = \cancel{V_{1y}} + a_y t$

$= 10 * 25 \times 10^{-3}$

$= 0.25$ m/s.

$x = V_{1x} t$

$1 = 40\, t$

$t = 25 \times 10^{-3}$ sec

$\vec{V}_2 = 40\hat{i} - 0.25\hat{j}$

$V = |\vec{V}_2| = \sqrt{40^2 + 0.25^2} =$

3) $\Delta y = V_{iy}t + \frac{1}{2}gt^2$

$= \frac{1}{2} * 10 * (0.025)^2$

$= 3.125 * 10^{-3} m$

$= 3.125 \ mm.$

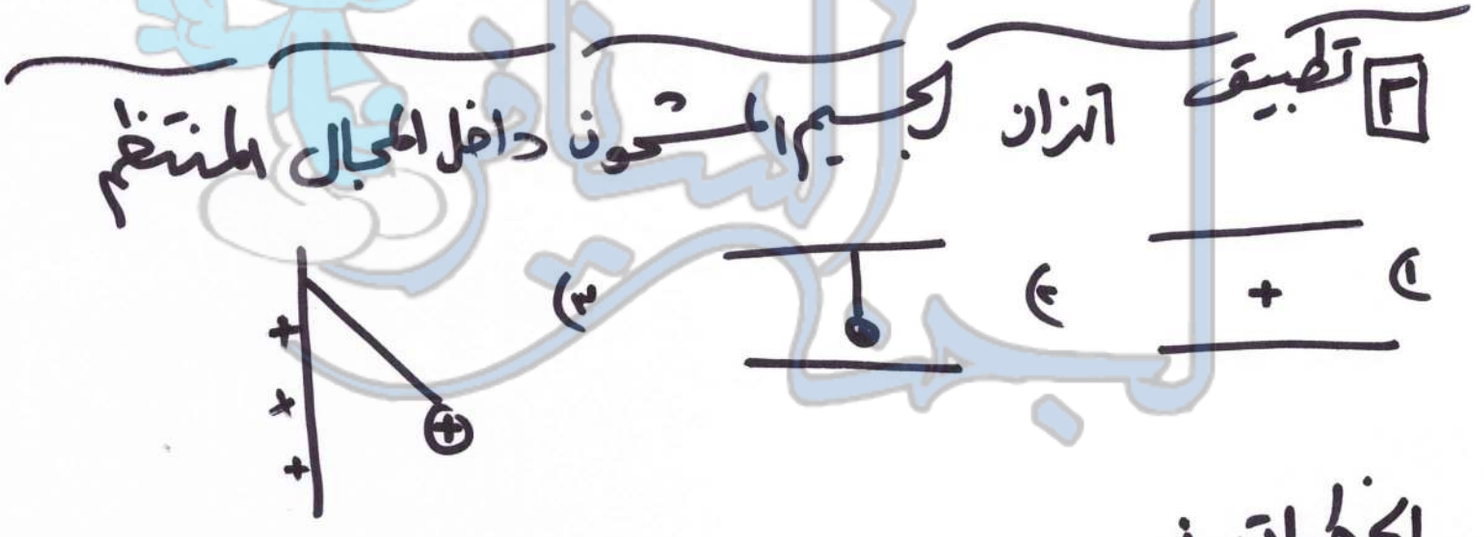
---

4) $t = 25 * 10^{-3} \ sec.$

$\Delta y = V_{iy}t + \frac{1}{2}at^2$

$\rightarrow \Delta x = V_{ix}t$

تطبيق اتزان الجسم المشحون داخل المجال المنتظم



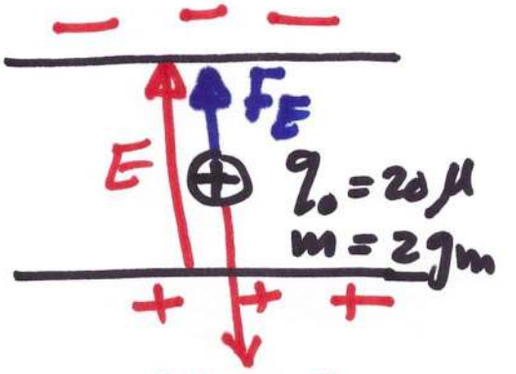
الخطوات :

١) نحدد اتجاه القوى المؤثرة .

٢) نحدد محاور مناسبة وعلل القوة غير منطبقة على المحاور .

٣) نطبق قوانين الاتزان : $\Sigma F_{\uparrow} = \Sigma F_{\downarrow}$, $\Sigma F_{\rightarrow} = \Sigma F_{\leftarrow}$

Ex: particle of mass 2gm and charge of 20μc, what is the magnitude and direction of $\underline{E}$ wher q_o is at equilibrium.



$$\Sigma F_{\uparrow} = \Sigma F_{\downarrow}$$

$$q_o E = w$$

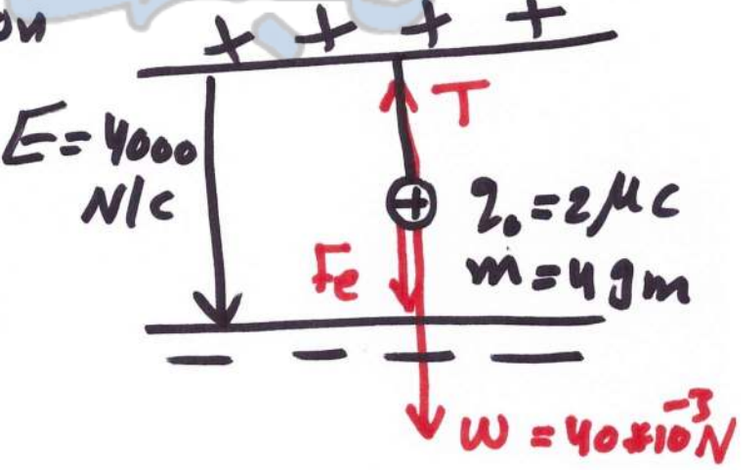
$$20 \times 10^{-6} E = 2 * 10^{-2}$$

$$\boxed{E = 1 * 10^{3}\ N/C}$$

$$= 2 * 10^{-3} * 10$$

$$= 2 * 10^{-2}\ N$$

Ex: what is the tension in the Cord.



$$\Sigma F_{\uparrow} = \Sigma F_{\downarrow}$$

$$T = F_e + w$$

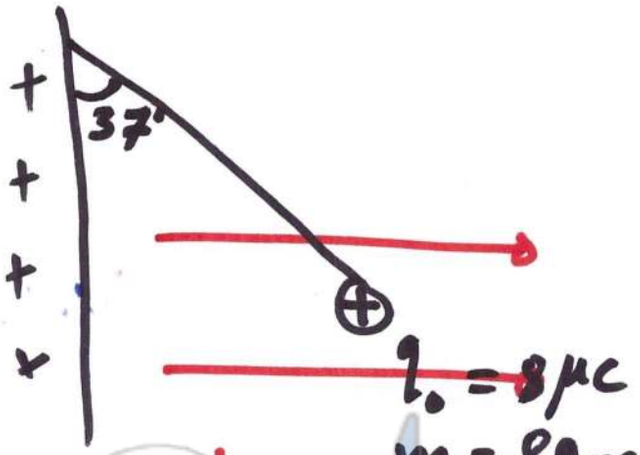
$$T = 48 * 10^{-3}\ N.$$

$$F_e = q_o E$$

$$= 2 \times 10^{-6} * 4 * 10^{3}$$

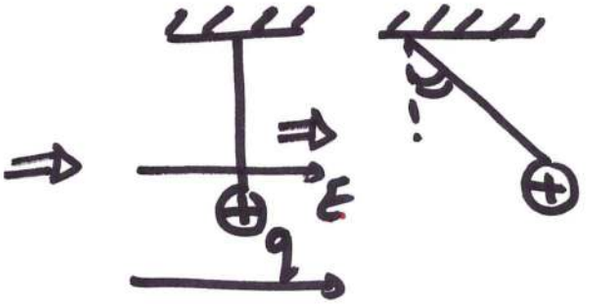
$$= 8 * 10^{-3}\ N$$

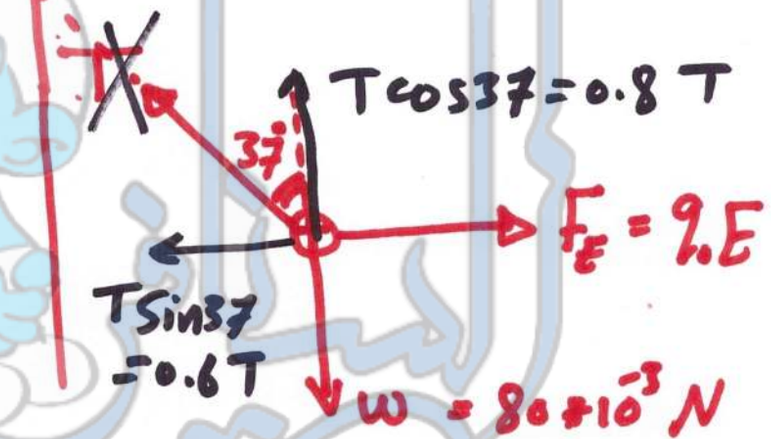
Ex: Find E



$q_0 = 8\mu c$

$m = 8gm$





$\Sigma F_\uparrow = \Sigma F_\downarrow$

$0.8\,T = 80*10^{-3}$

$\boxed{T = 0.1\ New}$

$\Sigma F_\rightarrow = \Sigma F_\leftarrow$

$q.E = T\,0.6$

$E = \dfrac{0.6*0.1}{8*10^{-6}}$

$= \dfrac{6*10^{-2}}{8*10^{-6}} = 7.5*10^{3}$ N/c

لو كانت سطحة

$E = 400\ N/c\,(-\hat{i})$

Electric Field due to distribution of charge.

المجال الناتج عن توزيع من الشحنات.

1) linear charge dist.

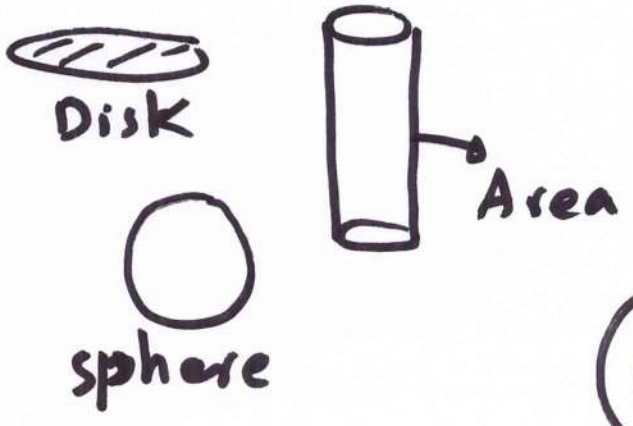


$\Rightarrow$ linear charge (λ) density
كثافة الشحنة الطولية.

$$q = \lambda L$$

$$dq = \lambda dL$$

2) Surface charge dist:-



$\Rightarrow$ Surface charge density
كثافة الشحنة السطحية
(σ)

$$q = \sigma A \Leftrightarrow dq = \sigma dA$$

3) Volume. charge dist.



$\Rightarrow$ Volume charge density
(ρ) كثافة الشحنة الحجمية

$$q = \rho Vol \Rightarrow dq = \rho dV$$

$D = 2r$ (r: radius, D: diameter)

محيط الدائرة $= 2\pi r$

مساحة الدائرة $= \pi r^2$

مساحة الكرة $= 4\pi r^2$

حجم الكرة $= \frac{4}{3}\pi r^3$

المساحة الجانبية للاسطوانة $= 2\pi rl$

حجم الاسطوانة $= \pi r^2 l$.



$q = \lambda L$

$dq = \lambda dx$

$\lambda = \lambda_0 x$

$$dE = \frac{k dq}{r^2} = \frac{k \lambda dx}{x^2}$$

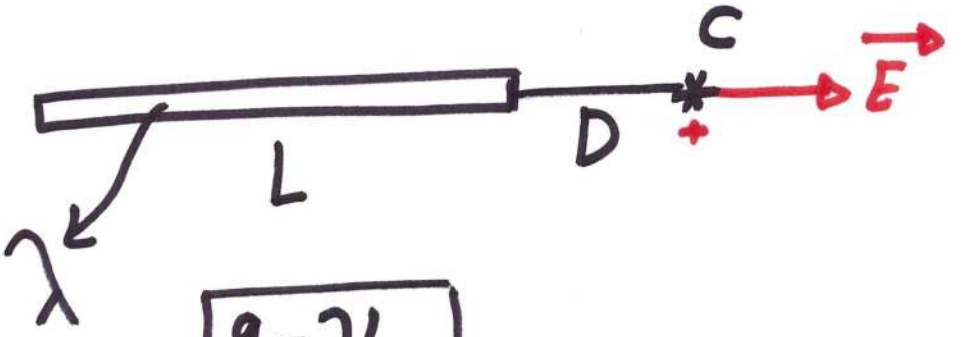
$$E = \int_{L+a}^{L+a} \frac{k\lambda dx}{x^2} = k\lambda \int_a^{L+a} \frac{dx}{x^2} = k\lambda \left[-\frac{1}{x}\right]_a^{L+a}$$

$$= k\lambda \left[\frac{-1}{L+a} - \frac{-1}{a}\right] = k\lambda \left[\frac{1}{a} - \frac{1}{L+a}\right]$$

$$= k\lambda \left[\frac{L+a-a}{a(L+a)}\right] = \frac{k\lambda L}{} \rightarrow q$$

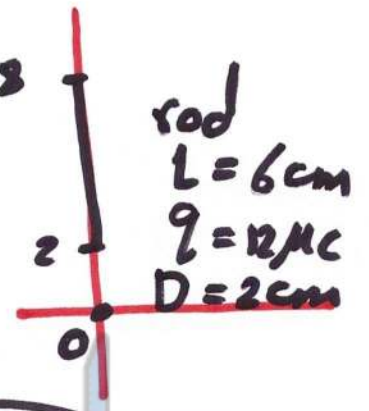
1 Rod:

a)



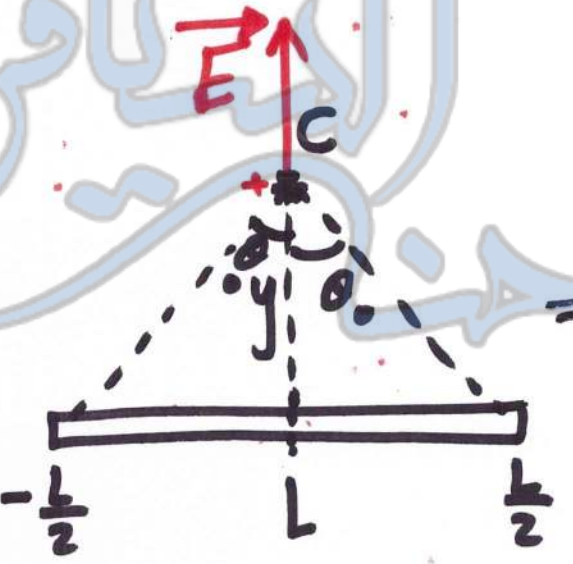
$$q = \lambda L$$

$$\Rightarrow E_c = \frac{k(\lambda L)}{D(D+L)} \rightarrow q$$



b)

finite rod



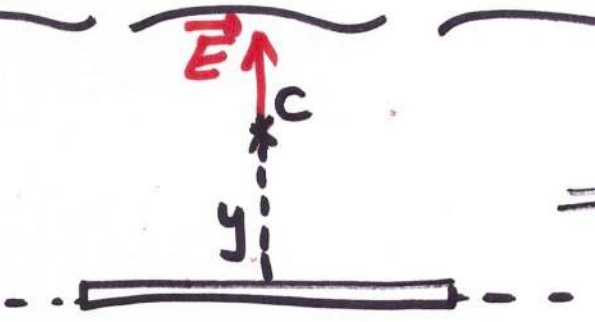
$$\Rightarrow dE = \frac{2k\lambda \sin\theta}{y}$$

or

$$E = \frac{2k\lambda L}{2y\sqrt{y^2 + L^2/4}}$$

c)

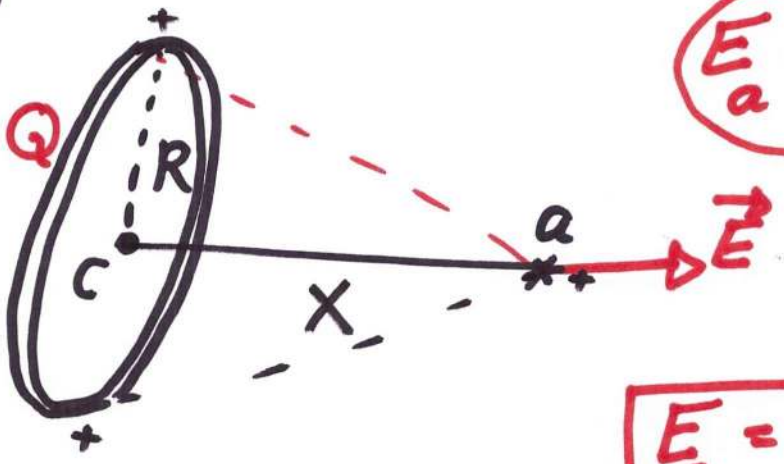
infinite rod



$$\Rightarrow E_c = \frac{2k\lambda}{y}$$

2 Ring

$$E_a = \frac{kQX}{(X^2+R^2)^{3/2}}$$

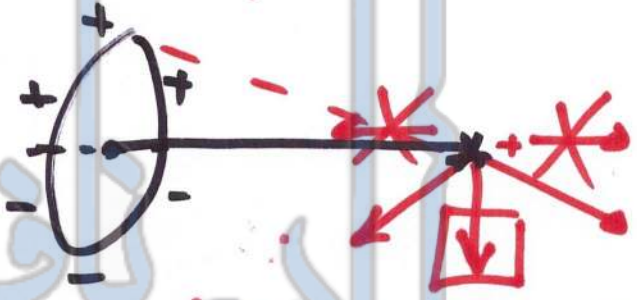


λ ↓

$Q = \lambda L$

$Q = \lambda(2\pi R)$

$E_c = 0$



3 Disk



$$E = 2\pi k\sigma\left[1 - \frac{X}{\sqrt{R^2+X^2}}\right]$$

σ

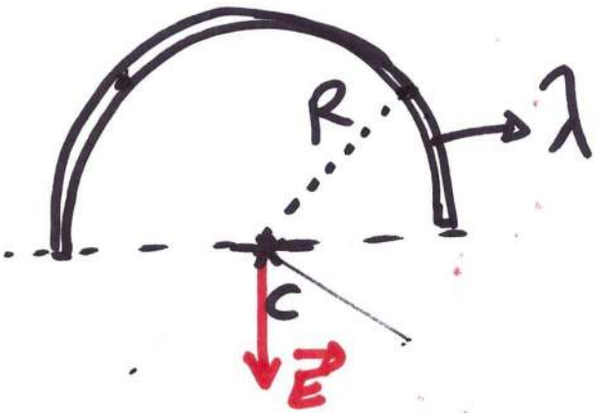
$Q = \sigma A$

$= \sigma \pi R^2$

$$E = 2\pi k\sigma\left[1 - \frac{X}{\sqrt{R^2+X^2}}\right]$$

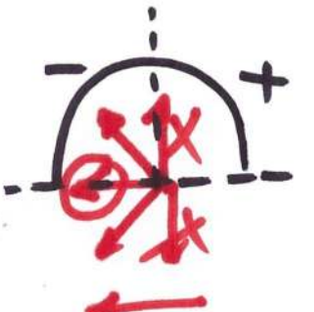
$\sigma = \frac{Q}{\pi R^2}$

5) Semi circle :



$Q = \lambda L$

$\boxed{Q = \lambda \pi R}$

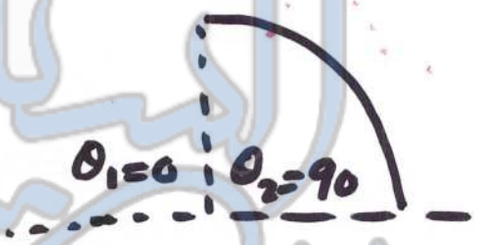




$$\boxed{E_c = \frac{k\lambda}{R}\left[\sin\theta_1 + \sin\theta_2\right]}$$

$\theta_1 = 90$

$\theta_2 = 90$



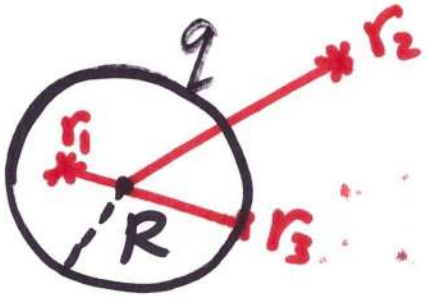
$$\Rightarrow \frac{k\lambda}{R}\left[0 + 1\right]$$



$\Rightarrow \theta_1 = 60$

$\theta_2 = 60$

6 Conducting sphere Spherical shell.



- at $r_1 < R$

$$E_{in} = 0$$

- at $r_2 > R$

$$E_{out} = \frac{kq}{r^2} = \frac{k\sigma 4\pi R^2}{r^2}$$

$$= \frac{\sigma R^2}{\epsilon_0 r^2}$$

$$q = \sigma(4\pi R^2)$$

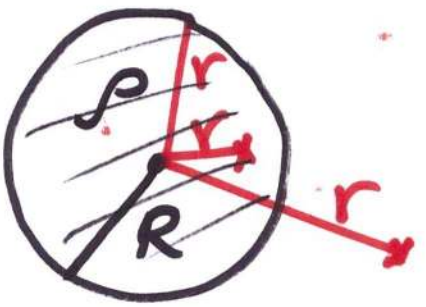
- at $r_3 = R$

$$E = \frac{kq}{R^2} = \frac{\sigma}{\epsilon_0}$$

7 Solid insulator sphere

ليست طارغه عازلة

① $r < R$ (in)



$$E_{in} = \frac{kqr}{R^3} = \frac{\rho r}{3\epsilon_0}$$

$$q = \rho\left(\frac{4}{3}\pi R^3\right)$$

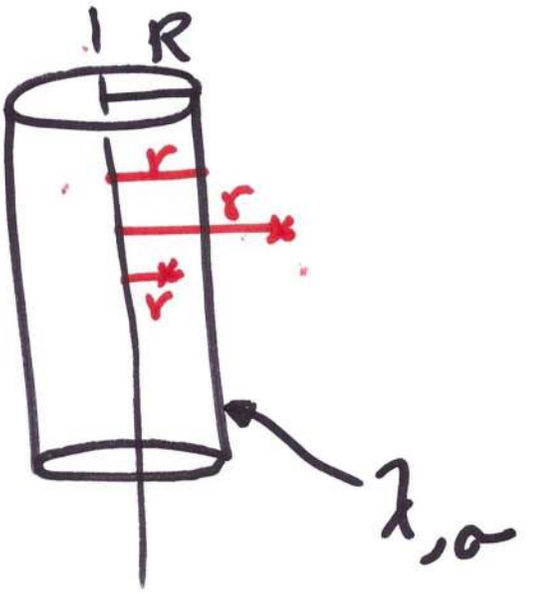
② $r \geq R$ (out)

$$E_{out} = \frac{kq}{r^2} = \frac{\rho R^3}{3\epsilon_0 r^2}$$

③ $r = R$

$$E = \frac{\rho R}{3\epsilon_0}$$

[8] Conducting infinite cylinder cylindrical shell.



1) $r < R$ (in)

$$\boxed{E_{in} = 0}$$

2) $r > R$ (out)

$$E_{out} = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2k\lambda}{r}$$

$E(2\pi r) = \frac{\lambda k}{\epsilon}$

3) $r = R$

$$E_{\text{على}} = \frac{\lambda}{2\pi\epsilon_0 R}$$

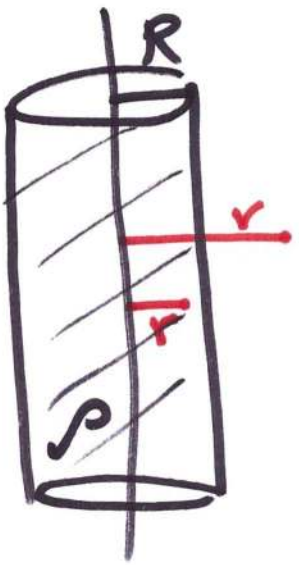
$$Q = Q$$

$$\lambda \cancel{L} = \sigma (2\pi R \cancel{L})$$

$$\sigma = \frac{\lambda}{2\pi R}$$

$$\Rightarrow E_{\text{على}} = \frac{\sigma}{\epsilon_0} \quad , \quad E_{out} = \frac{R\sigma}{\epsilon_0 r}$$

⑨ Infinite Solid insulating cylinder



1) $r < R$ (in)

$$E_{in} = \frac{\rho r}{2\epsilon_0}$$

2) $r > R$

$$E_{out} = \frac{\rho R^2}{2\epsilon_0 r}$$

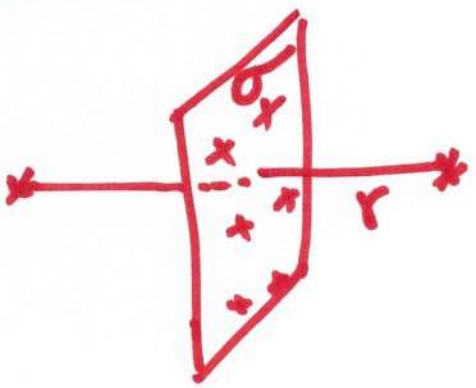
3) $r = R$

$$E = \frac{\rho R}{2\epsilon_0}$$

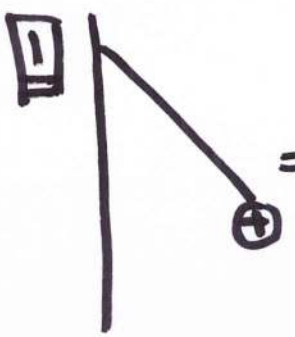
$E(2\pi rL) = \frac{\rho \pi r^2 L}{\epsilon_0}$

$E = \frac{\rho r}{2\epsilon_0}$

⑩ Infinite plate



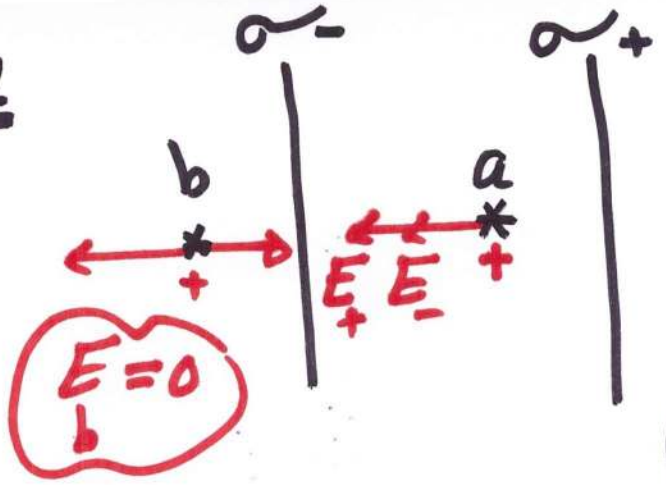
$$\Rightarrow E = \frac{\sigma}{2\epsilon_0}$$ صفيحة



⇒ find E | find σ

E = ✓ | $E = \frac{\sigma}{2\epsilon_0} \Rightarrow \sigma = 2\epsilon_0 E$

2



$$E_+ = \frac{\sigma}{2\varepsilon_0}$$

$$E_- = \frac{\sigma}{2\varepsilon_0}$$

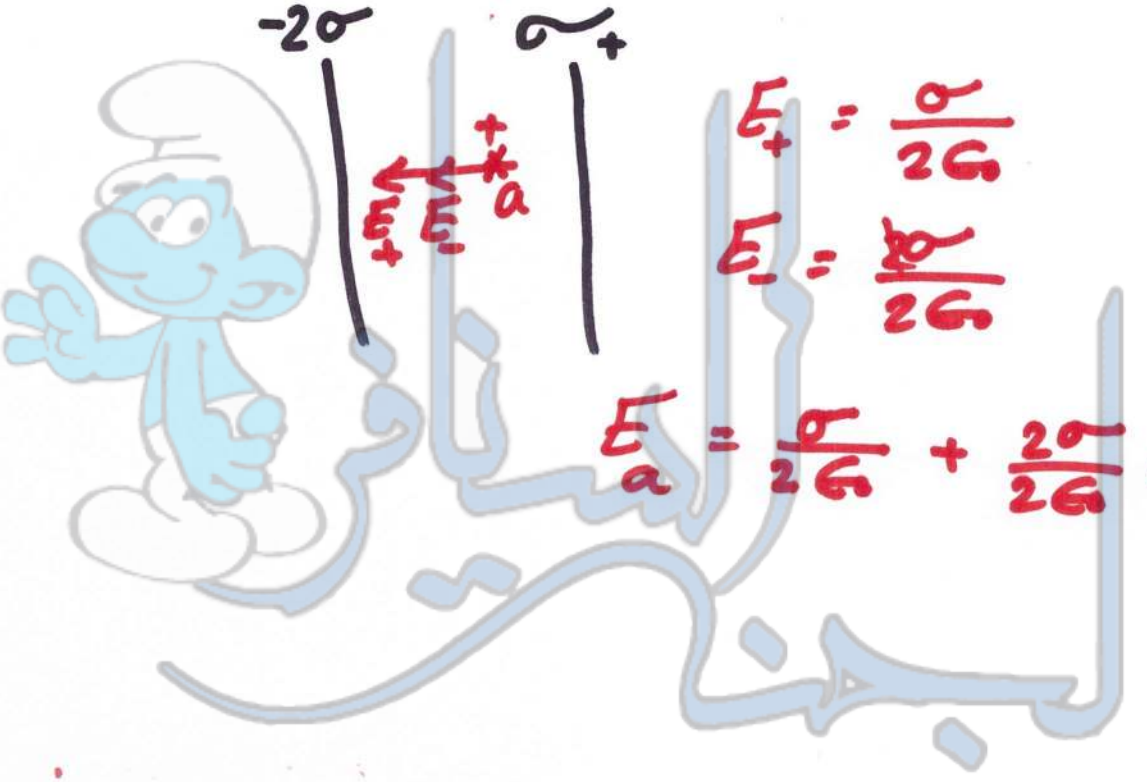
$$E_a = E_+ + E_- = \frac{\sigma}{\varepsilon_0}$$

3

$$E_+ = \frac{\sigma}{2\varepsilon_0}$$

$$E_- = \frac{2\sigma}{2\varepsilon_0}$$

$$E_a = \frac{\sigma}{2\varepsilon_0} + \frac{2\sigma}{2\varepsilon_0} = \frac{3\sigma}{2\varepsilon_0}$$




البجع الأزرق


البجعة السيئة

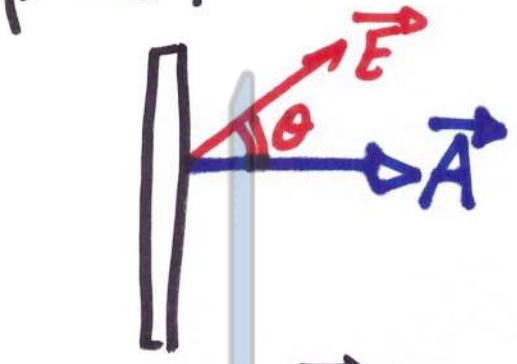
CH: 24 Electrical Flux and Gauss's law

Φ : عدد خطوط المجال التي تخترق سطحاً ما عمودياً عليه.

Flux

(Φ)

① سطح منتظم يخترقه مجال منتظم



$\Phi = \vec{E} \cdot \vec{A}$

$= EA\cos\theta$

EA

② سطح مغلق بداخله شحنات

$\Phi = \frac{\sum q_{in}}{\epsilon_0}$

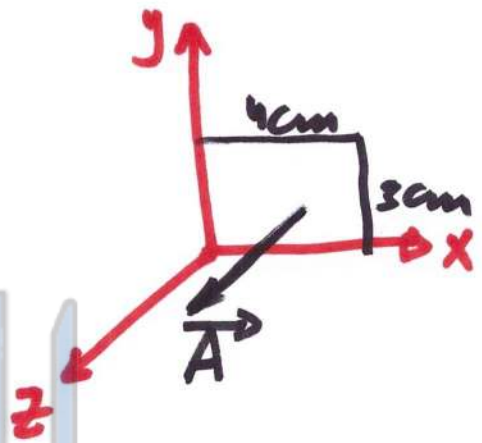


$= \frac{q_1 + q_2}{\epsilon_0}$

* السالب يعوض.

Ex: If $\vec{E} = 4\hat{i} + 2\hat{j} + 5\hat{k}$ and the Area is rectangular with dim. 4cm x 3cm lying on the xy ~~axis~~ plane. Find the electrical flux.

Sol:



$$A = 4 * 10^{-2} * 3 * 10^{-2}$$

$$A = 12 * 10^{-4}\ m^2$$

$$\vec{A} = 12 * 10^{-4}\ \hat{k}$$

$$\Phi = \vec{E} \cdot \vec{A}$$

$$= (4\hat{i} + 2\hat{j} + 5\hat{k}) \cdot (12 \times 10^{-4}\ \hat{k})$$

$$= 60 * 10^{-4}\ N \cdot m^2/C.$$

Ex: If the $\vec{E}$ is given by: $\vec{E} = 4\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{A}$ is given by: $\vec{A} = 3\hat{i} + 2\hat{j} + \hat{k}$, find

1) The net flux

$$\Phi = \vec{E} \cdot \vec{A}$$

$$= 12 + 2 - 2$$

$$= 12\ N \cdot m^2/C$$

2) angle between $\vec{E}$ and $\vec{A}$

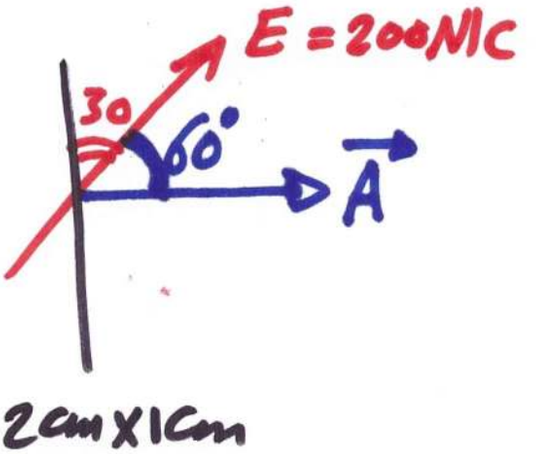
$$\vec{E} \cdot \vec{A} = E A \cos\theta_{EA}$$

$$12 = \sqrt{16+1+4}\ \sqrt{9+4+1} \cos\theta$$

$$\cos\theta = \frac{12}{\sqrt{21} * \sqrt{14}} = 0.7$$

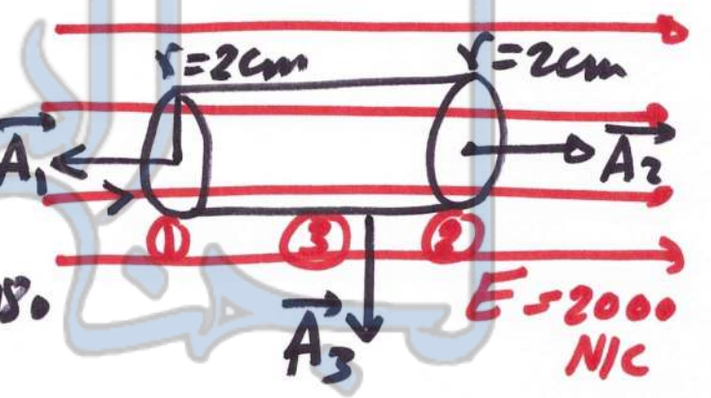
$$\theta = 45.6^\circ$$

Ex: In the figure, calculate the net flux



$$\Phi = EA\cos\theta_{EA}$$
$$= 200 \times (2*10^{-4}) \cos 60$$
$$= 0.02\ N.m^2/C.$$

Ex: Find the net Electrical Flux through the below cylinder:



Sol

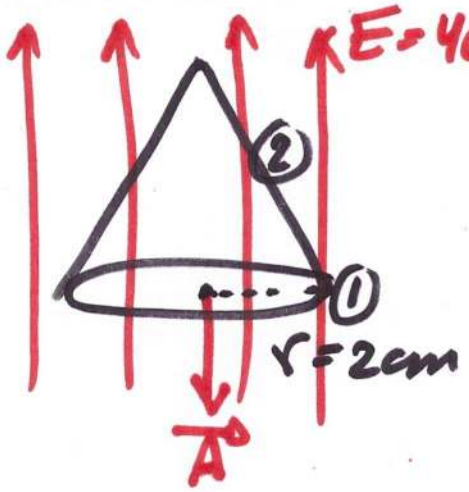
$$\Phi_1 = EA\cos\theta$$
$$= 2*10^3 * \pi (2\times10^{-2})^2 \cos 180$$
$$= -8\pi * 10^{-1}$$
$$= -0.8\pi\ N.m^2/C$$

$$\Phi_2 = EA\cos\theta$$
$$= 2\times10^3\ \pi (2*10^{-2})^2 \cos 0$$
$$= +0.8\pi\ N.m^2/C.$$

$$\Phi_3 = 2*10^3 * A_3 \cos 90 = 0$$

$$\Phi_{net} = -0.8\pi + 0.8\pi + 0 = 0$$

Ex:



Find the Flux through each surface.

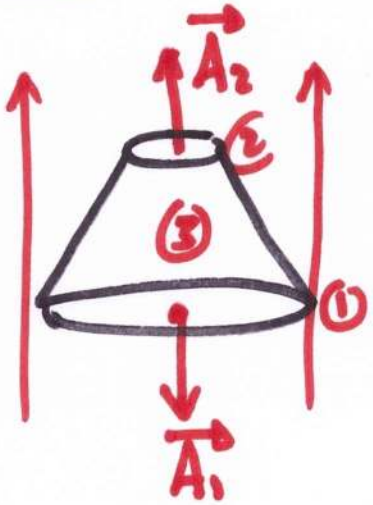
$$\Phi_1 = EA\cos\theta = 4*10^{2} * \pi (2*10^{-2})^2 \cos 180$$

$$= -16\pi \times 10^{-2} = -0.16\pi \ N\cdot m^2/C$$

$$\Phi_{net} = \Phi_1 + \Phi_2$$

$$0 = -0.16 + \Phi_2 \Rightarrow \Phi_2 = +0.16 N m^2/C .$$

Ex



$\Phi_1 = \dots\dots \cos 180 = -a$

$\Phi_2 = E A_2 \cos 0 = +b$

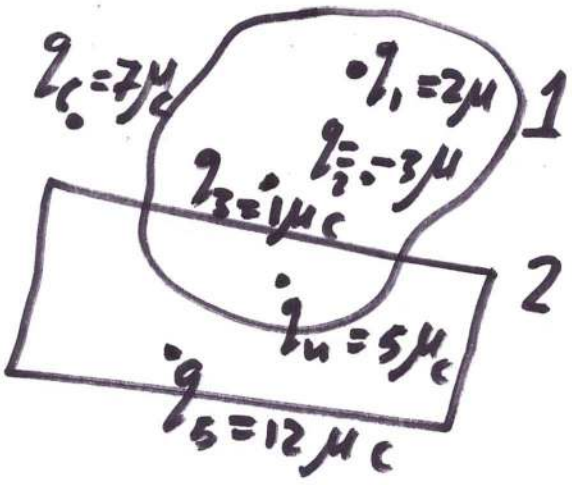
$\Phi_3 = ??$

$$\Phi_{net} = \Phi_1 + \Phi_2 + \Phi_3$$

$$0 = -(a) + (b) + \Phi_3$$

$$\boxed{\Phi_3 = -b + a} .$$

Ex: Find flux through each surface.



$$\Phi_1 = \frac{\sum q_{ins}}{\epsilon_0}$$

$$= \frac{2\mu + -3\mu + 1\mu + 5\mu}{\epsilon_0}$$

$$= \frac{5\times10^{-6}}{8.85\times10^{-12}} = 0.56\times10^{6}\ N\cdot m^2/C.$$

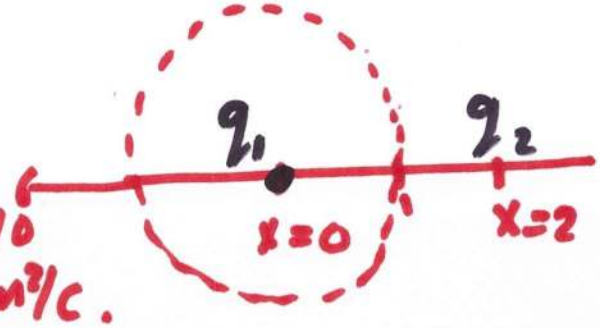
$$\Phi_2 = \frac{5\mu + 12\mu}{\epsilon_0} = \frac{17\times10^{-6}}{8.85\times10^{-12}} = 1.92\times10^{6}\ N\cdot m^2/C.$$

Ex: If we have two charges, $q_1 = 6\mu c$ at the origin, $q_2 = -4\mu c$ at $x = 2\,cm$.

Find the net flux through a sphere of radius $r = 1\,cm$, centered at the origin.

$$\Phi = \frac{\sum q_{ins}}{\epsilon_0}$$

$$= \frac{6\times10^{-6}}{8.85\times10^{-12}} = 0.68\times10^{6}\ N\cdot m^2/C.$$

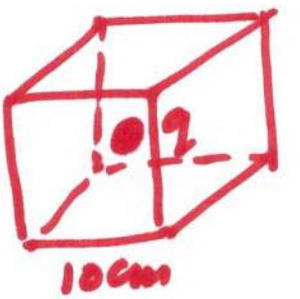


Ex: Cube of side 10 cm, Contains a charge at its center $q = 12\mu C$, Find:-

1) net flux.
2) Flux through each surface

1) $\Phi_{cube} = \frac{q}{\epsilon_0} = \frac{12 \times 10^{-6}}{8.85 \times 10^{-12}}$

$= 1.4 * 10^6 \ N.m^2/C.$



2) $\Phi_{face} = \frac{\Phi_{net}}{6} = 0.23 * 10^6 \ N.m^2/C.$

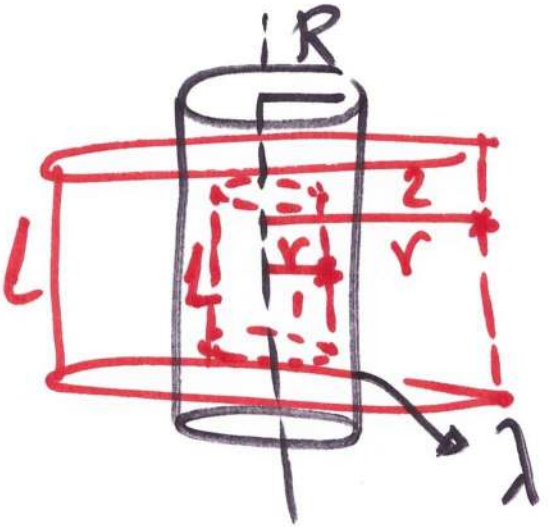
$$EA = \frac{\sum q_{ins}}{\epsilon_0} \Rightarrow \int E\,dA = \frac{q_{ins}}{\epsilon_0}$$

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$$q_{ins} = \begin{cases} \lambda L \rightarrow \int \lambda dx \\ \sigma A \rightarrow \int \sigma dA \\ \rho V \rightarrow \int \rho dv \\ q \\ 0 \end{cases}$$



Ex:



1) $EA = \frac{q_{ins}}{\epsilon_0}$

$E(2\pi r l) = \frac{Q}{\epsilon_0}$

$E_{in} = 0$

2) $r > R$

$EA = \frac{q_{ins}}{\epsilon_0} \Rightarrow E(2\pi r l) = \frac{\lambda l}{\epsilon_0}$

$$\boxed{E = \frac{\lambda}{2\pi\epsilon_0 r}}$$

Ex

1) $r < R$

$EA = \frac{q_{ins}}{\epsilon_0} \Rightarrow E(4\pi r^2) = \frac{\rho \frac{4}{3}\pi r^3}{\epsilon_0}$



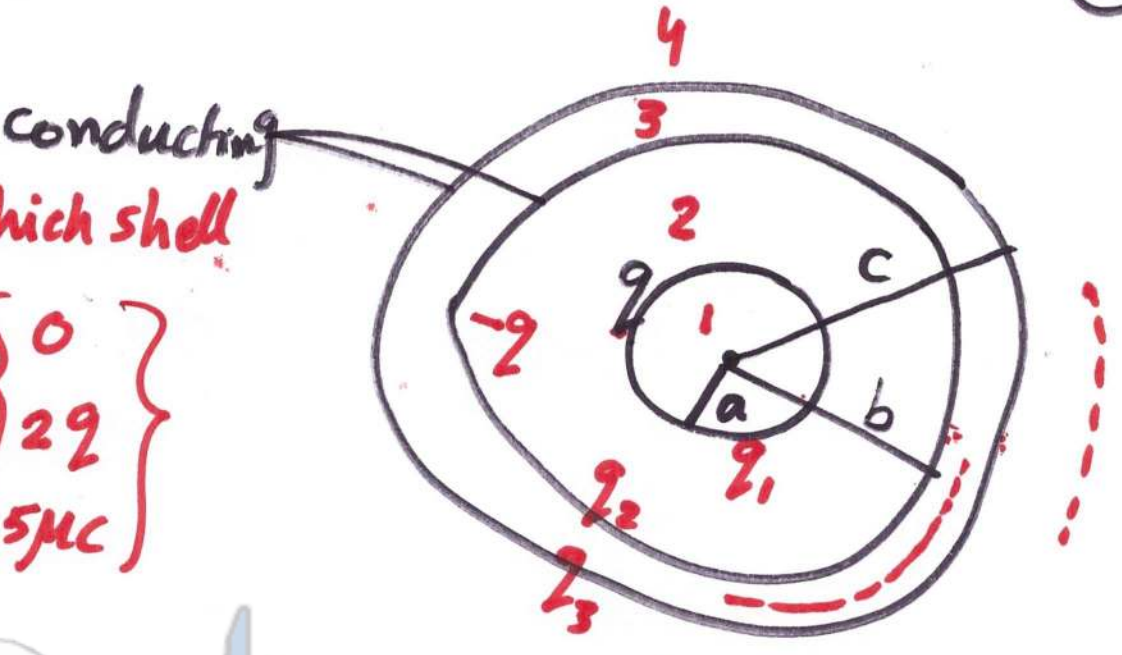
$$E = \frac{\rho r}{3\epsilon_0}$$

2) $r > R$

$EA = \frac{q_{ins}}{\epsilon_0} \Rightarrow E(4\pi r^2) = \frac{\rho(\frac{4}{3}\pi R^3)}{\epsilon_0}$

$$E = \frac{\rho R^3}{3\epsilon_0 r^2}$$

Ex:



$$q_{shell} = \begin{cases} 0 \\ 2q \\ 5\mu C \end{cases}$$

$q_{shell} = q_2 + q_3$

$0 = -q + q_3$

$\boxed{q_3 = q}$

$q_{shell} = q_2 + q_3$

$2q = -q + q_3$

$q_3 = 3q$

$\boxed{q_2 = -q_1}$

Find $\underline{E}$ every where.

① $r < a$: $E_1 = \frac{\rho r}{3\epsilon_0}$ (different) $\Big|$ $E_1 = 0$

② $a < r < b$: $E_2 = \frac{kq}{r^2} \Rightarrow E_2 = \frac{\rho R^2}{3\epsilon_0 r^2}$

③ $r > b$: $EA = \frac{q_{ins}}{\epsilon_0}$

$E(4\pi r^2) = \frac{(0)}{\epsilon_0}$

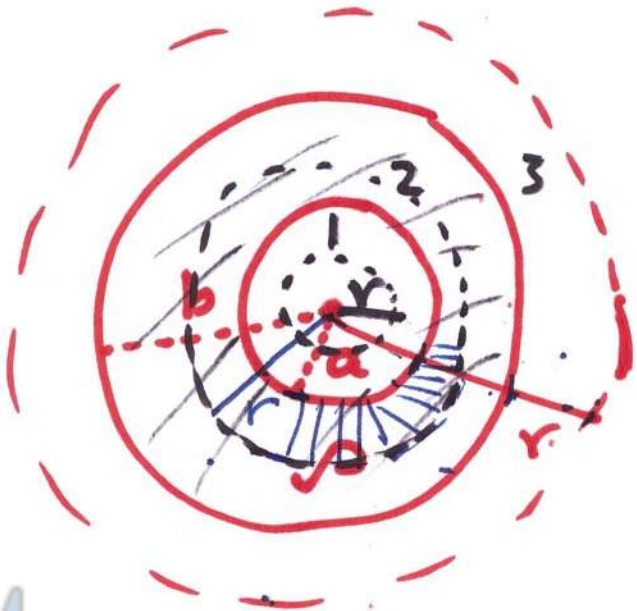
$E_3 = 0$

④ $r > c$

$E(4\pi r^2) = \frac{q_3}{\epsilon_0}$

$E = \frac{q}{4\pi\epsilon_0 r^2}$

Ex



1) $r < a$

$$EA = \frac{q_{ins}}{\epsilon_0} \Rightarrow E(4\pi r^2) = \frac{0}{\epsilon_0} \Rightarrow E_1 = 0$$

2) $a < r < b$

$$EA = \frac{q_{ins}}{\epsilon_0} \Rightarrow E(4\pi r^2) = \frac{\rho\left(\frac{4}{3}\pi r^3 - \frac{4}{3}\pi a^3\right)}{\epsilon_0}$$

$$E = \frac{\rho(r^3 - a^3)}{3\epsilon_0 r^2} = \frac{\rho\left(r - \frac{a^3}{r^2}\right)}{3\epsilon_0}$$

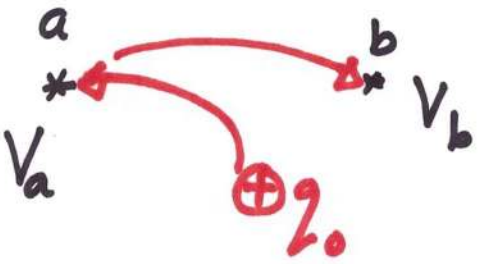
3) $r > b$

$$EA = \frac{q_{ins}}{\epsilon_0} \Rightarrow E(4\pi r^2) = \frac{\rho\left(\frac{4}{3}\pi b^3 - \frac{4}{3}\pi a^3\right)}{\epsilon_0}$$

$$\boxed{E_3 = \frac{\rho(b^3 - a^3)}{3\epsilon_0 r^2}}$$

CH:25 Electrical Potential

Volt ← (V) الجهد الكهربائي



Potential difference between a and b.

$$\Rightarrow \underset{\text{الجهد}}{V_{ab}} = V_a - V_b$$

U: Potential energy

$$\left.\begin{aligned} U_a &= q_o V_a \\ U_b &= q_o V_b \end{aligned}\right] \Rightarrow \underset{a\to b}{\Delta U} = U_b - U_a = q_o[V_b - V_a]$$

$$= q_o \underset{a\to b}{\Delta V}$$

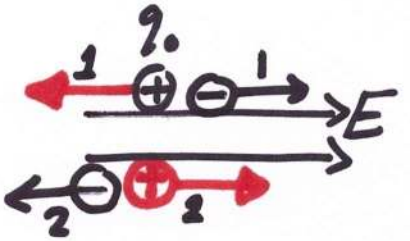
$$= q_o \underset{ba}{V}$$

[1] V due to Point charge

$$\underset{a\to b}{W} = \underset{a\to b}{\Delta U} = q_o \underset{a\to b}{\Delta V} = q_o \underset{ba}{V} \quad [\Delta K = 0]$$

[2] V due to Uniform E field:

① $\underset{a\to b}{W} = \underset{a\to b}{\Delta U} = q_o \underset{a\to b}{\Delta V}$ $[\Delta K = 0]$



② $\underset{a\to b}{W} = \underset{(\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2)}{\Delta K} = -\underset{a\to b}{\Delta U} = -q_o \underset{a\to b}{\Delta V}$

[3] V due to dist. of charges:

$$\underset{a\to b}{W} = \underset{a\to b}{\Delta U} = q_o \underset{a\to b}{\Delta V} \quad [\Delta K = 0]$$

1 Electrical Potential due to Point charge 2

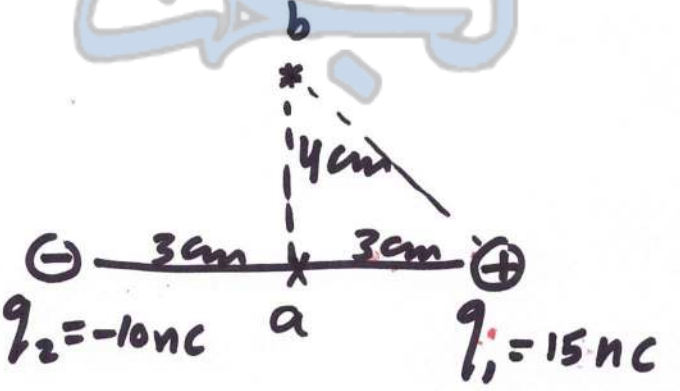
$$V = \frac{kq}{r}$$



* إشارة تعوض .
* الجهد ليس له اتجاه .
* لحساب الجهد الكلي: $V_{total} = V_1 + V_2 + V_3 + \dots$

Ex: In the figure, Find:

1) Potential difference between a and b.
2) Work needed to bring $q_o = 2\mu C$ from a to b.
3) " " " " a to ∞



Sol

1) $V_a = V_1 + V_2$

$$= \frac{9\times10^9 \times 15\times10^{-9}}{3\times10^{-2}} + \frac{9\times10^9 \times -10\times10^{-9}}{3\times10^{-2}}$$

$$V_a = 45\times10^2 - 30\times10^2 = 15*10^2 \text{ Volt}.$$

$$V_b = \frac{9\times10^9 \times 15\times10^{-9}}{5\times10^{-2}} + \frac{9\times10^9 * -10\times10^{-9}}{5\times10^{-2}} = (27-18)\times10^2$$

$$= 9*10^2 \text{ Volt}$$

$$V_{ab} = V_a - V_b$$

$$= 15 \times 10^2 - 9 \times 10^2 = 6 * 10^2 \text{ Volt.}$$

$$\Delta V_{a \to b} = V_{ba} = -V_{ab} = -6 * 10^2 \text{ Volt.}$$

---

2) $$W_{a \to b} = q_0 \, \Delta V_{a \to b}$$

$$= 2 * 10^{-6} * [-6 * 10^2]$$

$$= -12 * 10^{-4} \text{ J.}$$

---

3) $$W_{a \to \infty} = q_0 [V_\infty - V_a]$$

$$= 2 * 10^{-6} [0 - 15 \times 10^2]$$

$$= -30 * 10^{-4} \text{ J.}$$

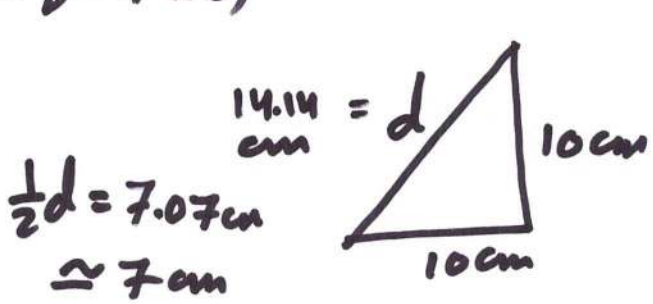
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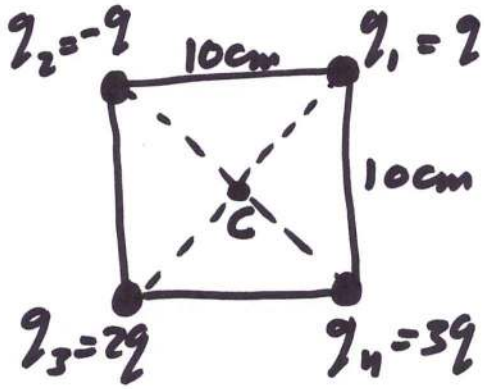
4 what is the potential at the Position of q_1.

$$V_{at\, q_1} = \frac{9 * 10^9 * -10 * 10^{-9}}{6 * 10^{-2}} = -15 * 10^2 \text{ Volt.}$$

Ex: Find the Electrical Potential at C

$(q = 7nc)$





$$V_C = V_1 + V_2 + V_3 + V_4$$

$$= \frac{9*10^9 * 7*10^{-9}}{7*10^{-2}} + \frac{9*10^9 * -7*10^{-9}}{7*10^{-2}} + \frac{9*10^9 * 14*10^{-9}}{7*10^{-2}}$$

$$+ \frac{9*10^9 * 21*10^{-9}}{7*10^{-2}}$$

$$V_c = \frac{9*10^9 * 10^{-9}}{7*10^{-2}} [14 + 21] = 45*10^2 \text{ Volt}$$

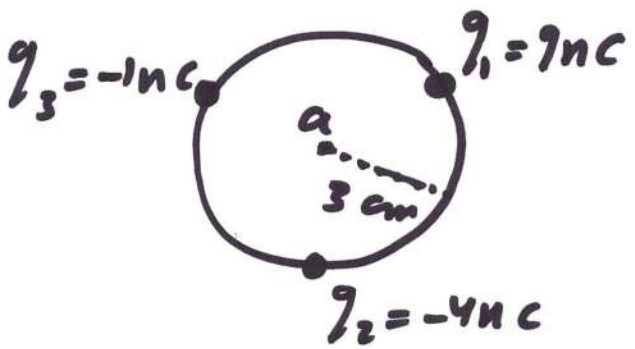
b) If we want the Work to bring an electron $q_e = -1.6*10^{-19} C$ from ∞ to C:

$$W = q_o [V_C - V_\infty]$$

$$= -1.6*10^{-19} [45*10^2 - 0]$$

$$= -72*10^{-17} J.$$

Ex: what is the Work needed to bring a proton from ∞ to a.



$$\Rightarrow W = q_o [V_a - V_\infty]$$

$$= +1.6 * 10^{-19} [V_a]$$

$$= 1.6 \times 10^{-19} * 12 * 10^{2} \quad = 19.2 * 10^{-21} \text{ J}.$$

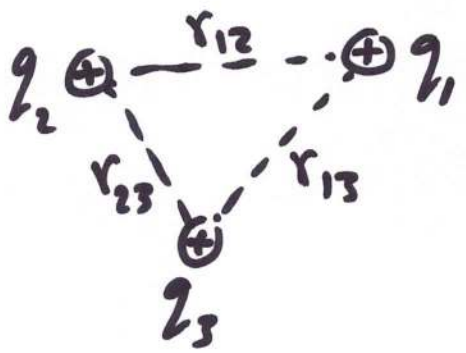
$$V_a = V_1 + V_2 + V_3$$

$$= \frac{9 \times 10^{9}}{3 \times 10^{-2}} [9 \times 10^{-9} + -4 \times 10^{-9} + -1 * 10^{-9}]$$

$$= 3 \times 10^{7} [4 \times 10^{-9}] = 12 \times 10^{-2} \text{ Volt}.$$

---

* Energy stored in q_1 (the one already in place):



$$U_1 = U_{12} + U_{13}$$

$$= \frac{k q_1 q_2}{r_{12}} + \frac{k q_1 q_3}{r_{13}}$$

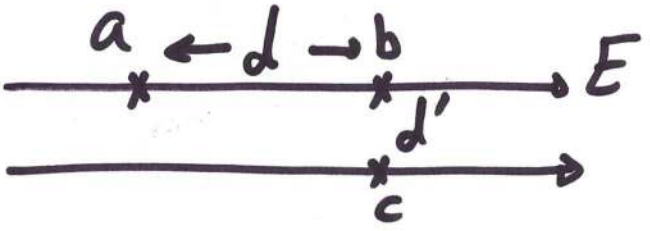
* Energy stored in the system:

$$U_{sys} = U_{12} + U_{13} + U_{23} \longrightarrow \frac{k q_2 q_3}{r_{23}}$$

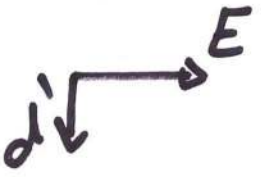
2 Electric Potential due Uniform $\underline{E}$:

$$V_{ba} = \underset{a\to b}{\Delta V} = -Ed\cos\theta_{Ed}$$

$$= -\vec{E}\cdot\vec{d}$$



* $\underset{b\to c}{\Delta V} = -E(d')\cos 90$



$$\underset{b\to c}{\Delta V} = 0$$

$$V_c - V_b = 0 \Rightarrow \boxed{V_c = V_b}$$

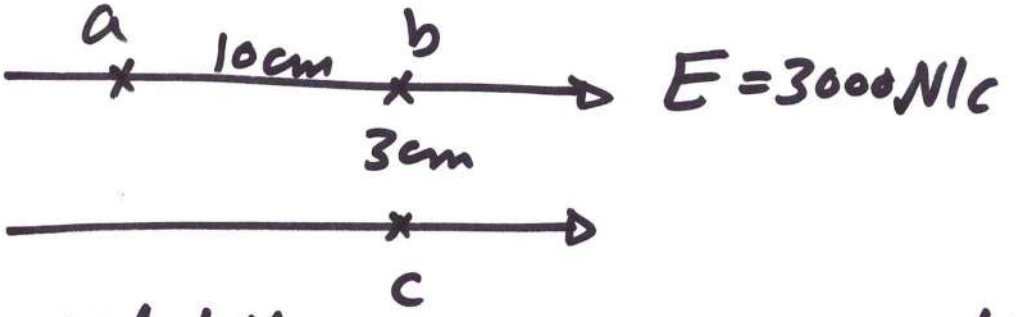
* $\underset{a\to c}{\Delta V} = \underset{a\to b}{\Delta V} + \cancel{\underset{b\to c}{\Delta V}}$

* if we know V_a ⇒ what is V_b:

$$\underset{a\to b}{\Delta V_b} = \checkmark \Rightarrow V_b - V_a = \checkmark$$

$$V_b = V_a + \checkmark .$$

Ex:



1) Find Potential difference between a and b.
   b & c
   a & c.

2) Find the Work needed to move $q_o = 2\mu c$ from a to b.

3) if q_o in Part(2) started from rest ($m = 2gm$), what is its final speed.

4) Find the Work needed to move $q_o = 5\mu c$ from b to a.

5) Find the Work needed to move $q_o = -1mc$ from a to b.

6) if $V_a = 20$ Volt, what is V_b.

7*) What is the Work needed to move q_o from a → a along the path a→b→c→a

Sol :

1) $V_{ab} = \Delta V_{b\to a} = -Ed\cos\theta_{a\to b}$



$= -3\times10^{3} * 10*10^{-2} * \cos180$

$= 300$ Volt .

* $\Delta V_{a\to b} = -300$ Volt.

* $\Delta V_{c\to b} = 0$

* $V_{ac} \Rightarrow \Delta V_{c\to a} = \Delta V_{c\to b} + \Delta V_{b\to a}$

$= 300$ Volt.

2) $W = -q_o \Delta V_{a\to b}$

$= -2*10^{-6}[-300] = 600*10^{-6}$ J

3) $\Delta K_{a\to b} = -q_o \Delta V_{a\to b}$

$\frac{1}{2}mV_b^2 - \frac{1}{2}mV_a^2 = 600*10^{-6}$

$\frac{1}{2} * 2*10^{-3} * V_b^2 = 600*10^{-6} \Rightarrow V_b = 0.77$ m/s.

④ $W_{b\to a} = q_o \Delta V_{b\to a}$

$= 5 * 10^{-6} [300]$

$= 1500 * 10^{-6}$ J .

⑤ $W_{a\to b} = q_o \Delta V_{a\to b}$

$= -1 * 10^{-9} * -300$

$= 300 * 10^{-9}$ J .

⑥ $\Delta V_{a\to b} = -300$

$V_b - V_a = -300$

$V_b - 20 = -300 \Rightarrow V_b = -280$ Volt.

⑦ $W_{a\to a} = q_o \Delta V_{a\to a}$

$= q_o [\Delta V_{a\to b} + \Delta V_{b\to c} + \Delta V_{c\to a}]$

$= q_o [-300 + 0 + 300]$

$=$ Zero .

[3] Potential due to distribution of charges (الجهد الناتج عن توزيع من الشحنات)

$$q = \lambda L \quad , \quad q = \sigma A \quad , \quad q = \rho V$$

$$dq = \lambda dx$$

Ex:



$$V = \frac{kq}{r} \Rightarrow dV = \int \frac{k\,dq}{r} = \int_{r=x} \frac{k\lambda\,dx}{}$$

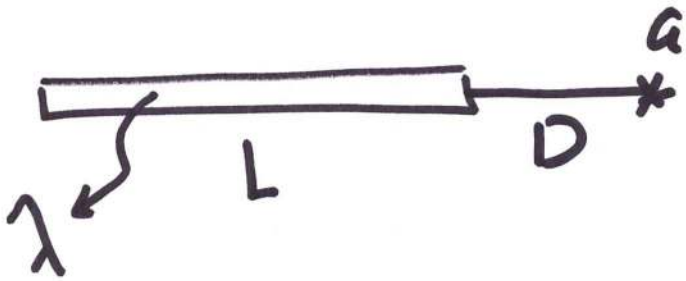
$$dV = k\lambda \int_{D}^{D+L} \frac{dx}{x} = k\lambda \ln x \Big|_{D}^{D+L}$$

$$= k\lambda \left[\ln(D+L) - \ln D\right]$$

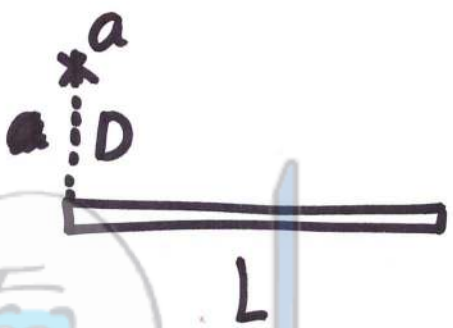
$$= k\lambda \ln\left(\frac{D+L}{D}\right)$$

(1) rod:

$q = \lambda L$



$$V_a = k\lambda \ln\left(\frac{D+L}{D}\right)$$

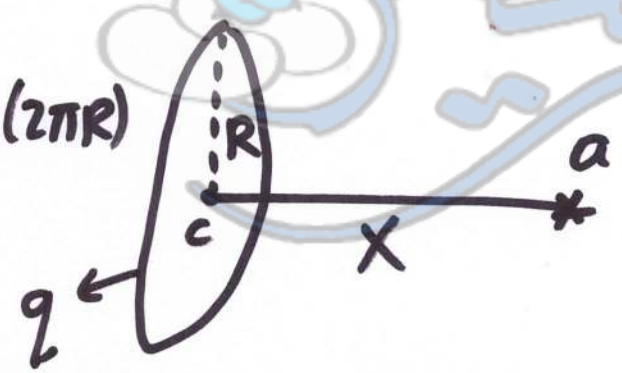


$$\Rightarrow V_a = k\lambda \ln\left[\frac{L+\sqrt{D^2+L^2}}{D}\right]$$

---

[2] ring:

$q = \lambda(2\pi R)$



$$V_a = \frac{kq}{\sqrt{R^2+x^2}}$$

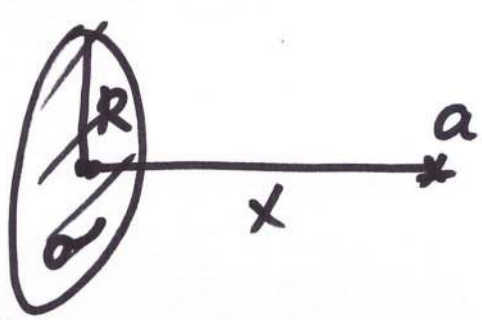
$$V_c = \frac{kq}{R}$$

---

[3] Disk

9×10^9

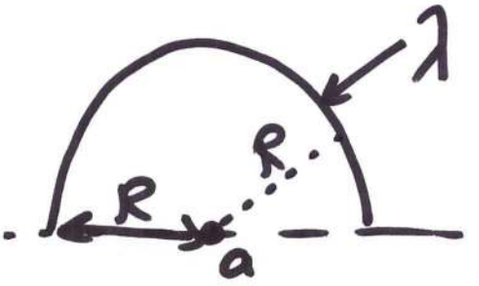
$$V_a = 2\pi k\sigma\left[\sqrt{R^2+x^2} - x\right]$$



$q = \sigma\pi R^2$

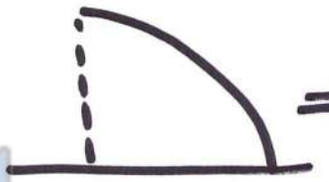
[4] Semicircle:

$q = \lambda \pi R$



$V_a = k\lambda\theta$

$3.14 = \pi$



$\Rightarrow \theta = \frac{\pi}{2} \Rightarrow V = \frac{k\lambda\pi}{2}$



$\Rightarrow V_a = V_1 + V_2 + V_3$.

$= k\lambda \ln\left(\frac{D+L}{D}\right) + k\lambda\theta + k\lambda \ln\left(\frac{D+L}{D}\right)$

$= k\lambda \left(\ln\left(\frac{R+2R}{R}\right) + k\lambda\pi + k\lambda \ln\left(\frac{R+3R}{R}\right)\right.$

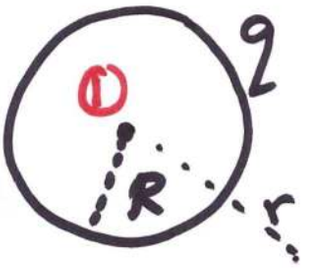
$= k\lambda \ln 3 + k\lambda\pi + k\lambda \ln 4$

$= (\vec{F})$

(5) Conducting sphere:-

1) $r < R$ (inside)

$$V_{in} = \frac{kq}{R} = V_{surface}$$



$q = \sigma 4\pi R^2$

2) $r > R$ (outside)

$$V_{out} = \frac{kq}{r}$$

---

(6) insulating sphere



1) $r < R \Rightarrow V_{in} = \frac{kq}{2R}\left(3 - \frac{r^2}{R^2}\right)$ $\quad q = \rho \frac{4}{3}\pi R^3$

2) $r \geq R \Rightarrow V_{out} = \frac{kq}{r}$

$$\Delta U = W = q_o \Delta V$$

(14) إذا اعطاك في السؤال $\vec{E}$ وطلب V :

(العادة)

$$\Delta V = -\int E_i dx - \int E_j dy - \int E_k dz - \int E dr$$

Ex: If $E = \frac{4}{r^2}$, Find V from ∞ to 5cm

sol

$$\Rightarrow V = -\int E dr$$

$$= -\int_{\infty}^{5cm} \frac{4}{r^2} dr = -4\left[\frac{-1}{r}\right]_{\infty}^{5cm}$$

$$= \frac{4}{5\times10^{-2}} - 0 = 80 \text{ Volt.}$$

Ex: If $\vec{E} = 3x^2yz\,\hat{i} - 5xz\,\hat{j} + 2zy\,\hat{k}$

Find V

$$V = -\int 3x^2yz\,dx - \int -5xz\,dy - \int 2zy\,dz$$

$$= -3yz\frac{x^3}{3} + 5xzy - 2y\frac{z^2}{2}$$

$$= -yzx^3 + 5xyz - yz^2. \quad (1,1,-2)$$

(15) إذا أعطاك $\vec{V}$ فكيف تحسب $\vec{E}$ ؟

$$\vec{E} = -\hat{i}\frac{\partial V}{\partial x} - \hat{j}\frac{\partial V}{\partial y} - \hat{k}\frac{\partial V}{\partial z} - \hat{r}\frac{\partial V}{\partial r}$$

Ex: If $V = 2x^2yz^3 + 5xz$, Find:-

1) $\vec{E}$ at $(1,2,-1)$
2) magnitude of $\vec{E}$ at $(1,2,-1)$
3) if $q_0 = 2\mu C$, Find Force.

Sol:

1) $\frac{\partial V}{\partial x} = 4xyz^3 + 5z \quad \left|\frac{\partial V}{\partial y} = 2x^2z^3\right| \quad \frac{\partial V}{\partial z} = 6x^2yz^2 + 5x$

$\Rightarrow \frac{\partial V}{\partial x} = -8 + -5 = -13$

$\frac{\partial V}{\partial y} = -2 \quad \left|\frac{\partial V}{\partial z} = 12 + 5 = 17\right.$

$\Rightarrow \vec{E} = -(-13)\hat{i} - (-2)\hat{j} - (17)\hat{k}$

$$\boxed{\vec{E} = 13\hat{i} + 2\hat{j} - 17\hat{k}}$$

2) $|\vec{E}| = \sqrt{13^2 + 2^2 + 17^2}$ N/C

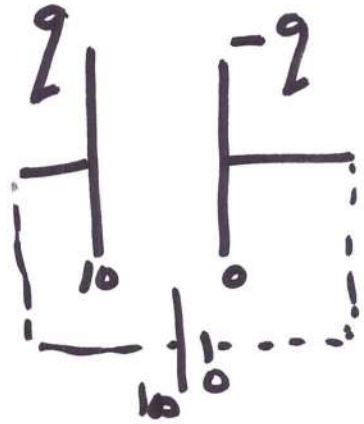
3) $\vec{F} = q_0\vec{E} = 2*10^{-6}(13\hat{i} + 2\hat{j} - 17\hat{k})$.

CH:26 Capacitors

C ≡ Capacitance
المواسعة الكهربائية

$$C = \frac{q}{\Delta V}$$

$$\Delta V = V_+ - V_-$$



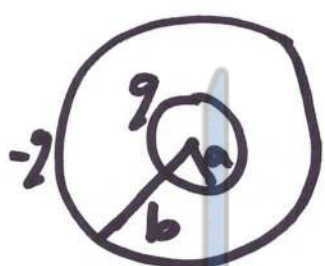
* أي جسمين متلامسين يصبح لهما نفس الجهد.



عام

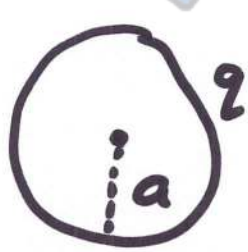
Farad

① Spherical capacitor:



$$C = \frac{4\pi\epsilon_0}{\frac{1}{a} - \frac{1}{b}}$$

② sphere:



$$C = 4\pi\epsilon_0 a$$

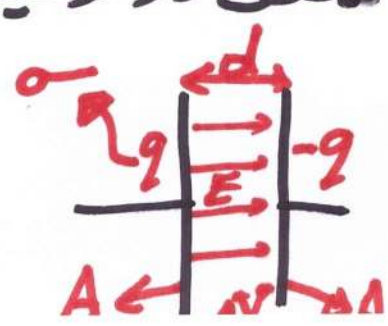
③ cylindrical capacitor



$$C = \frac{2\pi\epsilon_0 l}{\ln\left(\frac{b}{a}\right)}$$

④ Two Parallel plates Capacitor
مكثف ذو لوحين متوازيين

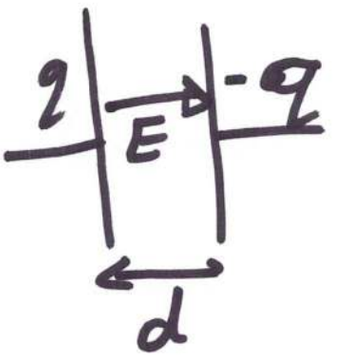
$$C = \frac{\epsilon_0 A}{d}$$



$$C = \frac{\epsilon_0 A}{d} \qquad E = \frac{\sigma}{\epsilon_0}$$

$$C = \frac{q}{\Delta V} \qquad q = \sigma A$$

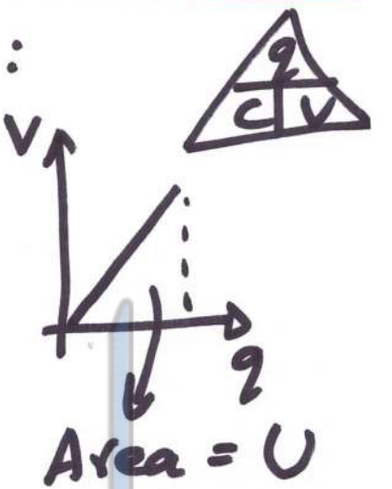
لوحين $\Delta V = Ed$ لوحين



---

* Energy stored in Capacitor :

$$U = \frac{1}{2} q V = \frac{1}{2} C V^2 = \frac{1}{2} \frac{q^2}{C}$$



$$W = \Delta U = U_f - U_i$$

---

u ≡ Energy density (كثافة الطاقة)

energy per Unit Volume.

الطاقة في وحدة الحجوم.

$$u = \frac{U}{\text{Volume}} = \frac{1}{2} \epsilon_0 E^2$$

Ad



Ex: Two Parallel Plates Capacitor has a surface charge density (σ) $= 2*10^{-6}$ C/m² and Area of 2 cm² ($A = 2*10^{-4}$ m²) with separation (البعد بين اللوحين) 10 cm ($d = 10*10^{-2}$ m),

Find:-

① Capacitance (C). $\Rightarrow C = \frac{\epsilon_0 A}{d}$

② Charge (q). $\Rightarrow q = \sigma A$

③ Potential difference across the Cap. $\Rightarrow V = \frac{q}{C}$

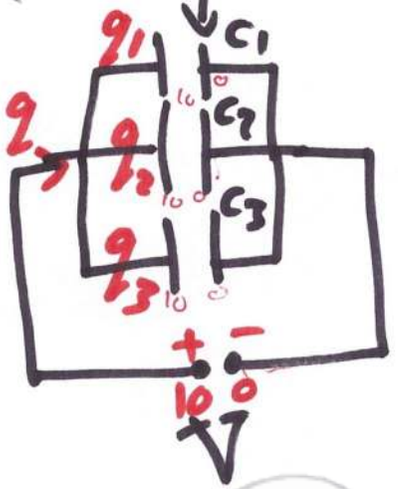
④ Electric field. $\Rightarrow E = \frac{\sigma}{\epsilon_0}$ or $E = \frac{V}{d}$

⑤ Energy stored in Cap. $\Rightarrow U = \frac{1}{2} q V = \frac{1}{2} C V^2$

⑥ energy density. (u) $\Rightarrow u = \frac{1}{2} \epsilon_0 E^2 = \frac{U}{Ad}$

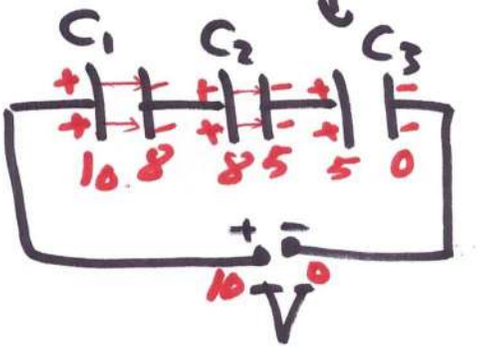
Connection of Capacitors

(التوازي) in Parallel





(توالي) in Series



- ما في تفرعات في الاصلاك

* اللوح الموجب مقابل ح باللوح السالب



* يوجد تفرع في الاصلاك، لكن لكل تفرع مكثف واحد فقط والفروع لها نفس البداية والنهاية

* اللوح الموجب مع الموجب والسالب مع السالب



* تتوزع الشحنة الكهربائية:

$$q_{eq} = q_1 + q_2 + q_3 + \dots$$

* تتوزع الشحنة على المكثفات:

$$q_{eq} = q_1 = q_2 = q_3 = \dots$$

* يتوزع الجهد الكهربائي

$$V_{eq} = V_{bat} = V_1 + V_2 + V_3 + \dots$$

* الجهد الكهربائي متساوي:

$$V_{bat} = V_{eq} = V_1 = V_2 = V_3 = \dots$$

* السعة المقلوبة

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

* السعة مجموعة:

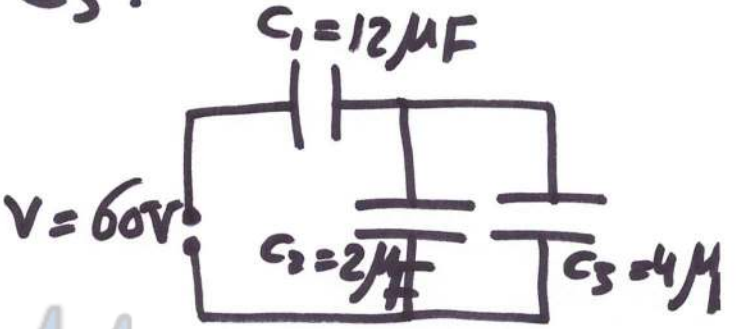
$$C_{eq} = C_1 + C_2 + C_3 + \dots$$

Ex: In the figure shown, Find:

① equivalent capacitance

② q, V for all capacitors.

③ energy stored in C_5.



* ارسم خريطة لسؤال وعين عليها المعلومة الاضافية
* نجد C_{eq}
* نجد جميع معلومات المعطى الاضافي
* في الخريطة نبدأ من المعطى الاضافي للوصول للمطلوب.

Sol:

C_2, C_3 توازي $\rightarrow C_4$

C_1, C_4 توالي $\rightarrow C_5$

$V_5 = 60V$

① $C_4 = C_2 + C_3$

$= 2\mu + 4\mu$

$= 6\mu F$

$\frac{1}{C_5} = \frac{1}{C_1} + \frac{1}{C_4}$

$\frac{1}{C_5} = \frac{1}{12\mu} + \frac{1}{6\mu}$

$\boxed{C_5 = 4\mu F}$

$\Downarrow$

$q_5 = C_5 V_5$

$\boxed{q_5 = 240\mu C}$

② $q_1 = q_4 = q_5 = 240\mu C$

$\rightarrow V_4 = \frac{q_4}{C_4} = 40$ Volt

$\rightarrow V_1 = \frac{q_1}{C_1} = 20$ Volt.

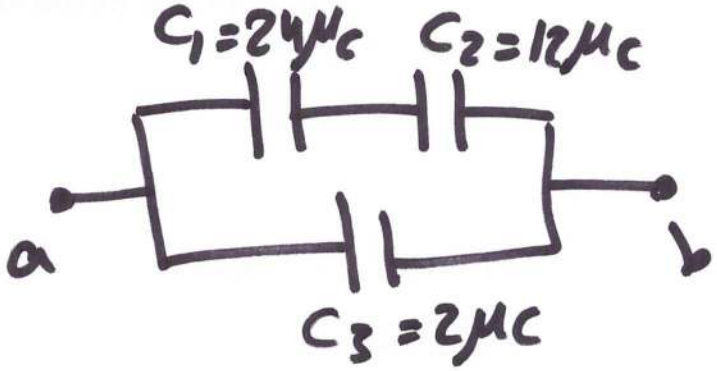
$V_2 = V_3 = V_4 = 40$ Volt

$\rightarrow q_3 = C_3 V_3 = 160\mu C$

$\rightarrow q_2 = C_2 V_2 = 80\mu C$

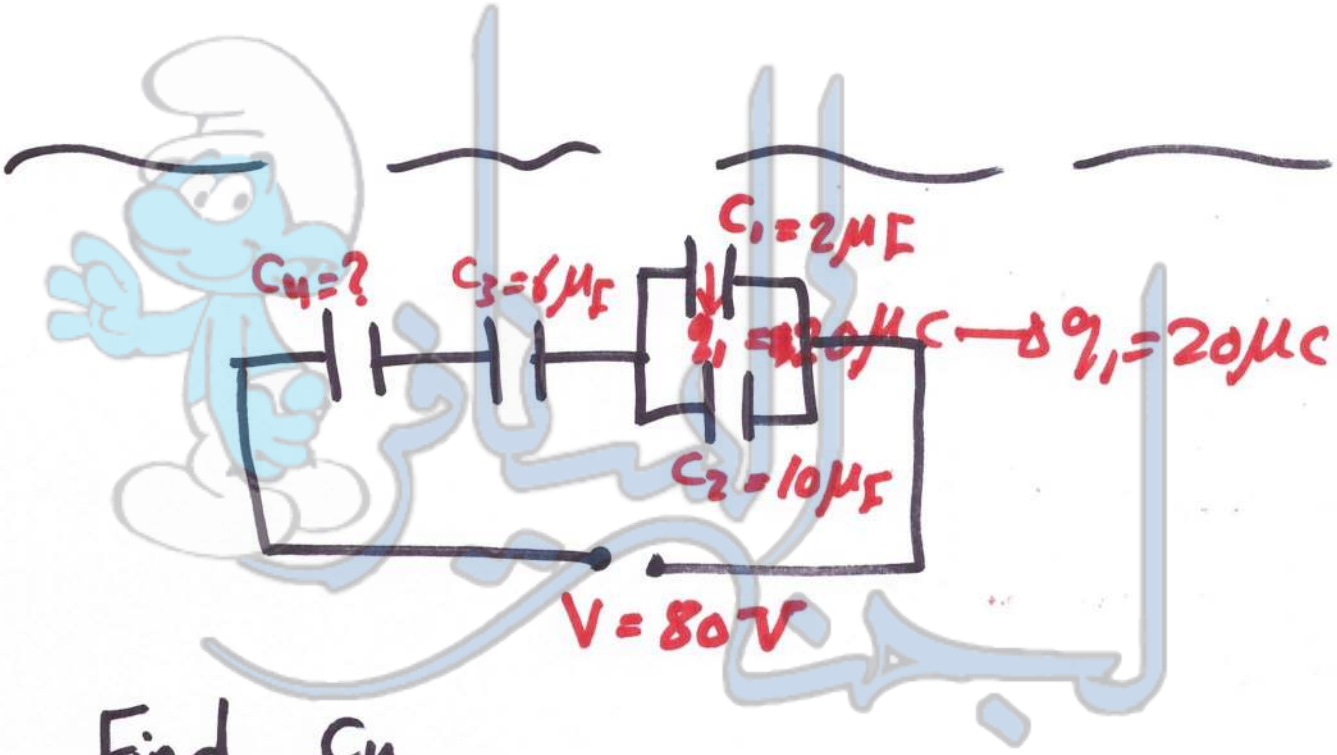
③ $U_5 = \frac{1}{2} q_5 V_5 = \frac{1}{2} \times 240\mu \times 60 = 7200\mu J$

Ex:



$V_{ab} = 120V$

Find : C_{eq}, V, Q for all C_s, U_3.



Find C_4.

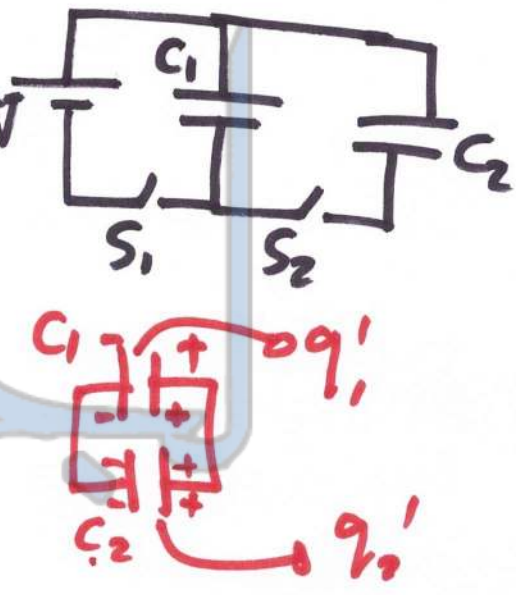
Q, V for all C's.

Ex: Capacitor has $C_1 = 6\mu F$, Connected to Voltage of 40 Volt. then disconnect from battary, and then Connected to another Capacitor $C_2 = 3\mu F$ initially Uncharged. Find q and V for C_1 and C_2 after Connected togather.

Sol:

$C_1 = 6\mu F$, $V_1 = 40$ Volt $\Rightarrow q_1 = 240\mu C$

$C_2 = 3\mu F$, $V_2 = 0$ $\Rightarrow q_2 = 0$



$V_1' = V_2'$

$\frac{q_1'}{6\mu} = \frac{q_2'}{3\mu}$

$\Rightarrow \boxed{q_1' = 2q_2'}$

$\sum q_{bef} = \sum q'_{after}$

$q_1 + q_2 = q_1' + q_2'$

$240\mu + 0 = q_1' + q_2'$

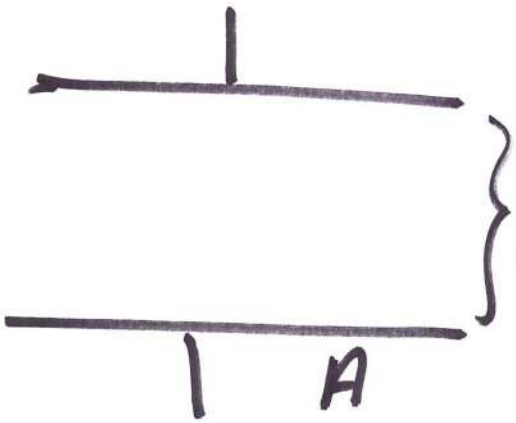
$\Rightarrow 240\mu = 2q_2' + q_2' = 3q_2'$

$\Rightarrow \boxed{q_2' = 80\mu C}$

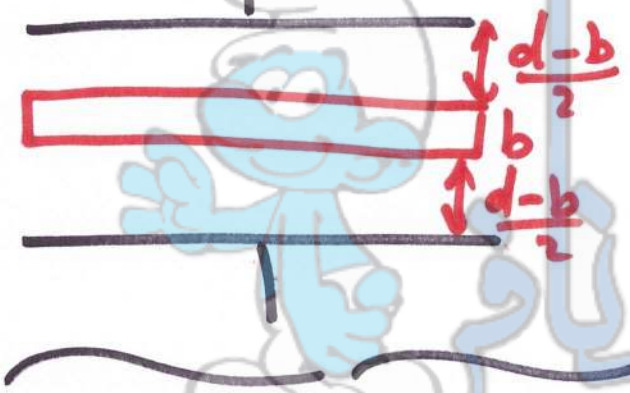
$\Rightarrow \boxed{q_1' = 160\mu C}$

$V_1' = \frac{160\mu}{6\mu} = 26.7\ V = V_2'$

metalic slap inside C



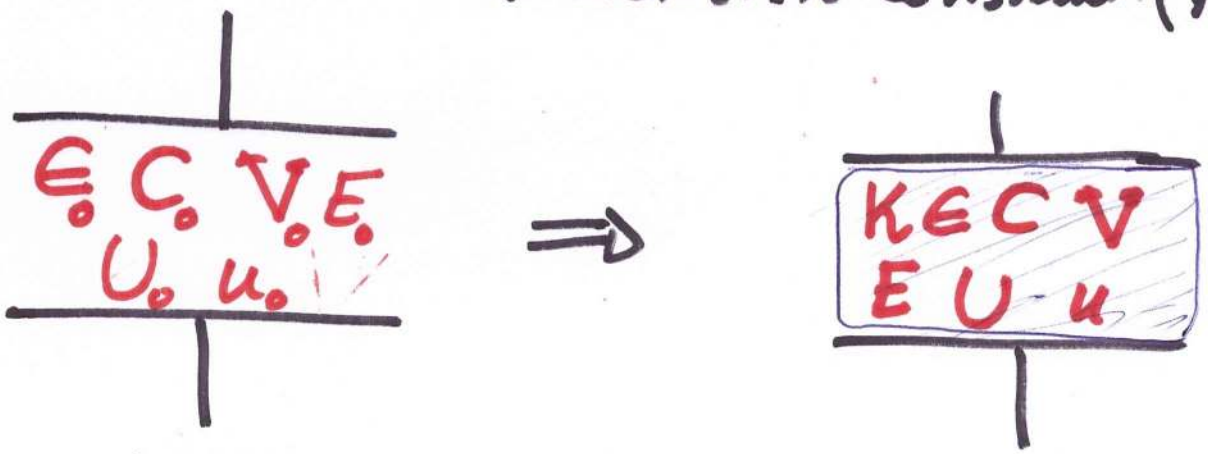
$$\Rightarrow C = \frac{\epsilon_0 A}{d}$$



$$\Rightarrow C = \frac{\epsilon_0 A}{(d-b)}$$

Insulatic material
(Dielectric material)

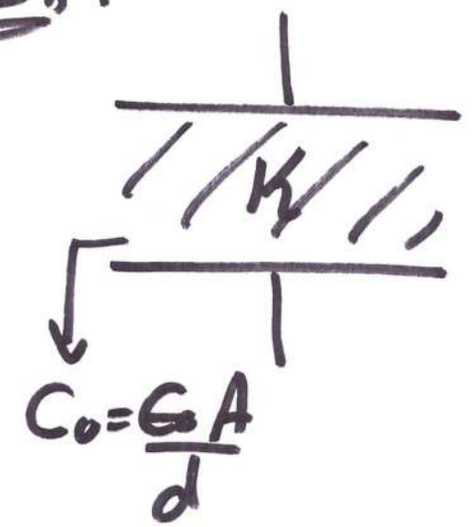
K: dielectric constant ($K = 1$ air, $K > 1$ other materials)



$$\epsilon = K\epsilon_0 \qquad C = KC_0$$

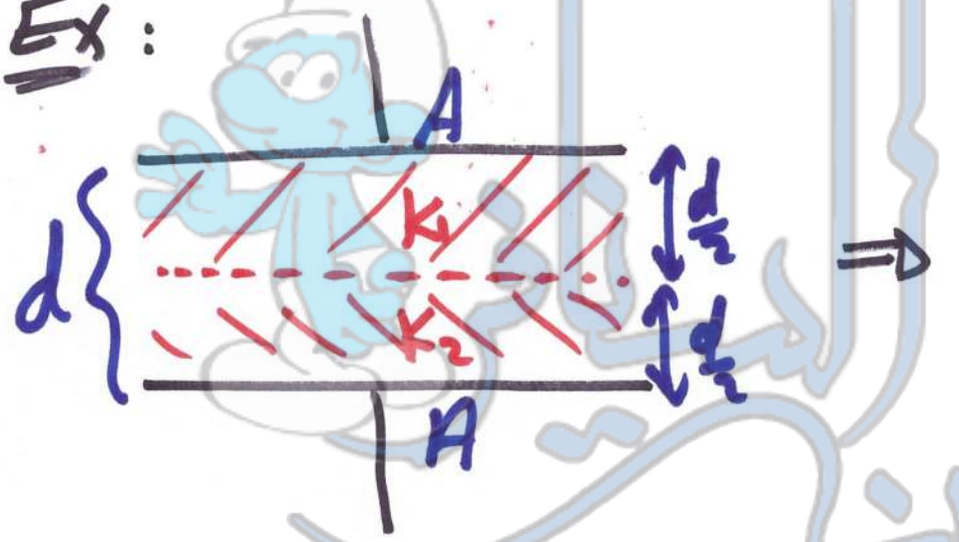
$$V = \frac{V_0}{K} \qquad E = \frac{E_0}{K} \qquad U = \frac{U_0}{K} \qquad u = \frac{u_0}{K}$$

Ex :



$$\Rightarrow C = \frac{K\epsilon_0 A}{d} = K C_0$$

Ex :



$$C_1 = \frac{2K_1\epsilon_0 A}{d} C_0$$

$$C_2 = \frac{2K_2\epsilon_0 A}{d} C_0$$

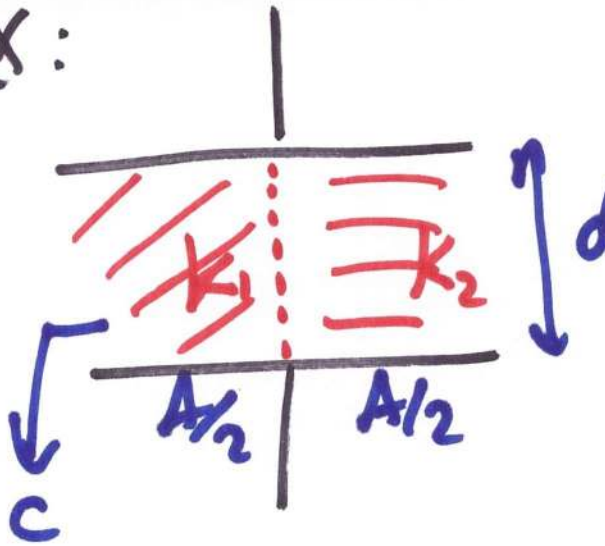
$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$= \frac{d}{2K_1\epsilon_0 A} + \frac{d}{2K_2\epsilon_0 A}$$

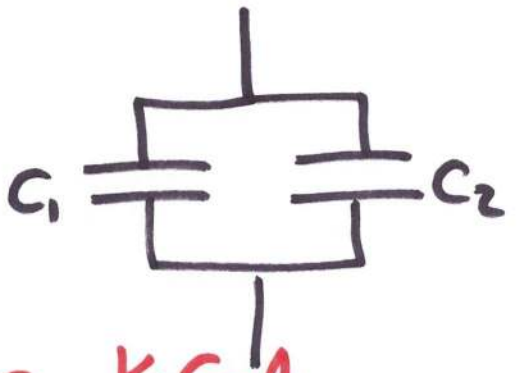
$$\frac{1}{C_{eq}} = \frac{dK_2 + dK_1}{2K_1K_2\epsilon_0 A} \Rightarrow C_{eq} = \frac{2K_1K_2}{(K_1+K_2)} \frac{\epsilon_0 A}{d}$$

$$\boxed{C_{eq} = \left(\frac{2K_1K_2}{K_1+K_2}\right) C_0}$$

Ex:



$\Rightarrow$



$$C_1 = \frac{K_1 \varepsilon_0 A}{\frac{d}{2}}$$

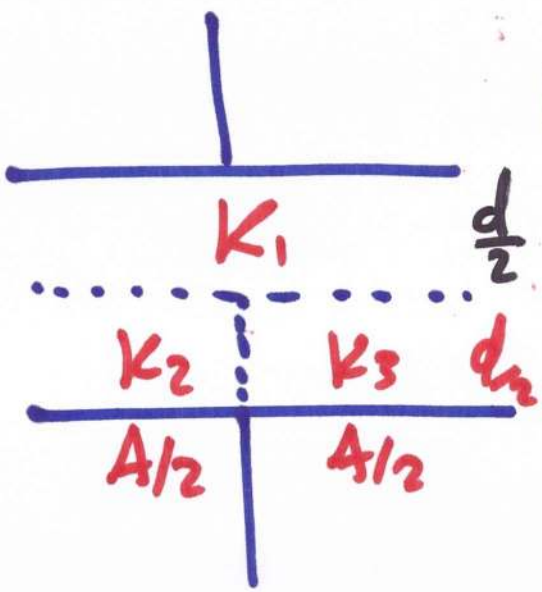
$$C_2 = \frac{K_2 \varepsilon_0 A}{2d}$$

$$C_{eq} = C_1 + C_2$$

$$= \frac{K_1 \varepsilon_0 A}{2d} + \frac{K_2 \varepsilon_0 A}{2d}$$

$$= \left(\frac{K_1 + K_2}{2}\right) \frac{\varepsilon_0 A}{d}$$

$$C_{eq} = \frac{K_1 + K_2}{2} C_0$$

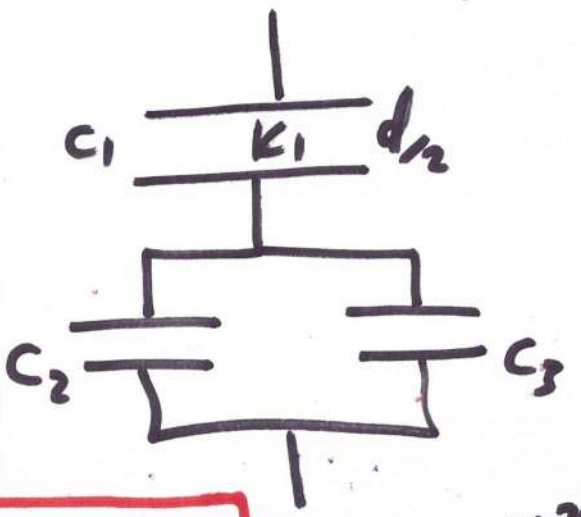


$$C_1 = \frac{K_1 \varepsilon_0 A}{d/2}$$

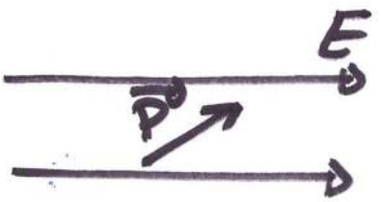
$$C_2 = \frac{K_2 \varepsilon_0 A}{2d/2}$$

$$= \frac{K_2 \varepsilon_0 A}{d}$$

$$C_3 = \frac{K_3 \varepsilon_0 A}{\frac{d}{2}}$$



Dipole $\vec{}$ inside $\vec{E}$





$\vec{P}$

$\vec{P} = q\vec{d}$

$P = qd$

* $\vec{\tau} = \vec{P} \times \vec{E}$

$\tau = PE \sin\theta_{PE}$

* $U = -\vec{P} \cdot \vec{E}$

$= -PE\cos\theta_{PE}$

$U_{max} = PE \quad (\theta = 180°)$

$U_{min} = -PE \quad (\theta = 0)$

$\Delta U = 2PE$

Q51 $q =$ ✓ $\quad q = (-1.2, 1.1) \quad \vec{E} = (\ )\hat{i} - (\ )\hat{j}$

$-q = (1.4, -1.3)$

a) $\vec{P} = q\vec{d}$

$= 2(-2.6\hat{i} + 2.4\hat{j})$

b) $\vec{\tau} = \vec{P} \times \vec{E}$

$\vec{\tau} \perp \vec{P}, \vec{E}$

c) $U = -\vec{P} \cdot \vec{E}$

$=$ J

d) $\Delta U = 2PE$

$$I_{av} = \frac{\Delta q}{\Delta t} \Rightarrow \Delta q = I \Delta t$$

$$I_{ins} = \frac{dq}{dt} \Rightarrow \Delta q = \int_{t_1}^{t_2} I\,dt$$

Ex: If $q = 4t^2 - 5t + 1$, Find the Current at $t = 2$ sec.

$$\Rightarrow I_{ins} = \frac{dq}{dt} = 8t - 5 \Rightarrow I = 11 \text{ Amp.}$$

$$I_{av} = \frac{\Delta q}{\Delta t} = \frac{q_2 - q_1}{t_2 - t_1} \qquad t = 0 \to 2 .$$

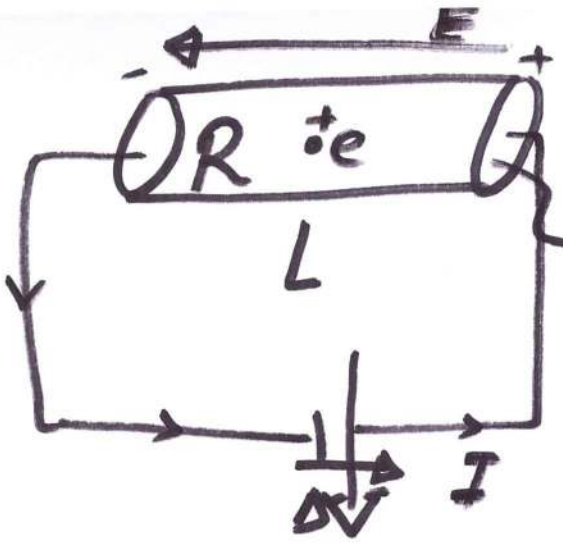
$q_1(t=0) = 1$

$q_2(t=2) = 4(4) - 5(2) + 1 = 7$

$$\Rightarrow I = \frac{7-1}{2-0} = \frac{6}{2} = 3\,A .$$

Ex: $I = 6t - 2$, Find charge from $t=0$ to $t=1$.

$$\Delta q = \int_0^1 (6t-2)\,dt = 3t^2 - 2t \Big|_0^1 = 3 - 2 = 1 \text{ Col.}$$



$A = \pi r^2$ (r = radius)

$Vol = AL$

$q = I * t$

$\Delta V_R = I * R$ — ohm's law.

$P_R = IV = I^2R = \frac{V^2}{R}$

$U = P * t$

$I = \frac{q}{t}$

$q = ne$

$I = n e v_d A$

$v_d = \frac{L}{t}$

$R = \frac{\rho L}{A}$

$\rho = \frac{1}{\sigma}$

$J = \frac{I}{A} = n' e v_d$ — $J = \sigma E$

$\Delta V_R = E * L$

$F = eE$

$\sigma = \frac{n' e^2 \tau}{m_e}$

q = charge inside the resistance.

I = Current

t = time of current moving

ΔV = Potential (voltage) difference accross the $\underline{R}$.

n = number of electrons Pass through the wire.

e = electron charge ($e = +1.6 * 10^{-19} C$)

n' = electron's density (no of $\underline{e}$ in Unit Volume) $[e/m^3]$

v_d = drift Velocity (سرعة انسياق)

A = Area of the wire. (m^2)

L = length of the wire.

J = Current density $[A/m^2]$

R = Resistance (ohm = Ω)

ρ = resistivity ($\Omega \cdot m$)

σ = Conductivity $(\Omega \cdot m)^{-1}$

E = Electric field.

P = Power (energy rate) $\Rightarrow$ watt

U = energy in $\underline{R}$.

m_e = mass of $\underline{e}$

τ = av. time interval between two successive ↑ collisions.

Ex: Resistance of length 10m (L) and radius of $(\frac{2}{\sqrt{\pi}}\,cm)$, $r \rightarrow A = 4*10^{-4}\,m^2$, has a conductivity of $4*10^{6}\,(\Omega \cdot m)^{-1}$ (σ), Connected accross a pot. diff of 20-V (ΔV) for 10-sec (t), Find:

1) resistivity 2) resistance 3) Current
4) charge 5) no of $\underline{e}$ passes through the wire. (n)
6) drift velocity 7) no of $\underline{e}$ Per unit Vol. (n')
8) Current density 9) Electric Field
10) Power 11) energy 12) Force on $\underline{e}$.
13) time between 2 successive Collisions (τ)

$\underline{\underline{R}}$ → L
→ A → r



$r_2 = 2r_1 \Rightarrow A_2 = 4A_1 \Rightarrow L_2 = \frac{1}{4} L_1 \Rightarrow R_2 = \frac{1}{16} R_1$

$A_2 = 2A_1 \longrightarrow L_2 = \frac{1}{2} L_1 \longrightarrow R_2 = \frac{1}{4} R_1$

$R_2 = 3R_1 \Rightarrow L_2 = \sqrt{3} L_1 \Rightarrow A_2 = \frac{1}{\sqrt{3}} A_1 \Rightarrow r_2 = \frac{1}{\sqrt[4]{3}} r_1$

$\Downarrow$

$\left(\frac{1}{3}\right)^{\frac{1}{4}}$

* $r_2 = 2r_1$

$A_1 = \pi r_1^2$

$A_2 = \pi r_2^2$

$= \pi (2r_1)^2$

$= 4\pi r_1^2$

$A_2 = 4A_1$

$V_1 = V_2$

$L_1 A_1 = L_2 A_2$

$L_1 \cancel{A_1} = L_2 4\cancel{A_1}$

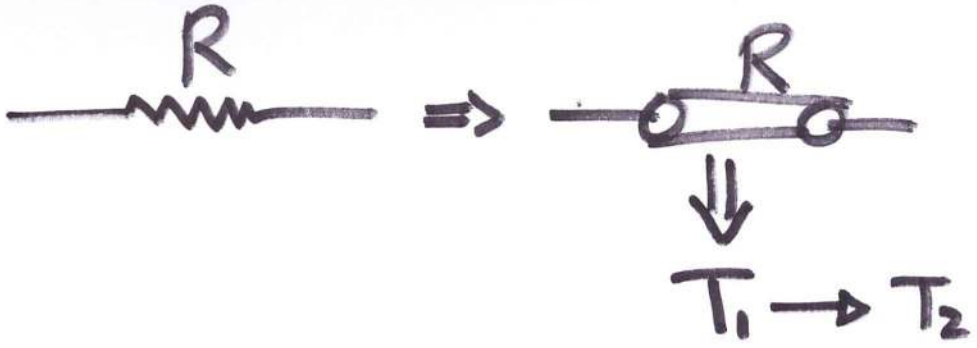
$L_2 = \frac{1}{4} L_1$

$R_1 = \frac{\rho L_1}{A_1}$

$R_2 = \frac{\rho L_2}{A_2}$

$= \frac{\rho L_1}{4 \times 4A_1}$

$= \frac{\rho L_1}{A} \times \frac{1}{16} = \underline{R_1}$



R_o : orginal resistance.

R : new resistance.

α : Temp. Coeff. of resistance. $\left(\frac{1}{C^\circ}\right)$

$$R = R_o[1 + \alpha(T_f - T_i)]$$

$$\rho = \rho_o[1 + \alpha(T_f - T_i)]$$

Ex : $\alpha = 2 * 10^{-5}/C$

$T_i = 50^\circ C$ $\qquad$ $T_f = ??$

$R_o = 10\Omega \longrightarrow R = 20\Omega$

$$20 = 10[1 + 2 * 10^{-5}(T_f - 50)]$$

$$T_f = 50050\, C^\circ.$$

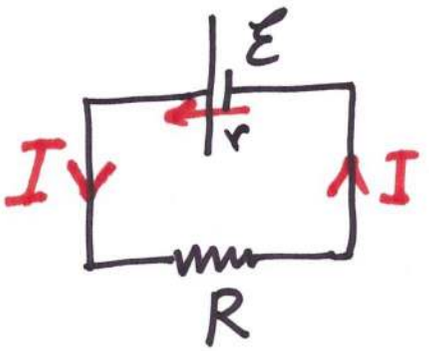
$\frac{\Delta R}{R} \Rightarrow R = R_o + R_o\alpha(\Delta T)$

$\Delta R = \alpha \Delta T$

$\frac{\Delta R}{R}$: fractional change in

CH:28 Electrical Circuits

الدارات الكهربائية



I: Current (A)

R: external resistance (Ω)

r: internal resistance (Ω)

$\mathcal{E}$: electromotive force (V)

Resistance:



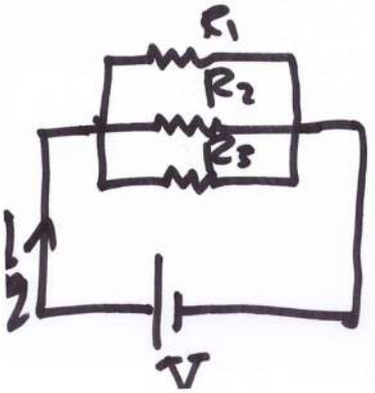
$$\Rightarrow \boxed{V_{ab} = I R} \quad \text{— ohm's law.}$$
$$(R)$$

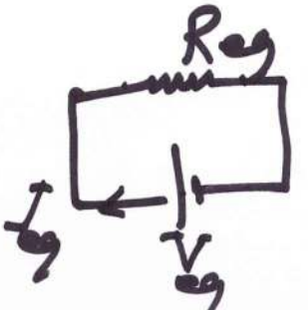
$$P_R = I * V_R = I^2 R = \frac{V^2}{R} .$$

$$U = P * t$$

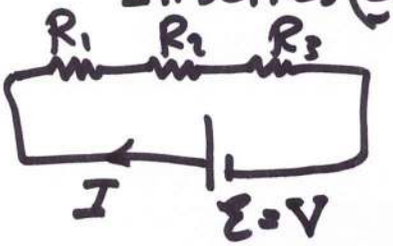
Resistors' Connection
توصيل المقاومات

In Parallel (التوازي)





In Series (التوالي)





التوالي

التيار مستمر

$$I_{eq} = I_1 = I_2 = I_3 = \dots$$

الجهد يتوزع

$$V = V_{eq} = V_1 + V_2 + V_3 + \dots$$ بطارية

المقاومة مجموعة

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

التوازي

التيار يتوزع:

$$I_{eq} = I_1 + I_2 + I_3 + \dots$$

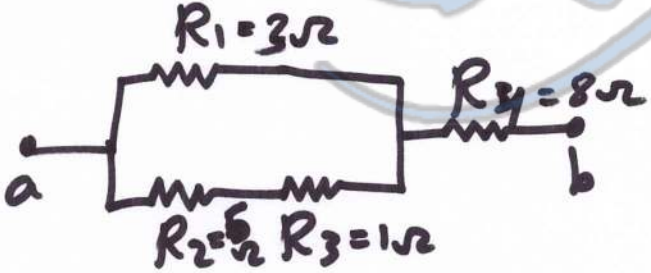
الجهد مستمر:

$$V_{eq} = V_1 = V_2 = V_3 = \dots$$

المقاومة مقلوبة:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Ex: Find the equivalent resistance (R_{eq}):



R_2, R_3 تواكي $\rightarrow R_5$

R_1, R_5 توازي $\rightarrow R_6$

R_4, R_6 تواكي $\rightarrow R_7 = R_{eq}$.

* $R_5 = R_2 + R_3$

$= 5 + 1$

$= 6\,\Omega$

$$\frac{1}{R_6} = \frac{1}{R_5} + \frac{1}{R_1}$$

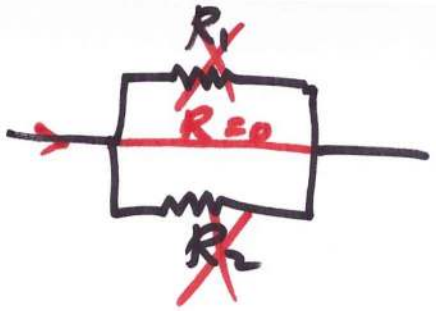
$$\frac{1}{R_6} = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$$

$R_6 = 2\,\Omega$

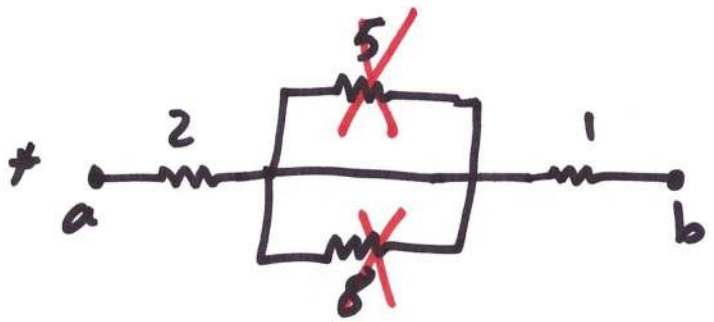
$R_7 = R_4 + R_6$

$= 8 + 2$

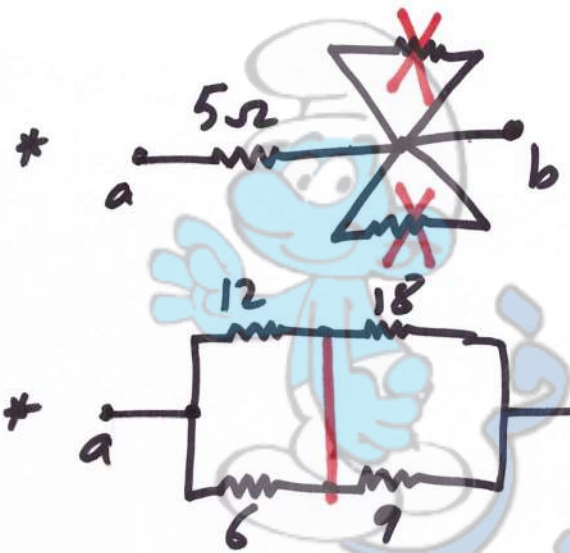
$R_7 = 10\,\Omega$



$\Rightarrow R_{eq} = 0$



$R_{ab} = 3\Omega$.



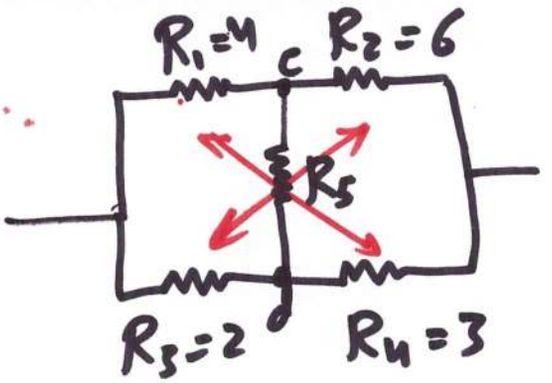
$R_{ab} = 5\Omega$

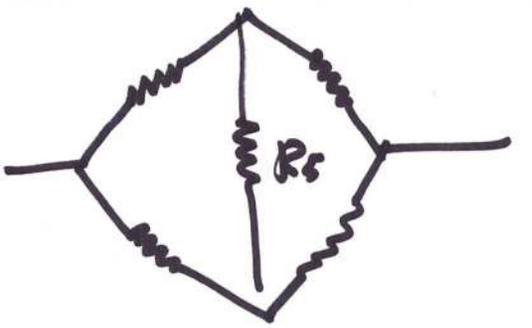


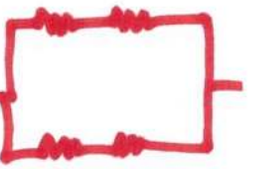
$\Downarrow$

$R_{ab} = 10\Omega$.

(4)



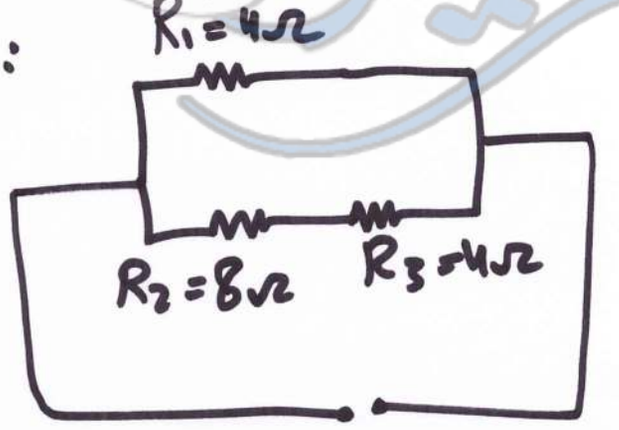


$R_1 * R_4 \stackrel{?}{=} R_2 * R_3 \Rightarrow R_5$ تلغى $\Rightarrow$ 





Ex:



Find : 1) R_{eq}.

2) Voltage and Current in each resistor.

3) Power in R_3

١) نرسم خريطة مع تعيين العقد الإضافي.

٢) نجد R_{eq}

٣) نجد جميع المعلومات عن العقد الإضافي

٤) نجد بقية المعلومات عن الخريطة

R_2, R_3 تواكي $\rightarrow R_4$

R_1, R_4 توازي $\rightarrow R_5 = R_{eq}$

$V_5 = 60$ Volt

(1) $R_4 = R_2 + R_3$

$= 8 + 4 = 12\ \Omega$

$\frac{1}{R_5} = \frac{1}{R_1} + \frac{1}{R_4}$

$= \frac{1}{4} + \frac{1}{12}$

$\boxed{R_5 = 3\Omega}$

(2) $V_5 = 60\ V$

$R_5 = 3\Omega$

$\Rightarrow I_5 = \frac{V_5}{R_5} = \frac{60}{3} = 20\ A.$

(5)

$V_5 = V_4 = V_1 = 60\ V$

$\rightarrow I_1 = \frac{V_1}{R_1} = \frac{60}{4} = 15A$

$\rightarrow I_4 = \frac{V_4}{R_4} = \frac{60}{12} = 5A$

$I_4 = I_2 = I_3 = 5A$

$\rightarrow V_3 = I_3 R_3 = 5 * 4 = 20\ Volt$

$\rightarrow V_2 = I_2 R_2 = 5 * 8 = 40\ Volt.$

(3) $P_3 = I_3^2 R_3$

$= (5)^2 * 4$

$= 25 * 4$

$= 100\ watt$

$P_1 = I_1^2 R_1$

$= (15)^2 * 4$

$= 900\ watt$

$P_2 = I_2^2 R_2$

$= (5)^2 * 8$

$= 200\ watt$

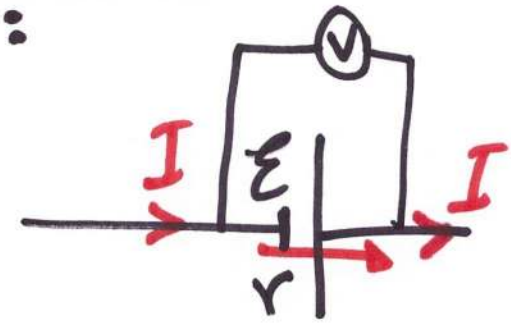
$P_5 = I_5^2 R_5$

$= (20)^2 * 3$

$= 400 * 3$

$= 1200\ watt$

Battery :



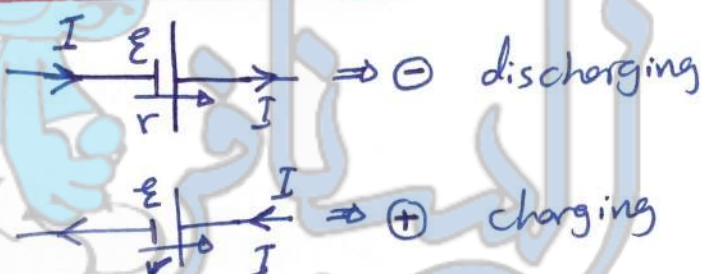
* V_ε : Potential difference accross the battery.



$$V_\varepsilon = \varepsilon \mp Ir$$

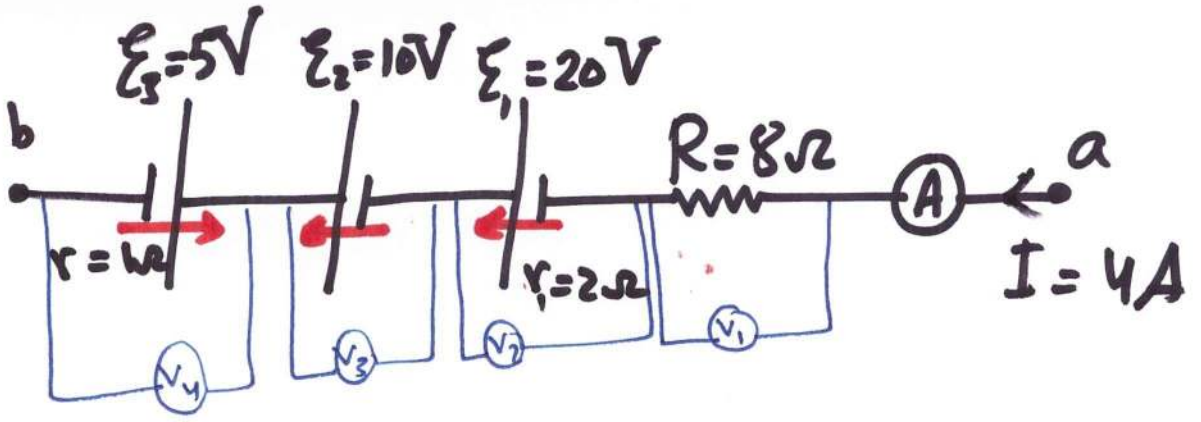
$V_\varepsilon = \varepsilon$ if

$r=0$	ideal battery
$I=0$	open circuit



* Power of battery : $P_\varepsilon = I\varepsilon$

* energy $(U) \Rightarrow U = P * t$.



What are the readings of (A) and (V)'s.

ملخص للقوانين

(V) = V

$\mathcal{E}$ → $V_{\mathcal{E}} = \mathcal{E} \mp IR$

R → $V = IR$

(watt) ≡ Power $[U = P \times t]$

$\mathcal{E}$ → $P = I\mathcal{E}$

R → $P_R = I^2R = \frac{V^2}{R} = IV$

(A) $= I = 4A.$

(V1) $= V_R = IR = 4 * 8 = 32$ Volt.

(V2) $= V_{\mathcal{E}_1} = \mathcal{E}_1 - I r_1$

$= 20 - 4 * 2$

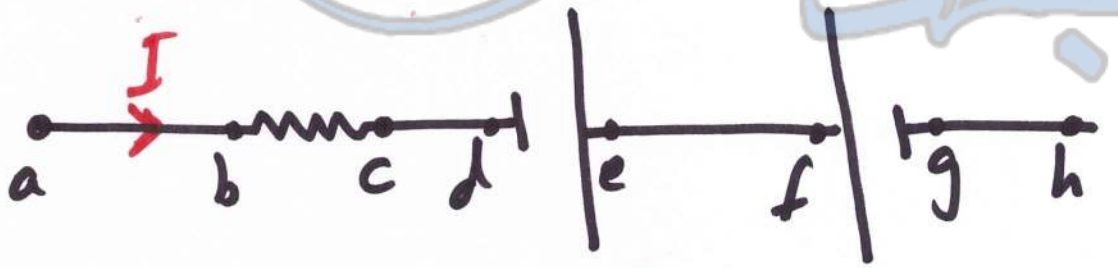
$= 12$ Volt.

(V3) $= V_{\mathcal{E}_3} = \mathcal{E} = 10$ Volt.

(V4) $= V_{\mathcal{E}_3} = \mathcal{E}_3 + I r_3$

$= 5 + 4 * 1$

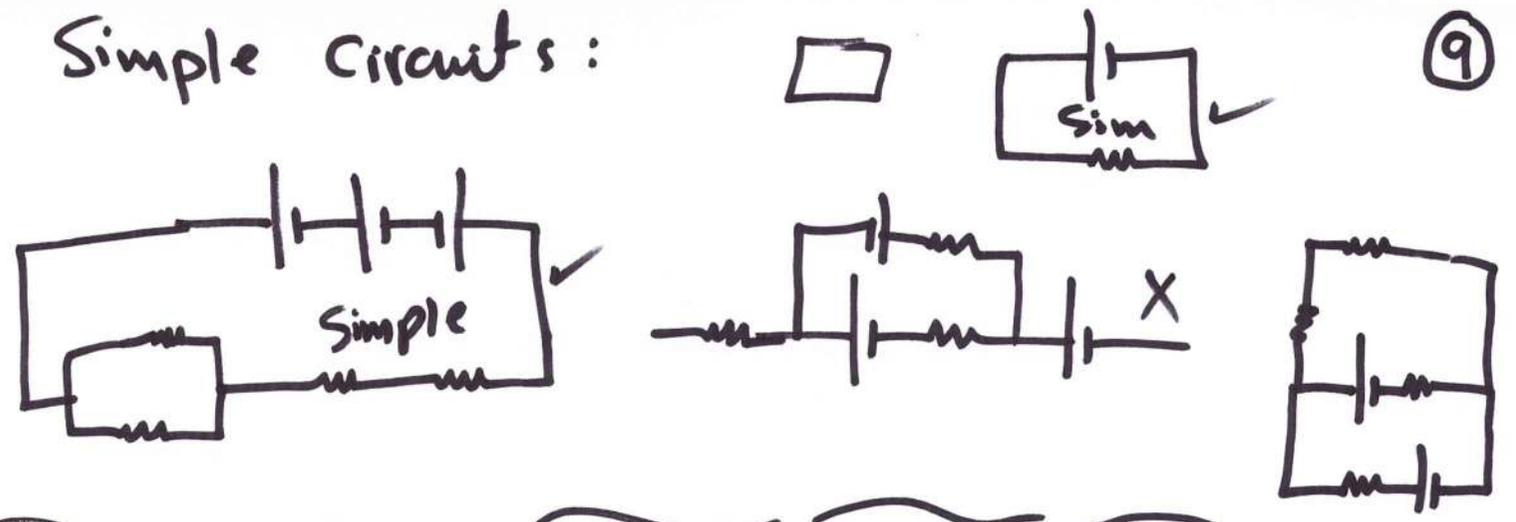
$= 9$ Volt.

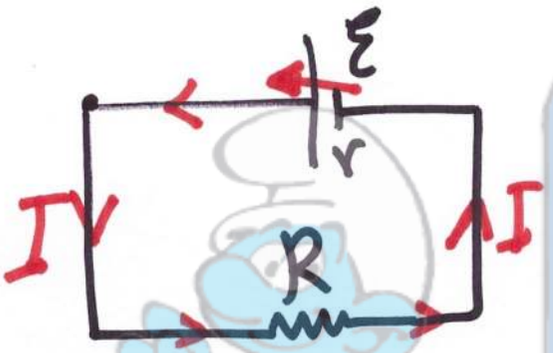


$I_a = I_b = I_c = I_d = I_e = I_f = I_g = I_b$

$V_a = V_b > V_c = V_d < V_e = V_f > V_g = V_h$

Simple Circuits:

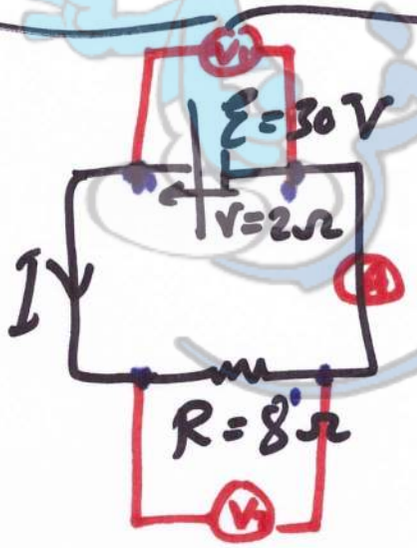




$$\Rightarrow \Sigma\varepsilon = I\,\Sigma R$$

R or r

Ex:



Find: 1) Reading of (A).
2) " " (V1), (V2)
3) Power disapated in R
4) " " " r.
5) Produced Power by ε
6) Consumed energy in R through 2-min.

1) $\Sigma\varepsilon = I\,\Sigma R$

$30 = I(2+8)$

$I = 3A \equiv (A)$

2) $V_\varepsilon = \varepsilon - Ir$

$= 30 - 3*2$

$= 24$ Volt

3) $V_2 = V_R = IR = 3*8$

$= 24$ Volt.

③ $P_R = I^2R = (3)^2 * 8 = 72$ watt.

④ $P_r = I^2r = (3)^2 * 2 = 18$ watt

⑤ $P_\mathcal{E} = I\mathcal{E} = 3 * 30 = 90$ watt.

⑥ $U_R = P_R * t = 72 * 120$

$= 8640$ J.

⑦ drop in $\mathcal{E}$ Potential :⇒ $Ir = 3 * 2 = 6$ Volt

Ex:



reading of (V) is 40V when S is open. what is the reading if the S is closed.

S is open:- $I = 0$ ⇒ (V) $= V_\mathcal{E} = \mathcal{E} - \cancel{Ir}$

$40 = \mathcal{E}$

S is closed: (V) $= V_\mathcal{E} = \mathcal{E} - Ir$

$= 40 - 4 * 1$

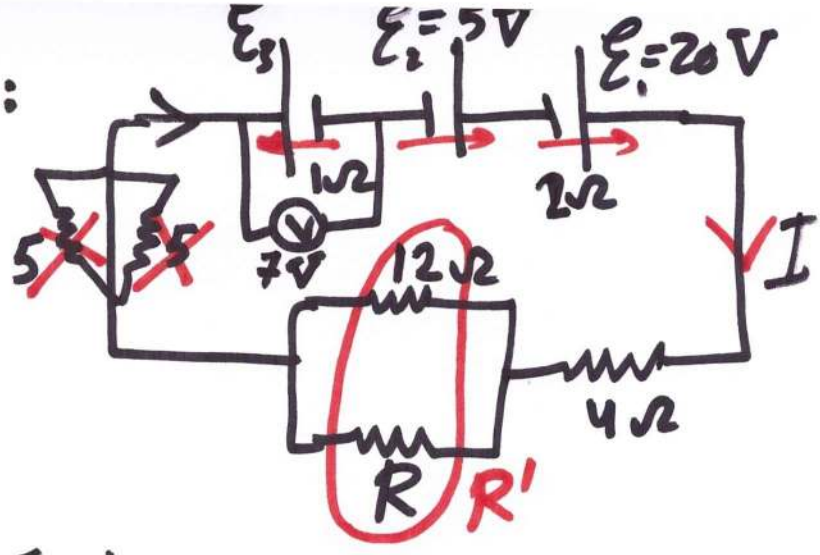
$= 36$ Volt.

$\Sigma\mathcal{E} = I\Sigma R$

$40 = I(1+3+6)$

$I = 4A$.

Ex :



* Power in $R=4\Omega$ is 16 watt.

Find ① I ② ε_3 ③ R

✓ ε ✗
R ✗
r ✓
I ✗

$P_R = I^2 R$

$16 = I^2 * 4$

$I^2 = 4$

$\boxed{I = 2\,A}$

$V_{\varepsilon_3} = \varepsilon_3 + Ir$

$7 = \varepsilon_3 + 2*1$

$\boxed{\varepsilon_3 = 5\ Volt}$

$\Sigma\varepsilon = I\Sigma R$

$20+5-5 = 2(2+1+R'+4)$

$20 = 2(7+R')$

$10 = 7+R'$

$\boxed{R' = 3\Omega}$

$\frac{1}{R'} = \frac{1}{R} + \frac{1}{12}$

$\frac{1}{3} - \frac{1}{12} = \frac{1}{R}$

$\boxed{R = 4\Omega}$

Find current in R. $\Rightarrow V_{R'} = IR'$

R. in series R'

$= 2*3$

$V_R = 6\ Volt$

$\Rightarrow I' = \frac{V_R}{R} = \frac{6}{4} = 1.5\,A$

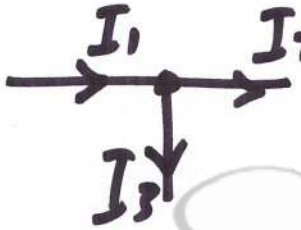
الدارات المعقدة (أكثر من حلقة)

قانونا كيرتشوف

خطوات حل الدارات المعقدة: -

① نحدد اتجاه التيارات إذا لم تكن محددة (بشكل عشوائي)

② نطبق في قانون كيرتشوف الأول:



$$\Sigma I_{in} = \Sigma I_{out}$$

* نستفيد من معطيات المسألة إن وجدت.

V_{ab} ، (V) ، P_ε ، P_R

③ نحدد مساراً مغلقاً لنا نتحرك عليه.

④ نطبق في قانون كيرتشوف الثاني:

$$V_{ab} = \Sigma IR - \Sigma \varepsilon \quad (V_{ab} = V_a - V_b)$$

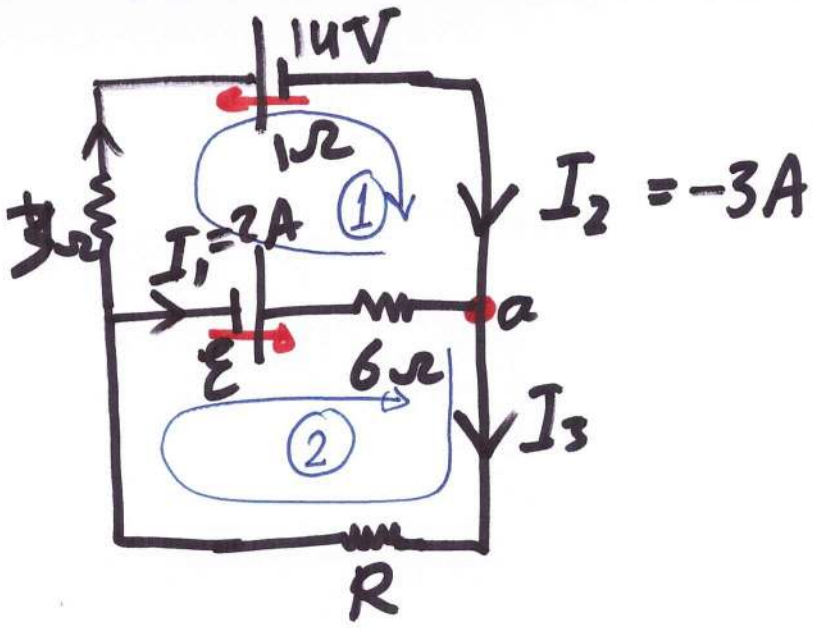
[معنا: + ، عكسنا: -]

* إذا وجدت I سالبة يعني اتجاه I عكس الاتجاه المفروض.

* $V_{aa} = V_{bb} = 0$

* إذا تغير أي شيء في الدارة لا تصدق حالة جديدة.

Ex:



Find:

1) I_3
2) $\mathcal{E}$
3) R

$$\Sigma I_{in} = \Sigma I_{out}$$

$$I_1 + I_2 = I_3$$

$$2 + -3 = I_3$$

$$\boxed{I_3 = -1A}$$

$$V_{aa} = (-6*2 + 3*I_2) - (-\mathcal{E} - 14)$$
$$+ 1*I_2$$

$$0 = -12 + 4*(-3) + \mathcal{E} + 14$$

$$-\mathcal{E} = -12 \quad -12 \quad +14$$

$$-\mathcal{E} = -10 \Rightarrow \boxed{\mathcal{E} = 10 \text{ Volt}}$$

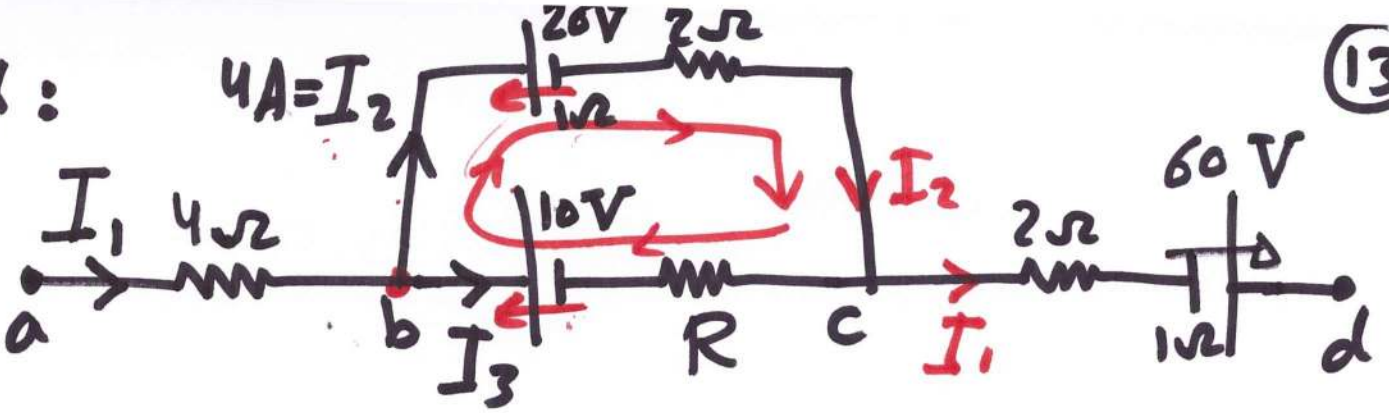
$$V_{aa} = (R*I_3 + 6I_1) - (\mathcal{E})$$

$$0 = -R + 6*2 - 10$$

$$R = 12 - 10$$

$$\boxed{R = 2\Omega}$$

Ex: (13)b



If $V_{ab} = 40$ V., Find: 1) I_3, 2) R

3) V_{dc}

Sol: ①

$$\Sigma I_{in} = \Sigma I_{out}$$

$$I_1 = I_2 + I_3$$

$$\boxed{I_1 = 4 + I_3}$$

$$V_{ab} = (\Sigma IR) - (\Sigma \mathcal{E})$$

$$40 = (4I_1) - (0)$$

$$\boxed{I_1 = 10\ A} \Longrightarrow \boxed{I_3 = 6A}$$

③ $V_{dc} = -(3\times 10) - (-60)$

$= -30 + 60$

$= 30$ Volt.

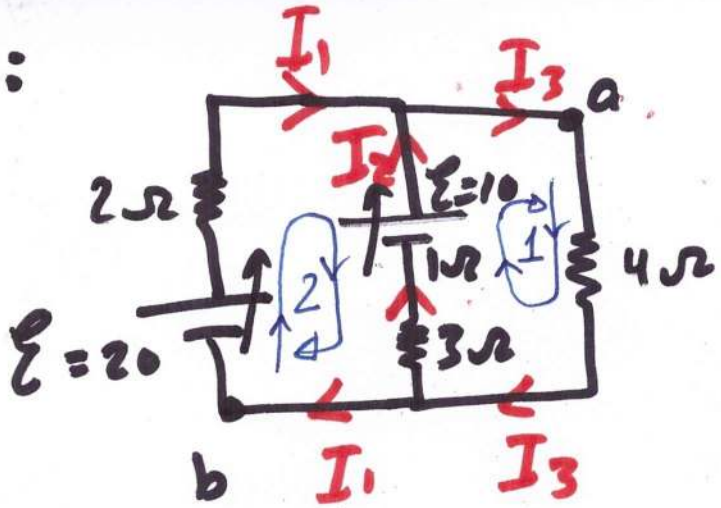
② $V_{cc} = (-6*R + 3*4) - (10 - 20)$

$0 = -6R + 12 - 10 + 20$

$6R = 22$

$\boxed{R = 3.7\ \Omega}$

Ex:



Find:

1) each current.
2) V_{ab}
3) if $V_b = 10V$, find V_a.

$$\Sigma I_{in} = \Sigma I_{out}$$

$$\boxed{I_1 + I_2 = I_3}$$

$$V_{aa} = (4I_3 + 4I_2) - (10)$$

$$0 = 4I_3 + 4I_2 - 10$$

$$10 = 4(I_1 + I_2) + 4I_2$$

$$\boxed{10 = 4I_1 + 8I_2} \quad \text{—— (1)}$$

$$V_{bb} = (2*I_1 - 4I_2) - (20 - 10)$$

$$0 = 2I_1 - 4I_2 - 10$$

$$\boxed{10 = 2I_1 - 4I_2} \quad \text{— (2)}$$

$$\Rightarrow \boxed{I_1 = 3.75\,A}$$

$$\boxed{I_2 = -0.625}$$

$$\Rightarrow I_3 = 3.75 + (-0.625)$$

$$\boxed{I_3 = 3.125\,A}$$

② $V_{ab} = (4 * I_3 \quad) - (\quad 0 \quad)$



$= 4 * 3.125$

$= 12.5$ Volt.

$V_{ab} = (2 * -3.75 \quad) - (-20 \quad)$

$= -7.5 + 20$

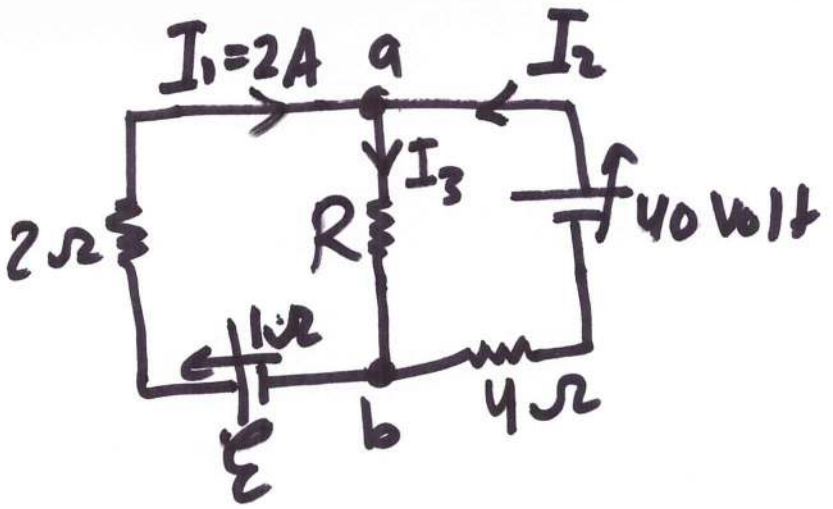
$= 12.5$ Volt.

③ $V_{ab} = V_a - V_b$

$12.5 = V_a - 10$

$\boxed{V_a = 22.5 \text{ Volt}}$

Ex:



Find
1) all I's.
2) R
3) ε

where $V_{ab} = 24V$.

Sol

$$\sum I_{in} = \sum I_{out}$$

$$I_1 + I_2 = I_3$$

$$\boxed{2 + I_2 = I_3}$$

$$V_{ab} = (-4 * I_2) - (-40)$$

$$24 = -4I_2 + 40$$

$$-16 = -4I_2$$

$$\boxed{I_2 = 4A} \Rightarrow \boxed{I_3 = 6A}$$

$$V_{ab} = (R * 6) - (0)$$

$$24 = 6R$$

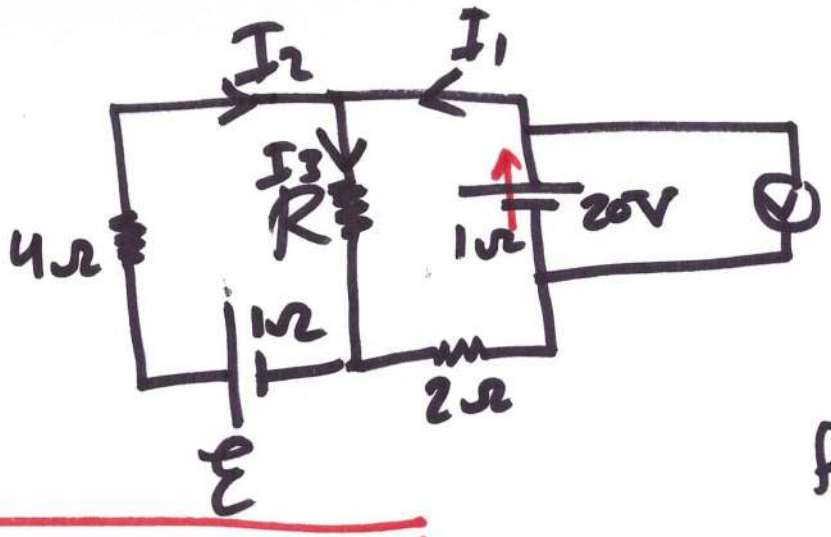
$$\boxed{R = 4\Omega}$$

$$V_{ab} = (-3 * 2) - (-\varepsilon)$$

$$24 = -6 + \varepsilon$$

$$\boxed{\varepsilon = 30V}$$

Ex:



(V) = 14V

$P_{4\Omega} = 64$ watt

find: I's, R, ε

$$I_1 + I_2 = I_3$$

$P_{4\Omega} = I_2^2 R$

$64 = I_2^2 \, 4$

$I_2^2 = 16$

$I_2 = 4A$

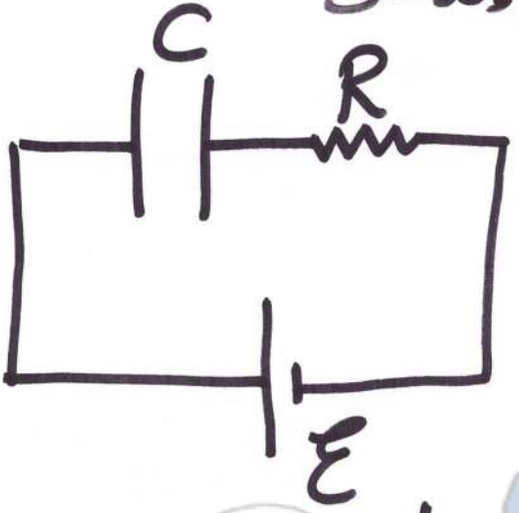
$V_\varepsilon = \varepsilon - I_1 r$

$14 = 20 - I_1 * 1$

$I_1 = 6A$ $\Longrightarrow$ $I_3 = 10A$

RC - circuit

دارة مقاومة ومكثف



$$\mathcal{E} = V_R + V_C$$

$$\mathcal{E} = IR + \frac{q}{C}$$

	$t=0$	long time $t=\infty$
I	I_{max}	0
q	0	q_{max}

$$I_{max} = \frac{\mathcal{E}}{R}$$

$$q_{max} = C\mathcal{E}$$

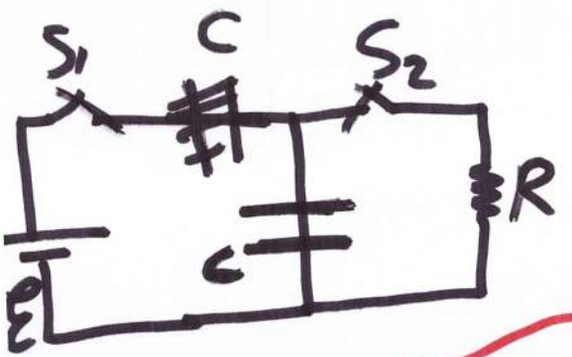
τ : Time Constant

ثابت الزمن بالعربي

$$\tau = RC$$

$$I = I_{max} e^{-t/RC}$$

$$q = q_{max}(1 - e^{-t/RC})$$



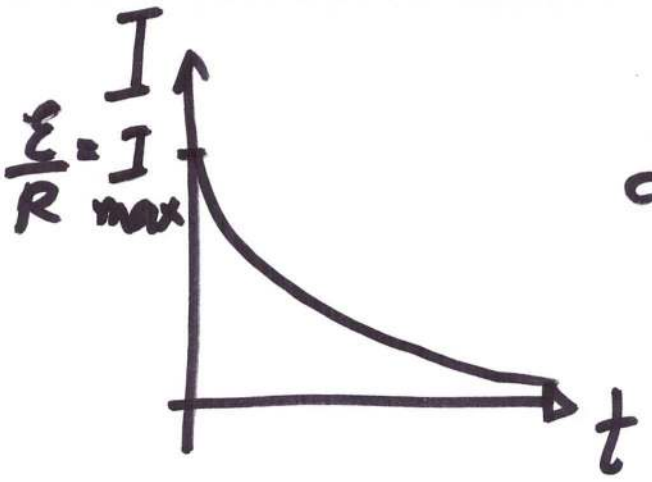
discharging

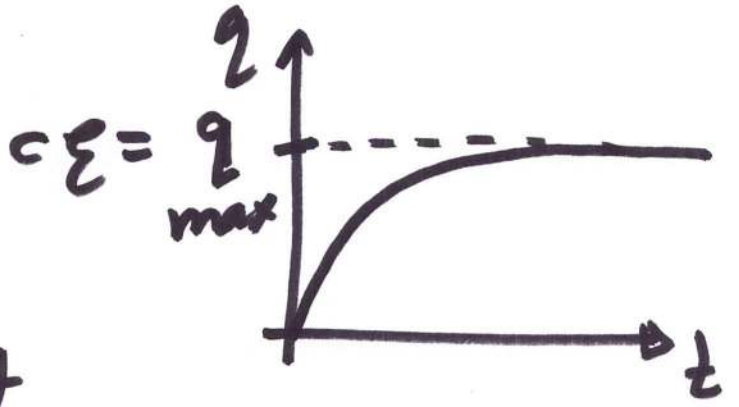


	$t=0$	$t=\infty$
I	I_{max}	0
q	q_{max}	0

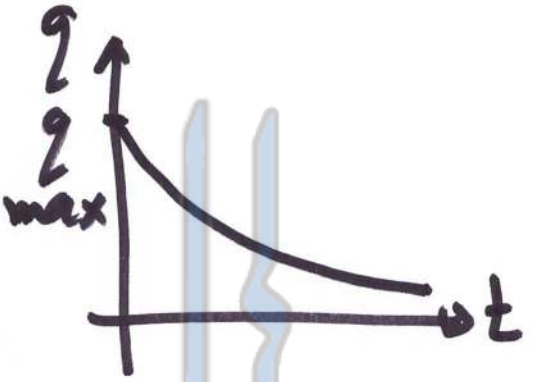
$$I = I_{max} e^{-t/RC}$$

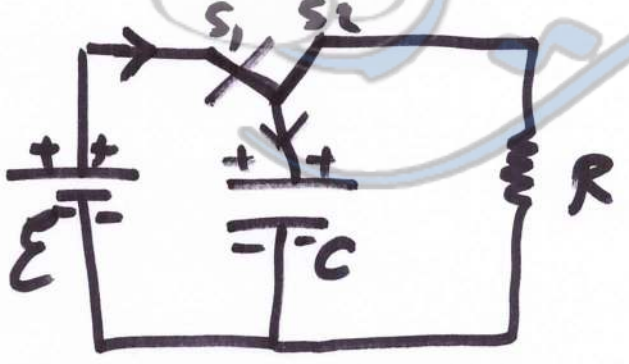
$$q = q_{max} e^{-t/RC}$$


I
$\frac{\mathcal{E}}{R} = I_{max}$
t

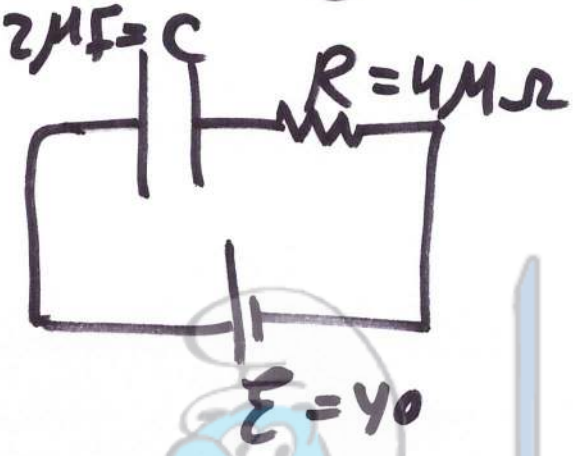

q
$C\mathcal{E} = q_{max}$
t


I
I_{max}
t


q
q_{max}
t


S_1
S_2
$\mathcal{E}$
C
R

Ex: RC circuit of $R = 4M\Omega$ and $C = 2\mu F$, R and C were connected to a battery $\mathcal{E} = 40$ Volt as figure:



Find:

1) max. charge and current.
2) Time Constant (τ)
3) charge and current at t= 4-sec.
4) Voltage of Capacitor and V. of resistor at t=4 sec.
5) charge and current after 4-time Const.
6) time needed to reach to half of max. Current (or half of resistor voltage)
7) time needed to reach to 30% of max charge.
8) energy in Capacitor after 4 sec

$C = 2\mu F \quad R = 4M\Omega \quad \mathcal{E} = 40$ Volt.

1) $q_{max} = C\mathcal{E} = 2\mu * 40 = 80\mu C.$

$$I_{max} = \frac{\mathcal{E}}{R} = \frac{40}{4*10^6} = 10\mu A.$$

2) $\tau = RC = 4*10^6 * 2*10^{-6} = 8\text{-sec}.$

3) $q = q_{max}(1 - e^{-t/RC})$

$= 80*10^{-6}(1 - e^{-4/8})$

$= 31.5*10^{-6} C$

$I = I_{max} e^{-t/RC}$

$= 10*10^{-6} * e^{-4/8}$

$= 6.1*10^{-6} A$

4) $V_C = \frac{q}{C} = \frac{31.5\mu}{2\mu} = 15.75$ V.

$V_R = IR = 6.1*10^{-6} * 4*10^6 = 24.4$ Volt

$V_R = \mathcal{E} - V_C = 40 - 15.75 = 24.25$ Volt.

5) $q = 80*10^{-6}[1 - e^{-4\tau/\tau}]$

$= 78.5*10^{-6} C.$

$I = 10*10^{-6} e^{-4\tau/\tau}$

$= 0.18\mu A.$

6) $I = I_{max}\, e^{-t/\tau}$

$\frac{1}{2} I_{max} = I_{max}\, e^{-t/\tau}$

$\frac{1}{2} = e^{-t/\tau} \Rightarrow \ln 0.5 = \ln e^{-t/8}$

$-0.69 = -\frac{t}{8}$

$\boxed{t = 5.52 \text{ sec}}$

7) $q = q_{max}(1 - e^{-t/\tau})$

$\frac{30}{100} q_{max} = q_{max}(1 - e^{-t/8})$

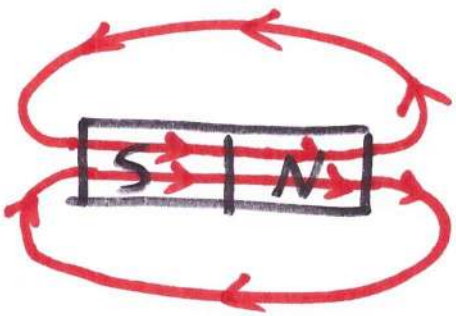
$0.3 = 1 - e^{-t/8}$

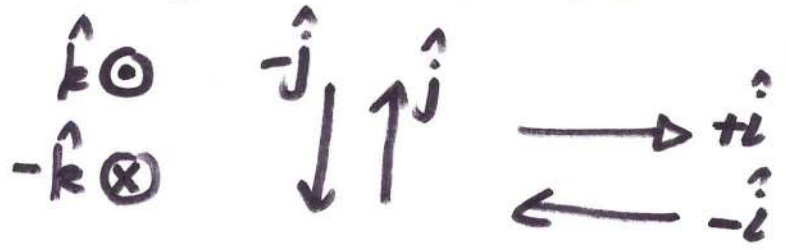
$-0.7 = -e^{-t/8}$ $(\ln)$

$\ln 0.7 = \ln e^{-t/8}$

$-0.36 = -\frac{t}{8}$

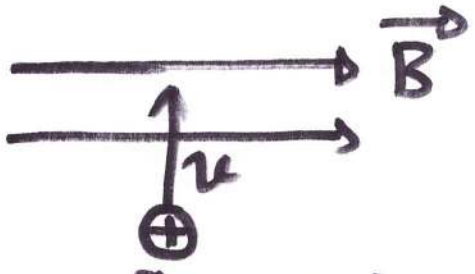
$\boxed{t = 2.88 \text{ sec}}$





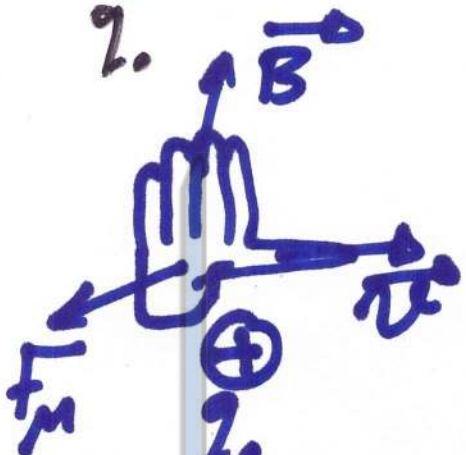
$$\vec{F}_M = q\vec{v}\times\vec{B}$$

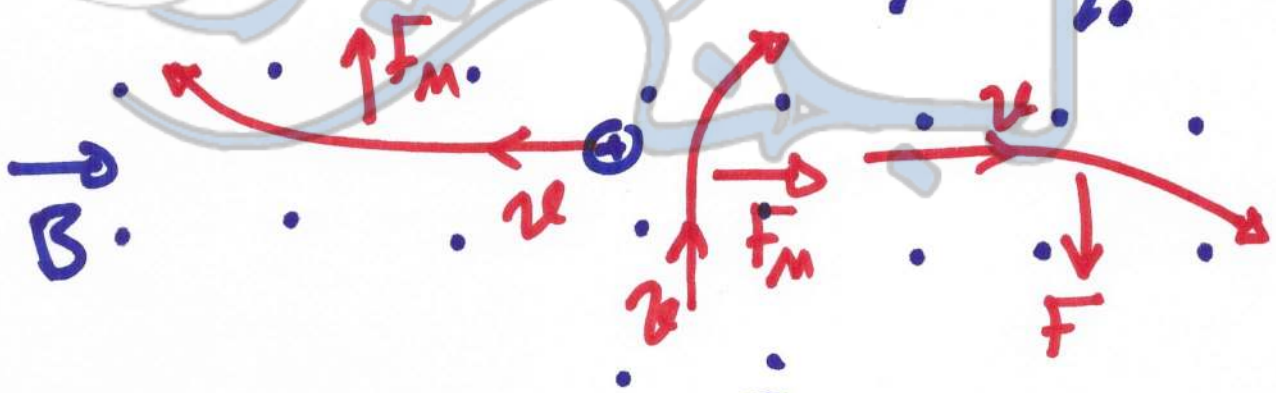
$$F = q.vB\sin\theta_{vB}$$



$F=0 \rightarrow v=0$ (rest)

$\theta = 0°, 180°$





B

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k} \quad \| \quad \vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$$

$$\vec{A}\times\vec{B} = \begin{vmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$|\vec{A}\times\vec{B}| = |\vec{A}||\vec{B}|\sin\theta_{AB}$$

$$= (AB - AB_y)\hat{i} - (AB - AB_x)\hat{j} + (AB - AB_x)\hat{k}$$

Ex: If $\vec{v}$ is given by:

$$\vec{v} = 2\hat{i} - 3\hat{j} + \hat{k}$$

and $\vec{B} = \hat{i} + 2\hat{j} + 2\hat{k}$. $q_0 = 2 * 10^{-4}\,C$

Find: 1) Force as a vector.

2) magnitude of the force.

1)

$$\vec{v} \times \vec{B} = \begin{vmatrix} \hat{i} & -\hat{j} & \hat{k} \\ 2 & -3 & 1 \\ 1 & 2 & 2 \end{vmatrix}$$

$$= (-6 - 2)\hat{i} - (4 - 1)\hat{j} + (4 + 3)\hat{k}$$

$$\vec{v} \times \vec{B} = -8\hat{i} - 3\hat{j} + 7\hat{k}$$

$$\vec{F}_M = q_0 \vec{v} \times \vec{B} = 2 * 10^{-4}[-8\hat{i} - 3\hat{j} + 7\hat{k}]$$

$$= (-16\hat{i} - 6\hat{j} + 14\hat{k}) * 10^{-4}\ New.$$

2) $|\vec{F}_M| = \sqrt{16^2 + 6^2 + 14^2} * 10^{-4} = 22 * 10^{-4}\,N.$

* If: find $\theta_{vB} \Rightarrow |\vec{F}| = q_0 v B \sin\theta_{vB}$

$|\vec{v}| = v = \sqrt{2^2+3^2+1^2} = \sqrt{14}$

$|\vec{B}| = B = \sqrt{1^2+2^2+2^2} = \sqrt{9} = 3$

$\Rightarrow 22 * 10^{-4} = 2 * 10^{-4} * \sqrt{14} * 3 \sin\theta_{\vec{v}\vec{B}}$

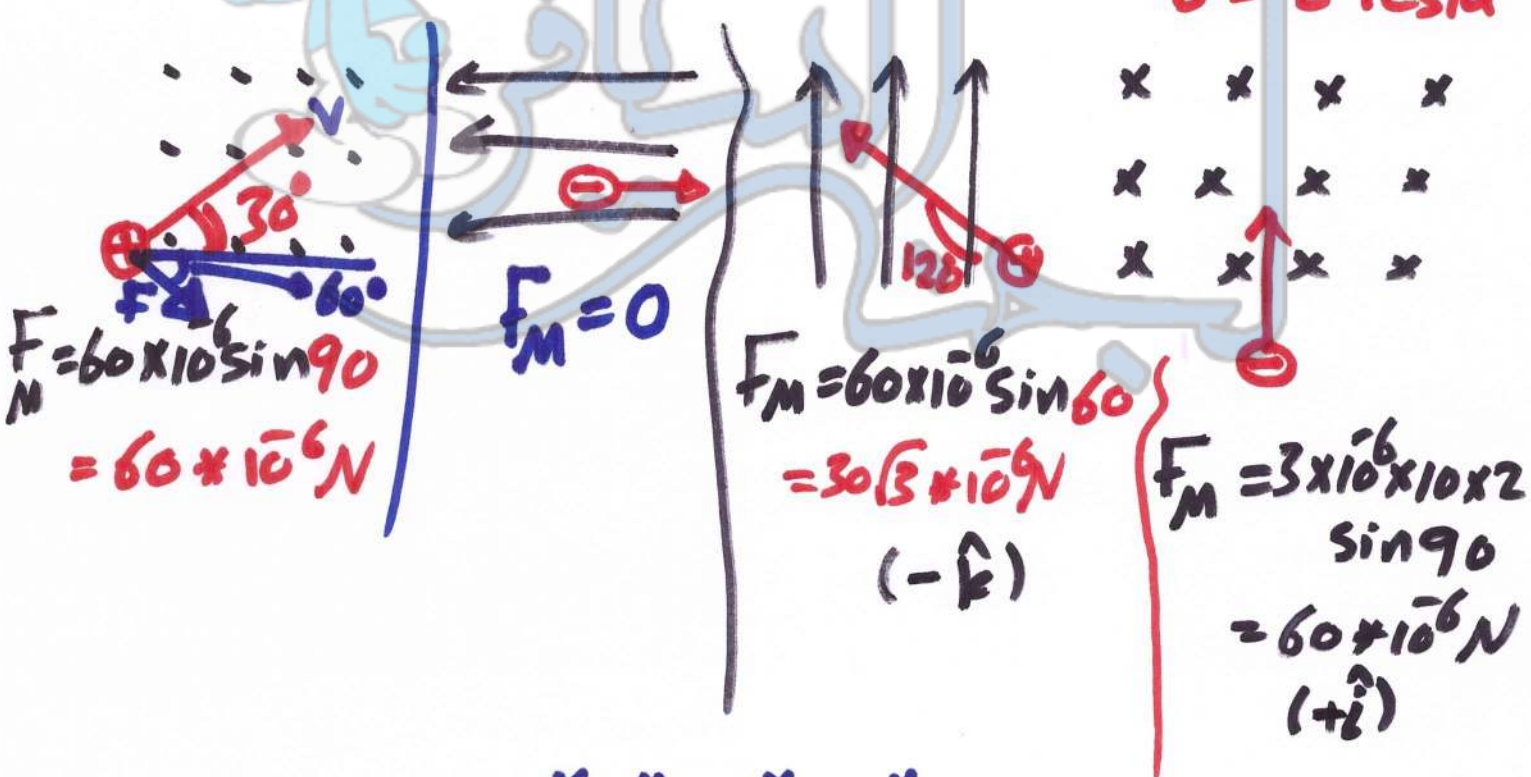
$\sin\theta = 0.98 \Rightarrow \boxed{\theta = 78.5}$

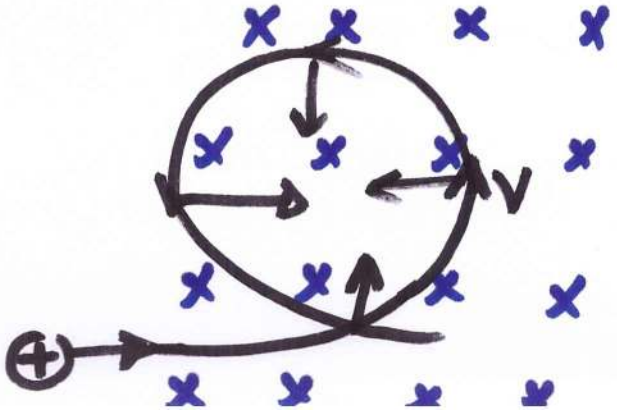
Ex: In the figures, Find $\vec{F}_M$:

$|q| = 3 \times 10^{-6} C$

$v = 10 m/s$

$B = 2$ Tesla





Circular motion.

F_m toward Center

$$F_c = \frac{mv^2}{r}$$

$$a_c = \frac{v^2}{r}$$

a_c ↓ centripetal accel. التسارع المركزي

$$qvB = \frac{mv^2}{r}$$

$$\boxed{r = \frac{mv}{qB}}$$

$$v = \frac{2\pi r}{T_p}$$

T_p : Period الزمن الدوري

$$T_p = \frac{2\pi r}{v} \Rightarrow T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}$$

$f \equiv$ frequency التردد $\Rightarrow f = \frac{1}{T_p} = \frac{v}{2\pi r} = \frac{qB}{2\pi m}$

* $\omega \equiv$ angular freq. $\Rightarrow \omega = 2\pi f = \frac{v}{r} = \frac{qB}{m}$

Ex: If $q = 4\mu C$ entered a Uniform mag. field $B = 4\,T$ with Const speed $v = 1 a m/s$ and if $m = 2 * 10^{-5}\,kg$, Find:

1) magnetic force.
2) centripatal force

$$F_m = F_c = \begin{cases} qvB\sin\theta \checkmark \\ \frac{mv^2}{r} \times \end{cases}$$

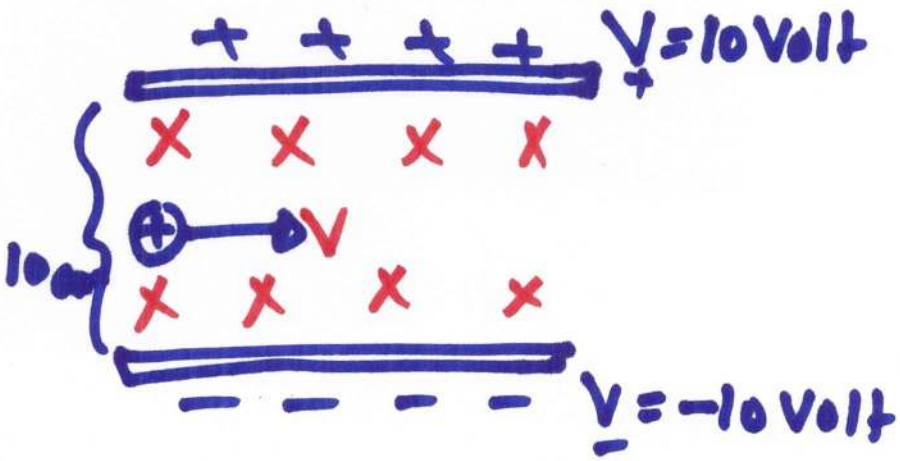
3) accel. $\Rightarrow a_c = \frac{F_c}{m}$

4) radius. $\rightarrow r = \frac{mv}{qB}$

5) Period $T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}$

6) freq. $\rightarrow f = \frac{1}{T}$

7) ang. freq. (velocity) $\rightarrow \omega = 2\pi f = \frac{2\pi}{T}$



$B = 20\,T$

$q_o = 4 * 10^{-6}\,C$

$v = 20\,m/s$

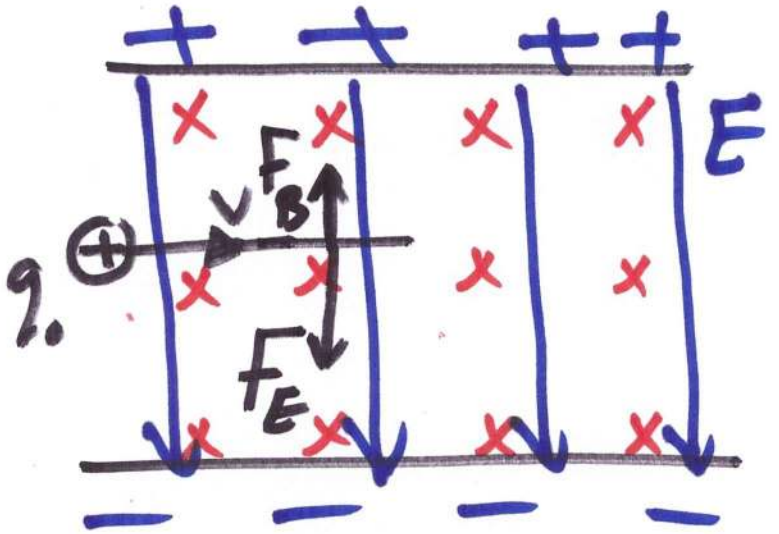
① E

② F_E

③ $F_B = F_m$

④ [illegible] to F_m

⑤ V to ensure q moves in a stright [illegible]



① $\Delta V = Ed$

$(10 - -10) = E\ 10\times10^{-2}$

$E = \frac{20}{10\times10^{-2}} = 200\ N/C$

② $F_E = qE = 4\times10^{-6} * 200 = 800*10^{-6}\ N.$ $(-\hat{j})$

③ $F_B = qVB\sin\theta = 4\times10^{-6}\times20\times20\sin90$

$= 1600*10^{-6}\ N$ $(+\hat{j})$

④ $\vec{F}_{lorentz} = \vec{F}_E + \vec{F}_B$

$F_{lor} = F_B - F_E = 800*10^{-6}(+\hat{j})$

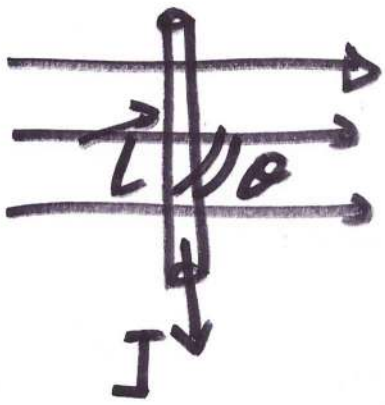
⑤ $F_E = F_B$

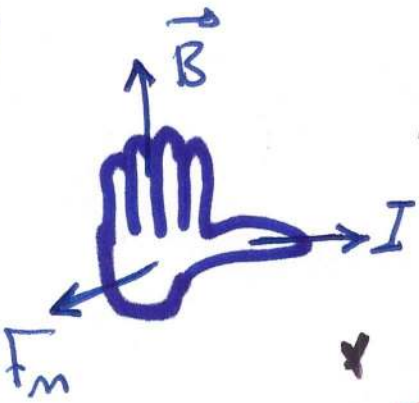
$qE = qVB$

$V = \frac{E}{B} \Rightarrow V = \frac{200}{10} = 10\ m/s.$

$$\vec{F}_B = I\,\vec{l}\times\vec{B}$$

$$F_B = I\,l\,B\sin\theta_{IB}$$





$F = 0$

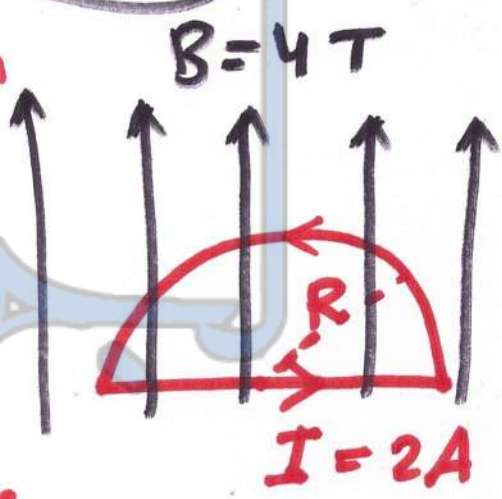
$\theta = 0°, 180°$.



Ex: $R = 10\,cm$, $B = 4\,T$

Find:

1) Net force on the current loop
2) Force on the stright wire
3) " " Bented wire



① $\vec{F}_{Net} = 0$

② $\vec{F} = I\,l\,B\sin\theta$

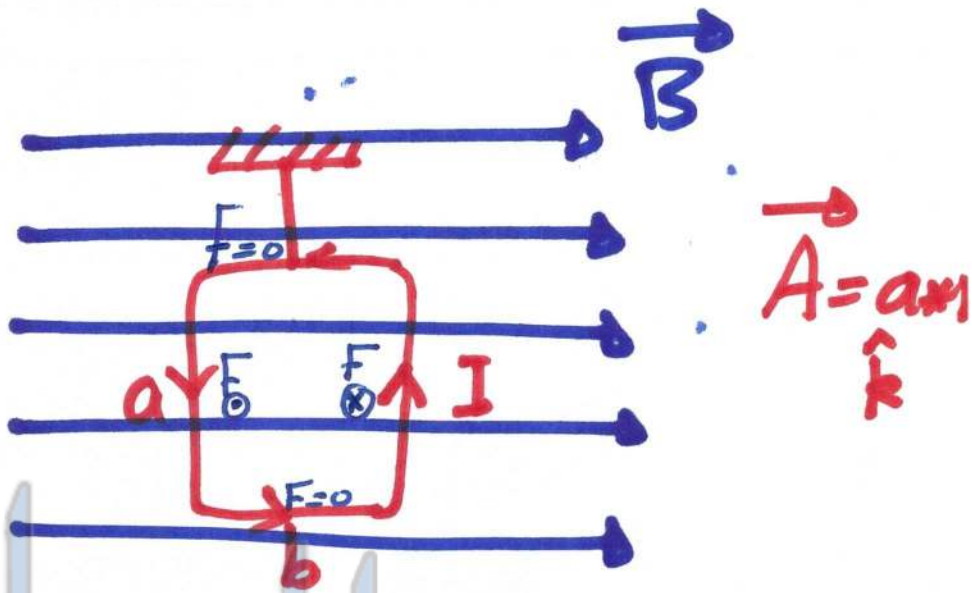
$= 2 * 20\times10^{-2} * 4\sin 90$

$= 1.6\,(N)\cdot[2\,j\,B]\,\hat{k}$

③ $\vec{F}_{arc} = 2\,I\,R\,B\,(-\hat{k})$

$= 1.6\,(-\hat{k})$ New.

Torque: $(\underline{\underline{\tau}})$



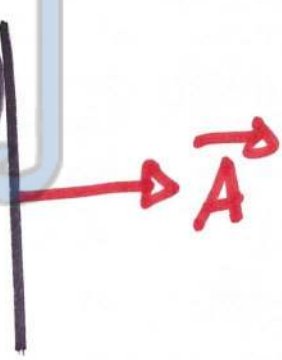
$$\vec{\tau} = NI\vec{A} \times \vec{B}$$

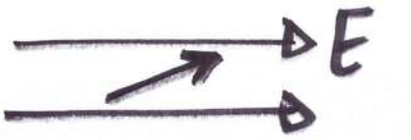
$$\tau = NIAB \sin\theta_{AB}$$

* $\vec{\mu} = NI\vec{A}$ [dipole magnetic moment]

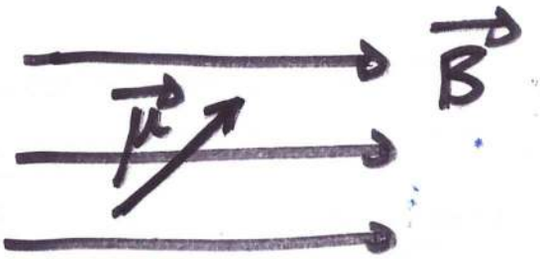
$$\Rightarrow \vec{\tau} = \vec{\mu} \times \vec{B}$$

$$\tau = \mu B \sin\theta_{\mu B}$$





$$U = -\vec{P}\cdot\vec{E}$$



U: Potential energy of magnetic dipole moment in $\vec{B}$

$$U = -\vec{\mu}\cdot\vec{B}$$

$$= -\mu B\cos\theta_{\mu B}$$

$$U_{max} = \mu B \quad (\theta = 180^\circ)$$

$$U_{min} = -\mu B \quad (\theta = 0)$$

CH:29 انتهى

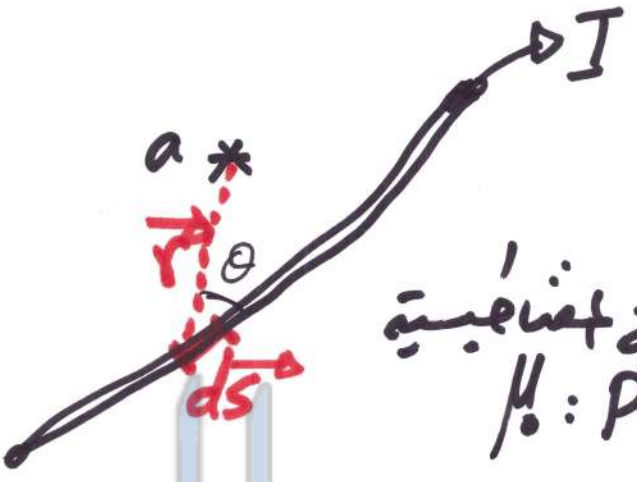


CH:30 Sources of mag. Field

1 Biot - Savart law :-

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I\, d\vec{s} \times \hat{r}}{r^2}$$

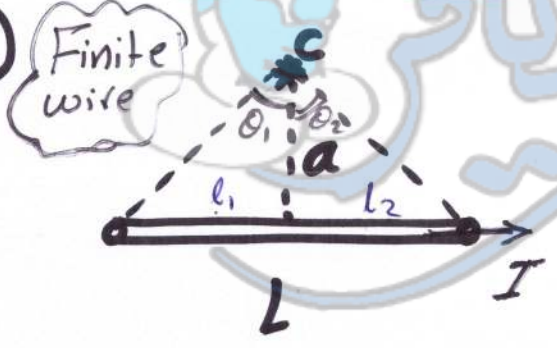
$$\vec{B} = \frac{\mu_0}{4\pi} I \int \frac{d\vec{s} \times \hat{r}}{r^2}$$



μ_0 : Permeability of free space

$\mu_0 = 4\pi * 10^{-7}$ T.m/A

(a) Finite wire



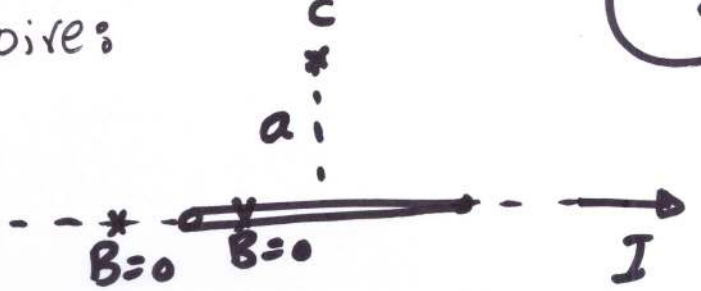
$$B_c = \frac{\mu_0 I}{4\pi a} [\sin\theta_1 - \sin\theta_2]$$

$$\sin\theta_1 = \frac{l_1}{\sqrt{l_1^2 + a^2}}$$

$$\sin\theta_2 = \frac{l_2}{\sqrt{l_2^2 + a^2}}$$

(b) Infinite wire:

$$B_c = \frac{\mu_0 I}{2\pi a}$$



ⓒ circular wire



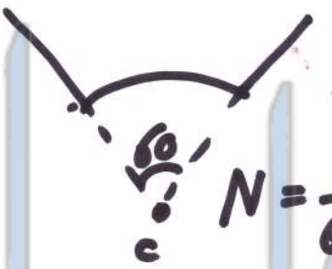
N: # of turns

$$B = \frac{\mu_0 N I}{2a} = \frac{\mu_0 I \theta}{4\pi a}$$

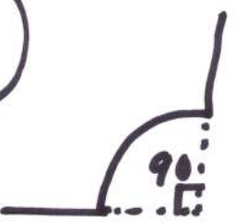
$$N = \frac{\theta}{2\pi}$$



$N = \frac{1}{2}$



$N = \frac{1}{6}$



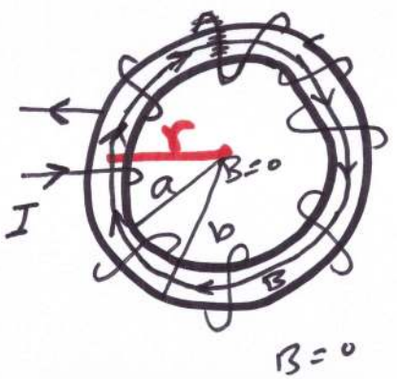
$N = \frac{1}{4}$

ⓓ



$$B_b = \frac{\mu_0 I a^2 N}{2(a^2+x^2)^{3/2}}$$

ⓔ Toroid:



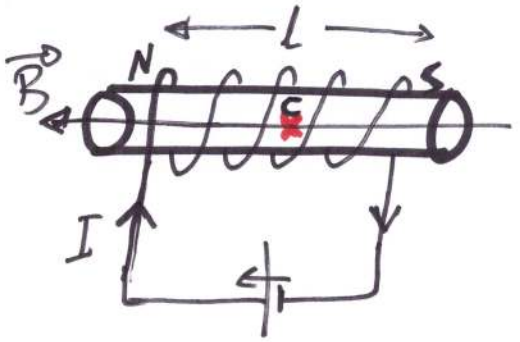
B = 0

$$B = \frac{\mu_0 N I}{2\pi r}$$

$a < r < b$

(f) Solenoid





$$B_c = \frac{\mu_0 N I}{L} = \mu_0 n I$$

$\frac{N}{L} = n$: # of turns per unit length

Find the net $\vec{B}$ at C:

Ex:



$B = 4 * 10^{-5}\ T$

$I = 6A$

$B_1 \equiv \otimes\ (-\hat{k})$

$B_2 \equiv \otimes\ (-\hat{k})$ straight

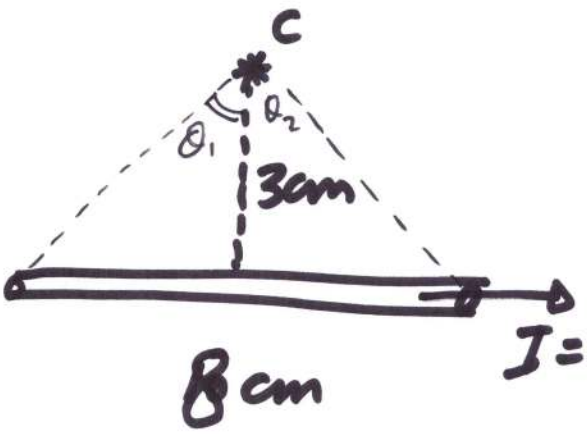
١) الاتجاهات
٢) الحسابات
٣) المحصلة

$$B_1 = 4 * 10^{-5}\ T$$

$$B_2 = \frac{\mu_0 I}{2\pi a} = \frac{4\pi \times 10^{-7} * 6}{2\pi\, 3 \times 10^{-2}} = 4 * 10^{-5}\ T.$$

$$B_c = B_1 + B_2 = 8 * 10^{-5}\ T \otimes (-\hat{k}).$$

Ex:



$\sin\theta_1 = \frac{4\,cm}{5\,cm} = 0.8$

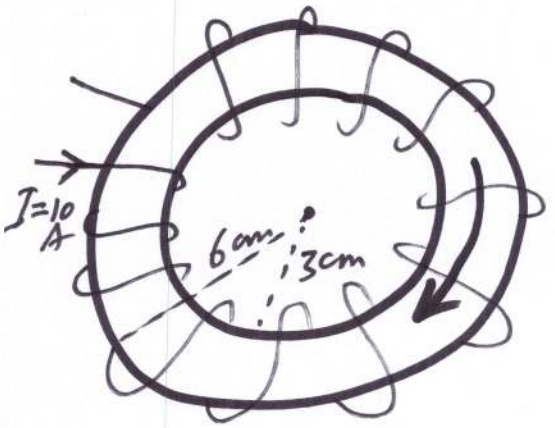
$\sin\theta_2 = \frac{-4\,cm}{5\,cm} = -0.8$

$$B_c = \frac{\mu_0 I}{4\pi a}(\sin\theta_1 - \sin\theta_2)$$

$$= \frac{4\pi \times 10^{-7} * 10}{4\pi 3 * 10^{-2}}(0.8 - (-0.8))$$

$$= \frac{16}{3} * 10^{-5}\ T$$

Ex: Find B at 5cm from the center:



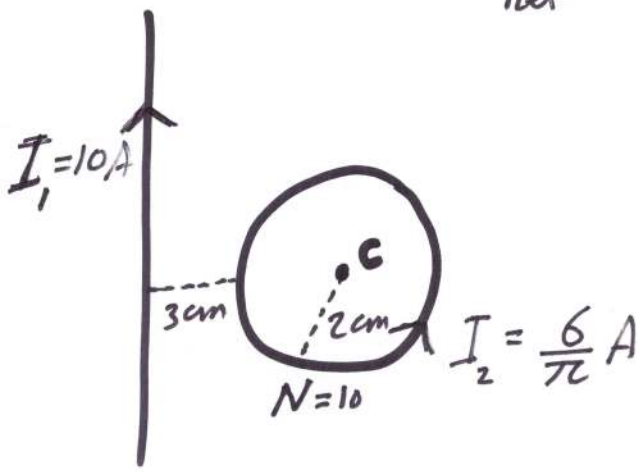
N = 10

$$B = \frac{\mu_0 N I}{2\pi r}$$

$$= \frac{4\pi \times 10^{-7} * 10 * 10}{2\pi 5 * 10^{-2}}$$

$$= 40 * 10^{-7}\ T$$

Ex: Find B_{net} at C :-



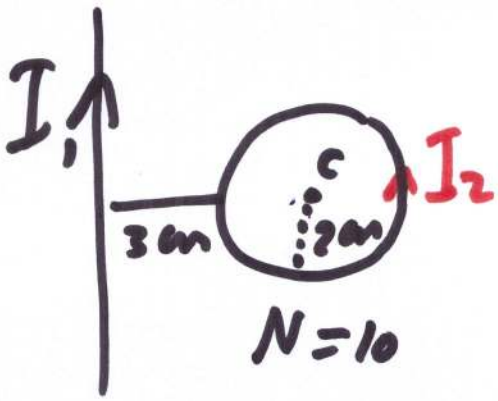
* C $B_1 \otimes$
$B_2 \odot$

$$B_1 = \frac{\mu_0 I}{2\pi a} = \frac{4\pi \times 10^{-7} * 10}{2\pi 5 * 10^{-2}} = 4 * 10^{-5}\ T$$

$$B_2 = \frac{\mu_0 I N}{2a} = \frac{4\pi * 10^{-7} * 10 * 6}{2 * 2 \times 10^{-2} \pi} = 60 * 10^{-5}\ T$$

$$B_{net} = B_2 - B_1 = 56 * 10^{-5}\ T \odot (+\hat{k})$$

Ex: In the previous example, Find I_2 if B at C is $1 * 10^{-5}\ T\ (-\hat{k})$



* C $B_1 \otimes = 4 * 10^{-5}\ T$

$B_c \otimes = 1 * 10^{-5}\ T$

$B_2 \odot = 3 * 10^{-5}\ T$

if $B_c = 0$

$B_1 = B_2$

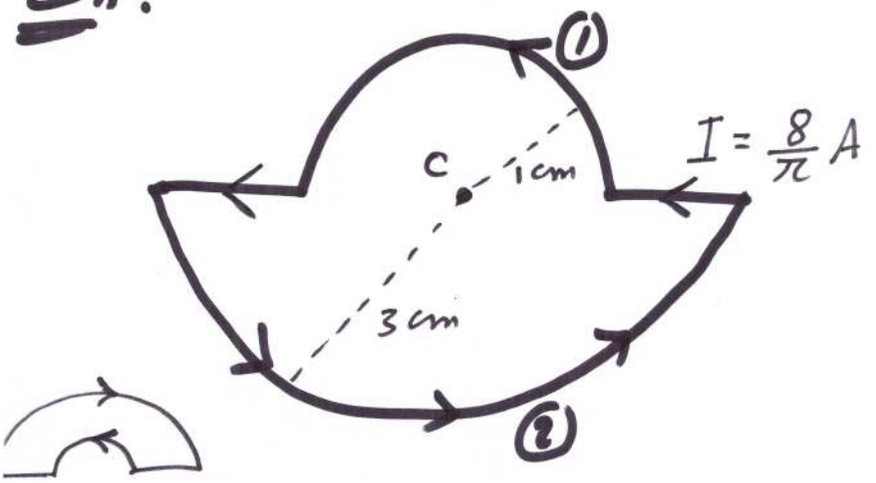
$\frac{\mu_0 I_1}{2\pi a_1} = \frac{\mu_0 I_2 N}{2\pi a_2}$

$\frac{4\pi}{\pi 5 \times 10^{-2}} = \frac{I_2 * 10}{2\ cm}$

$I_2 = \frac{0.4}{\pi}$ A

$B_2 = \frac{\mu_0 I N}{2\pi a} \Rightarrow 3 * 10^{-5} = \frac{4\pi \times 10^{-7} I_2\ 10}{2 * 2 * 10^{-2}} \Rightarrow I_2 = \frac{0.3}{\pi}$ A

Ex:



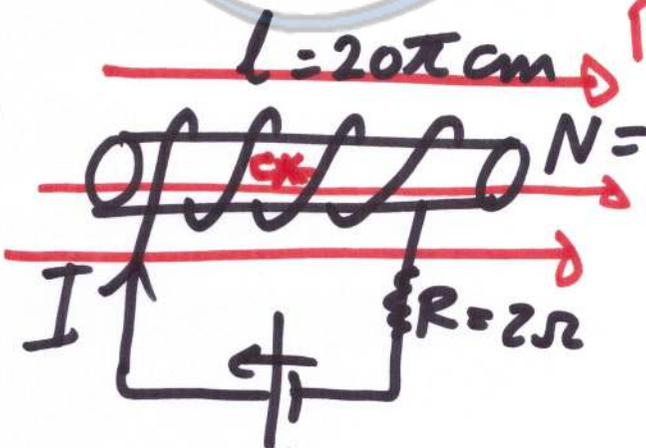
C
$B_1 \odot$
$B_2 \odot$

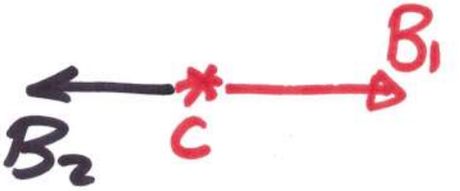
$$B_1 = \frac{\mu_0 N I}{2a} = \frac{4\pi \times 10^{-7} * 1 * 8}{2 \times 1 \times 10^{-2} * 2 * \pi} = 8 * 10^{-5} T$$

$$B_2 = \frac{4\pi \times 10^{-7} * 1 \times 8}{2 \times 3 \times 10^{-2} \; 2 \times \pi} = \frac{8}{3} \times 10^{-5} T$$

$$B_c = B_1 + B_2 = \frac{32}{3} * 10^{-5} T \odot (+\hat{k}) .$$

Ex:





$$I = \frac{\mathcal{E}}{R} = 10A$$

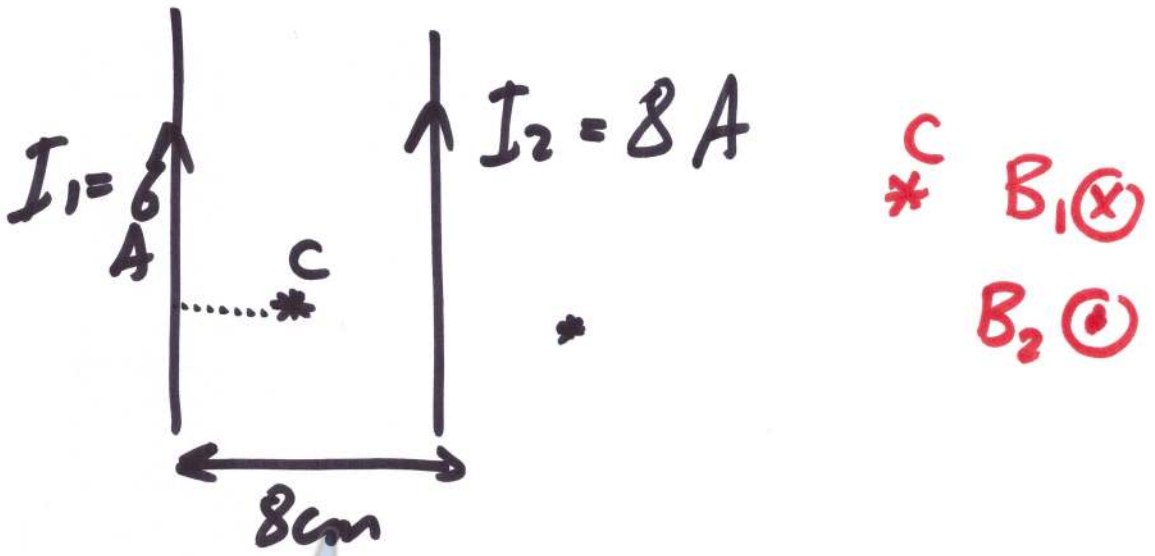
$$B_c = B_1 - B_2$$

$$= 25 \times 10^{-5} - \frac{4\pi \times 10^{-7} \times 10 \times 10}{20\pi \times 10^{-2}}$$

$$= 25 \times 10^{-5} - 20 * 10^{-5}$$

$$= 5 * 10^{-5} T \; (+\hat{k})$$

Ex:

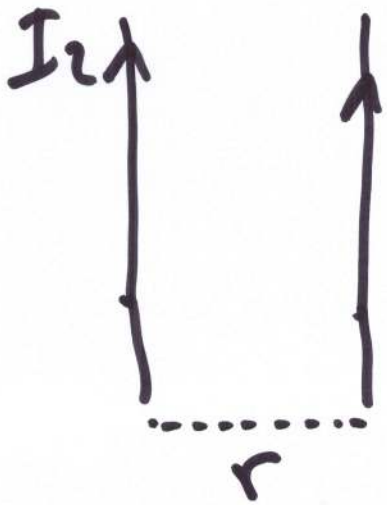


$$B_1 = \frac{4\pi \times 10^{-7} * 6}{2\pi\, 4\times 10^{-2}} = 3\times 10^{-5}\ T$$

$$B_2 = \frac{4\pi \times 10^{-7} \times 8}{2\pi * 4\times 10^{-2}} = 4\times 10^{-5}\ T$$

$$B_c = B_2 - B_1 = 1 * 10^{-5}\ T \odot (+\hat{k}).$$

نقطة يتعادد عندها:



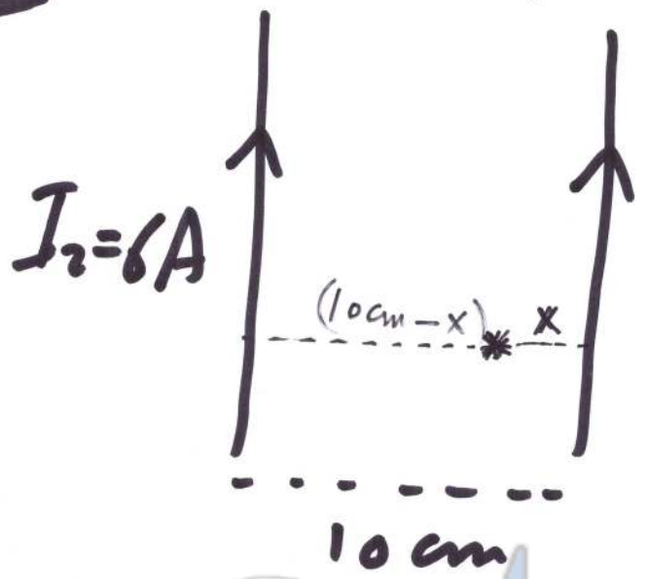
$B_1 = B_2$

متعاكسان

بينهما أقرب للأصغر ⇐ ↑ ↑

خارجهما أقرب للأصغر ⇐ ↓ ↑

Ex: Find the Point of equilibrium (where $B_{net} = 0$)



$$B_1 = B_2$$

$$\frac{\mu_0 I_1}{2\pi a_1} = \frac{\mu_0 I_2}{2\pi a_2}$$

$$\frac{4}{x} = \frac{6}{10\times10^{-2} - x}$$

$$\Rightarrow 40\times10^{-2} - 4x = 6x$$

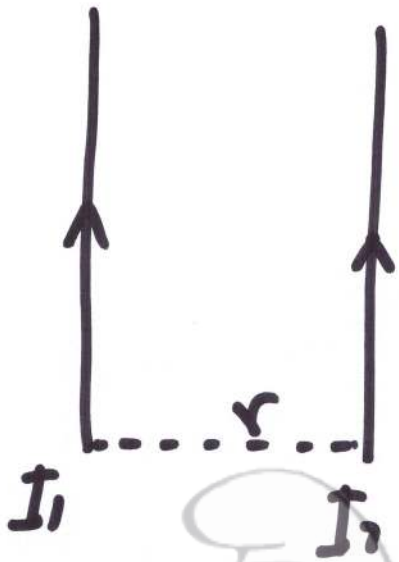
$$10x = 40\times10^{-2}$$

$$x = 4\times10^{-2}\ m = 4\ cm.$$

Force between the two wires:-

* ↑↑ ⇐ تجاذب

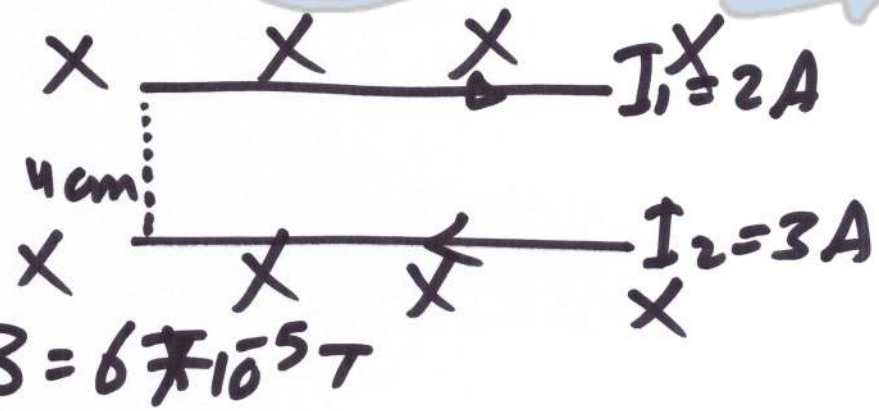
* ↓↑ ⇐ تنافر



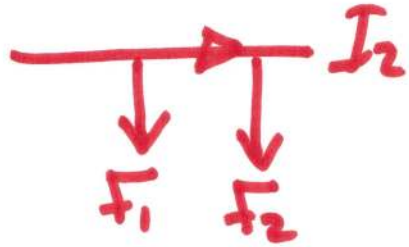
$$F = \frac{\mu_0 I_1 I_2 L}{2\pi r}$$

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi r}$$

Ex: what is the net Forces on I_2:



$B = 6 * 10^{-5} T$

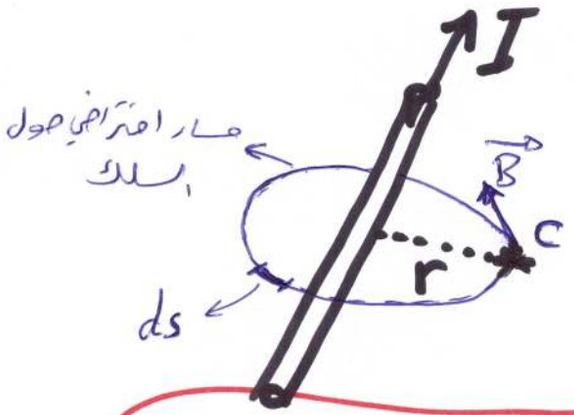


$$\frac{F_1}{L} = \frac{4\pi \times 10^{-7} * 2 * 3}{2\pi \times 4 \times 10^{-2}} = 3 * 10^{-5} \text{ N/m}.$$

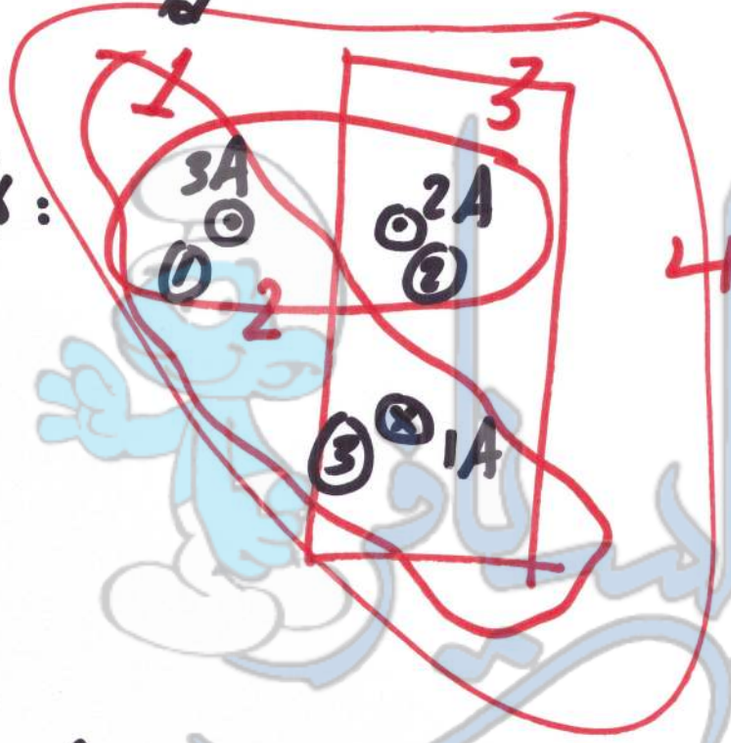
$$\frac{F_{net}}{L} = \frac{F_1 + F_2}{L} = 21 \times 10^{-5} \frac{N}{m} (-\hat{j})$$

$$\frac{F_2}{1} = I_2 B \sin\theta = 3 * 6 * 10^{-5} \sin 90 = 18 * 10^{-5} \text{ N/m}$$

§3 Ampere's law :-



$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{ins}$$



Ex : Find the line integral of $\vec{B} \cdot d\vec{s}$ ($\oint \vec{B} \cdot d\vec{s}$) for each line.

1) $\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{ins} = 4\pi * 10^{-7} * (3-1)$

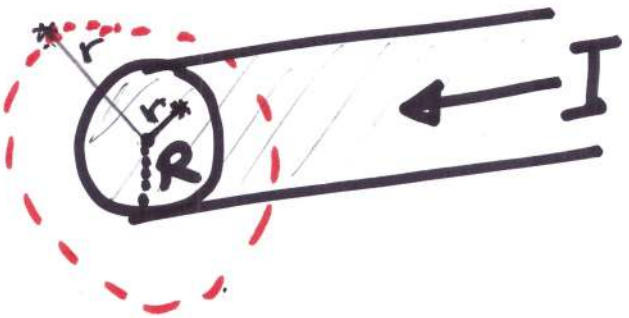
$= 8\pi * 10^{-7}$ T.m

2) $\oint \vec{B} \cdot d\vec{s} = \mu_0 (3+2) = 5\mu_0$ T.m

3) $\oint \vec{B} \cdot d\vec{s} = \mu_0 (2-1) = \mu_0$ T.m.

4) $\oint \vec{B} \cdot d\vec{s} = \mu_0 (3+2-1) = 4\mu_0$ T.m.

Ex: Find $\vec{B}$ at 1) $r \geq R$
2) $r < R$



1) $r \geq R$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{ins}$$

$$\int ds = S_{total} = 2\pi r$$

$$B(2\pi r) = \mu_0 I$$

$$\boxed{B_{out} = \frac{\mu_0 I}{2\pi r}}$$

2) $r < R$



$$B * S = \mu_0 I_{ins}$$

$$B * (2\pi r) = \mu_0 I'$$

$$B = \frac{\mu_0 I r^2}{2\pi R^2}$$

$$\boxed{B = \frac{\mu_0 I r}{2\pi R^2}}$$

$$\frac{I}{I'} = \frac{\pi R^2}{\pi r^2}$$

$$\boxed{I' = I \frac{r^2}{R^2}}$$

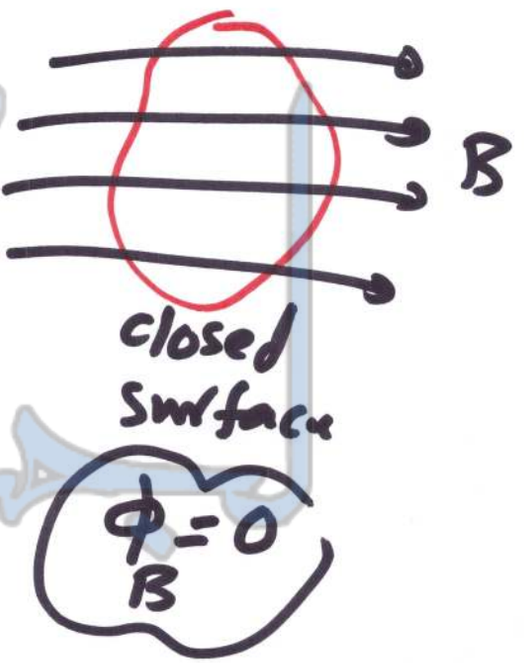
§5: Gauss's law of magnetism

ϕ_B: magnetic Flux التدفق المغناطيسي

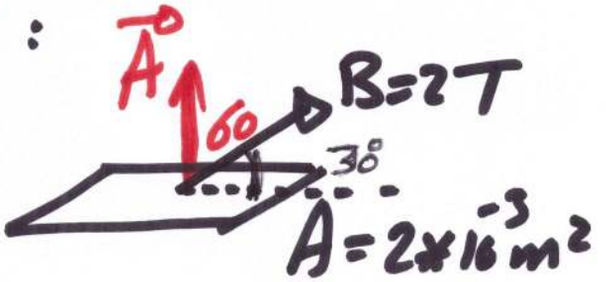
$$\phi_B = \int \vec{B} \cdot d\vec{A} = \vec{B} \cdot \vec{A} = BA\cos\theta_{BA}$$

* The net magnetic flux (ϕ_B) through any closed surface is always Zero.

$$\phi_B = \oint \vec{B} \cdot d\vec{A} = 0$$



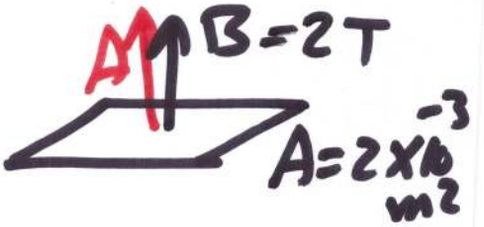
Ex:



$\phi = BA\cos\theta$

$= 2 * 2\times10^{-3} \cos 60$

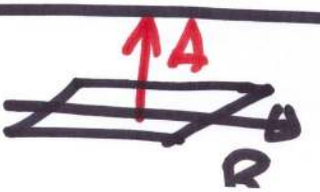
$= 2 * 10^{-3}$ Weber



$\phi = 2 * 2\times10^{-3} \cos 0$

$\phi_{max} = 4 * 10^{-3}$ Wb.

$\phi = 0$



Ex: If $\vec{B} = 4\hat{i} + 3\hat{j} + 5\hat{k}$

$\vec{A} = 4\hat{i} + 2\hat{j}$

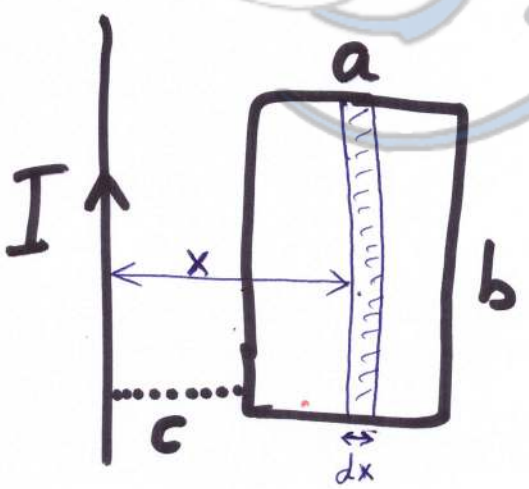
Find Flux (magnetic) and θ_{BA}

$$\phi = \vec{B} \cdot \vec{A} = 16 + 6 = 22 \text{ Wb}.$$

$$\phi = BA\cos\theta$$

$$22 = \sqrt{4^2+3^2+5^2} * \sqrt{4^2+2^2} \cos\theta_{BA}$$

Ex:



$x : c \rightarrow c + a$

$dA = b * dx$

$$|\vec{B}| = \frac{\mu_0 I}{2\pi x}$$

$$\phi = \int \frac{\mu_0 I}{2\pi x} \cdot b\,dx$$

$$= \frac{\mu_0 I b}{2\pi} \int \frac{dx}{x}$$

$$= \frac{\mu_0 I b}{2\pi} \ln x \Big]_c^{c+a}$$

$$\phi = \frac{\mu_0 I b}{2\pi} \ln\left(\frac{c+a}{c}\right).$$

CH:30

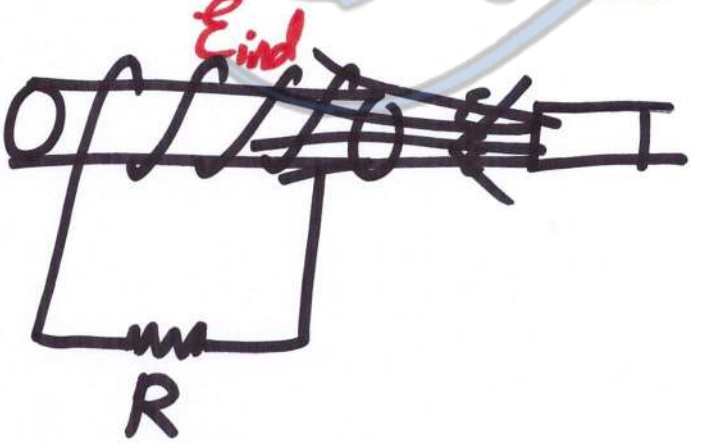
CH:31 Faraday's law

$$\phi_B = \int \vec{B}\cdot d\vec{A} = \vec{B}\cdot\vec{A} = BA\cos\theta_{BA}$$

$$\Delta\phi = \begin{cases} \Delta B\, A\cos\theta \rightarrow (\Delta B = B_2 - B_1) \\ B\,\Delta A\cos\theta \rightarrow (\Delta A = A_2 - A_1) \\ B\,A\,\Delta\cos\theta \rightarrow (\Delta\cos\theta = \cos\theta_2 - \cos\theta_1) \end{cases}$$

$$\phi_2 - \phi_1 = B_2A_2\cos\theta_2 - B_1A_1\cos\theta_1$$

$$d\phi = d(BA\cos\theta) = A\cos\theta\, dB$$



$$\mathcal{E}_{ind} = -N\frac{d\phi}{dt}$$

$$\mathcal{E}_{ind} = -N\frac{\Delta\phi}{\Delta t}$$

$$I_{ind} = \frac{|\mathcal{E}_{ind}|}{R}$$

*magnitude of $\mathcal{E}_{ind}$ is $|\mathcal{E}_{ind}|$

Ex: A plane of dimentions 10cm x 6cm, and a Uniform mag. field B = 4T directed outof page Per.R to the plane. Find the

1) electromotive force (induced).

2) induced Curret (R = 2Ω).

if the B is dropped to zero through 0.2 sec.

B = 4T ↓ 0T , A = 60 x 10^-4 , θ = 0 , Δt = 0.2 sec.

1) $\Delta\phi = \Delta B\ A \cos\theta$

$= (0 - 4)\, 60 * 10^{-4} \cos 0$

$= -240 \times 10^{-4}$ Wb.

$$\mathcal{E}_{ind} = -N \frac{\Delta\phi}{\Delta t} = -1 \times \frac{-240 \times 10^{-4}}{0.2} = +120 * 10^{-3} \text{ wb/s}$$

2) $$I_{ind} = \frac{|\mathcal{E}|}{R} = \frac{120 * 10^{-3}}{2} = 60 * 10^{-3} A.$$

Ex: If $B = B_{max} e^{-\alpha t}$, Find the induced $\mathcal{E}$ through a surface of Area A parralled to the field. ($\theta = 0$)

Sol

$$\Phi = BA\cos\theta$$

$$= A B_{max} e^{-\alpha t}$$

$$\frac{d\Phi}{dt} = AB_{max}(-\alpha e^{-\alpha t})$$

$$= (-\alpha A B_{max} e^{-\alpha t})$$

$$\mathcal{E}_{ind} = -N\frac{d\Phi}{dt} = \alpha A B_{max} e^{-\alpha t}.$$

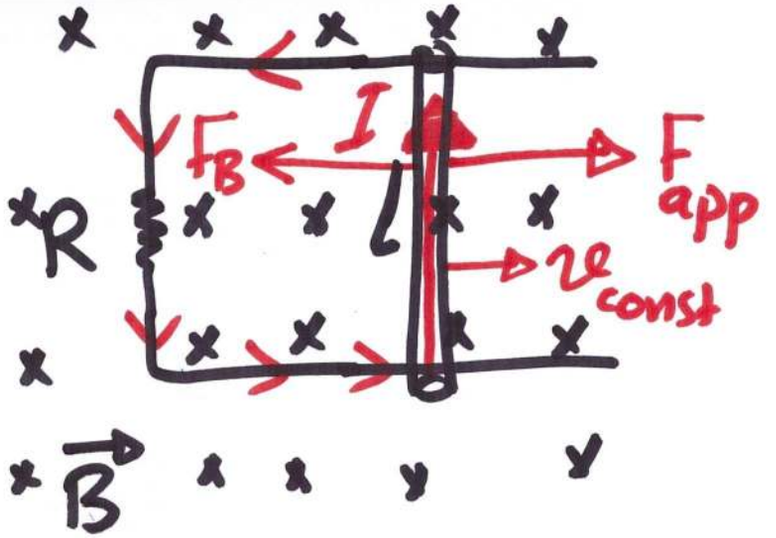
(ins)

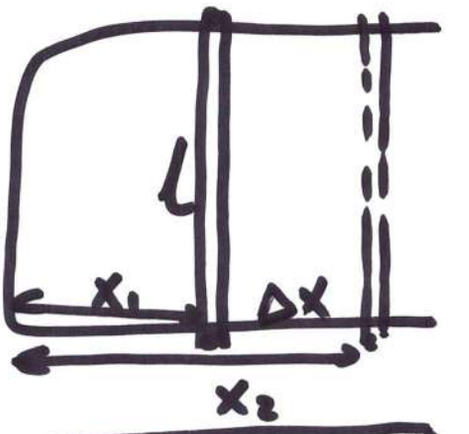
* $\mathcal{E}_{av} = -\frac{\Delta\Phi}{\Delta t} = \frac{\Phi_2 - \Phi_1}{t_2 - t_1}$

from $t=0 \to 2$

$$\Phi_1 = A B_1$$

$$\Phi_2 = A B_2$$





$$\Delta A = \ell * \Delta x$$

$$\Delta\Phi = B\,\Delta A \cos\theta$$

$$\mathcal{E}_{in} = -\frac{\Delta\Phi}{\Delta t}$$

$$\mathcal{E}_{ind} = -\ell v B$$

$$F_{app} = F_B = I\ell B \sin_{\ell B}$$

$$\frac{|\mathcal{E}|}{R} = I$$

$$\frac{\Delta x}{\Delta t}$$

$$\text{Power} = F_{app}\, v = I\ell B v$$

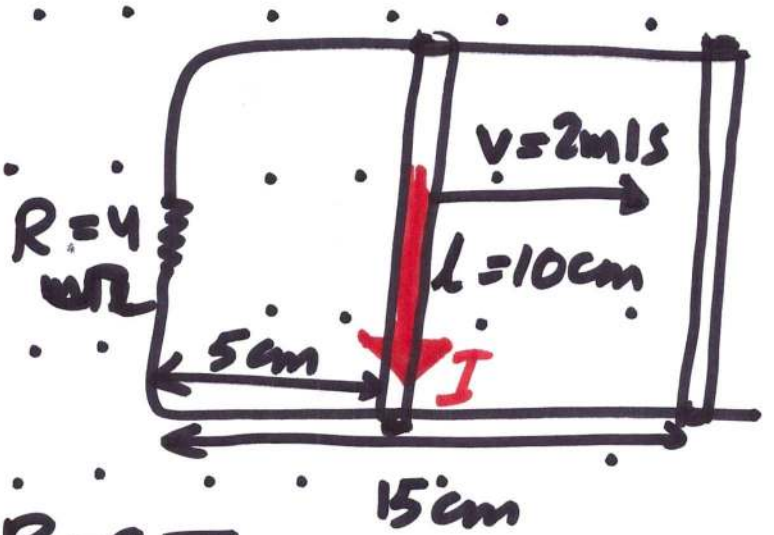
$$\text{Power} = \frac{B^2\ell^2 v^2}{R} = \mathcal{E}^2/R$$

If $V_{ini} \neq 0$ (V is not constant)

$$\Rightarrow v = v_{initial}\, e^{-t/\tau} \qquad \left(\tau = \frac{mR}{B^2\ell^2}\right)$$

Ex In the figure, Find:

1) ΔX
2) Δt
3) ΔA
4) $\Delta \phi$
5) $\mathcal{E}$
6) I
7) F_{app} , F_B
8) Power.



Sol

1) $\Delta X = 15 - 5 cm = 10 \times 10^{-2} m$

2) $V = \frac{\Delta X}{\Delta t} \Rightarrow \Delta t = \frac{10 \times 10^{-2}}{2} = 5 * 10^{-2} sec.$

3) $\Delta A = L * \Delta X = 10 \times 10^{-2} * 10 \times 10^{-2} = 100 * 10^{-4} m^2.$

4) $\Delta \phi = B \Delta A \cos\theta = 2 * 1 \times 10^{-2} \cos 0 = 2 \times 10^{-2} Wb$

5) $\mathcal{E}_{ind} = -\frac{\Delta \phi}{\Delta t} = -\frac{2 \times 10^{-2}}{5 \times 10^{-2}} = -0.4$ Volt.

6) $I_{in} = \frac{|\mathcal{E}|}{R} = \frac{0.4}{4} = 0.1 A [-\hat{j}]$

F_{app}, $+i$, $-i$, F_B

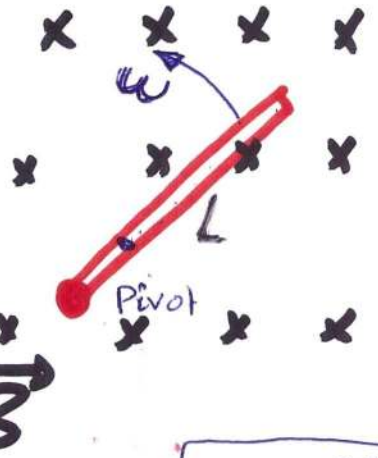
7) $F_{app} = F_B = I l B = 0.1 * 10 \times 10^{-1} * 2 = 2 \times 10^{-1} N$

8) Power $= \mathcal{E}^2 / R = 0.16 / 4 = 4 * 10^{-2}$ watt.

$$\mathcal{E} = \frac{1}{2} B\omega l^2$$

$$I_{in} = \frac{|\mathcal{E}|}{R}$$

$$T = \frac{2\pi}{\omega}$$



$$\omega = \frac{2\theta}{r}$$

ω : angular Velocity (rad/sec)

T : Period

Ex 31.6

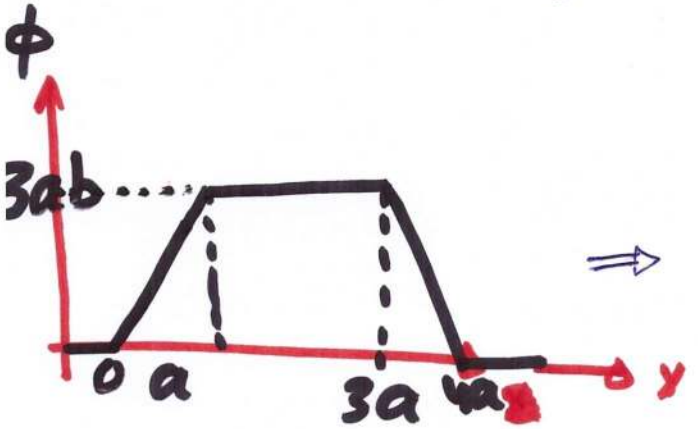


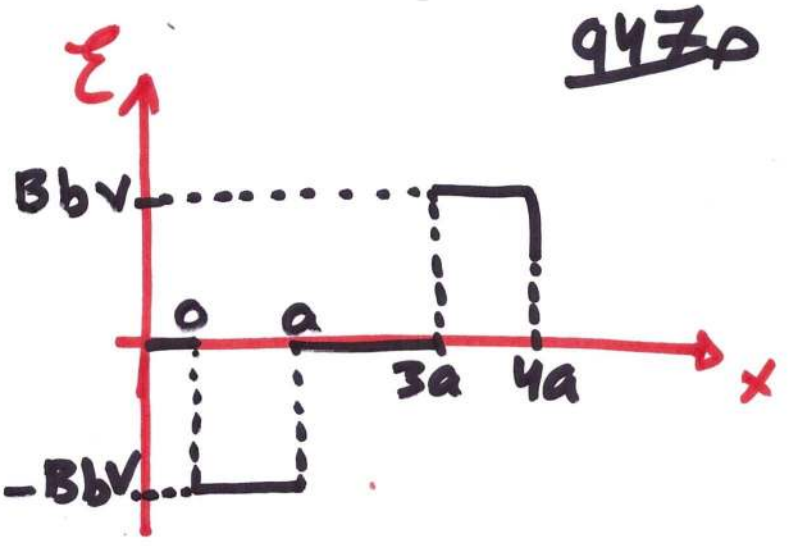
$$I = \frac{|\mathcal{E}|}{R}$$

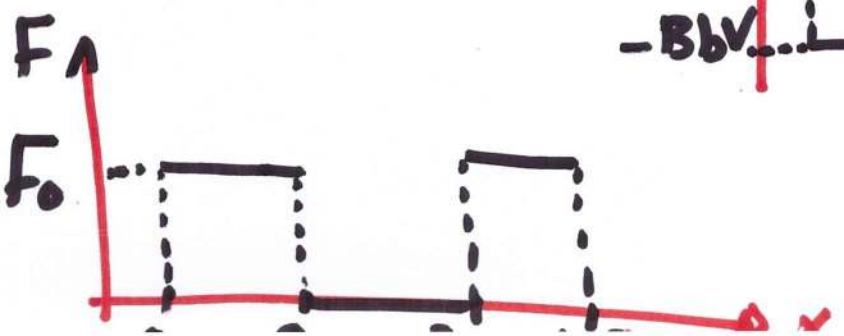
$$F = I l * B$$

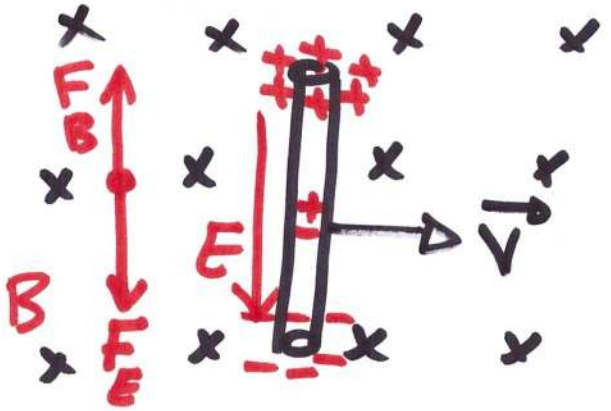
$$F_0 = \frac{2 l^2 B^2}{R}$$

94Zo









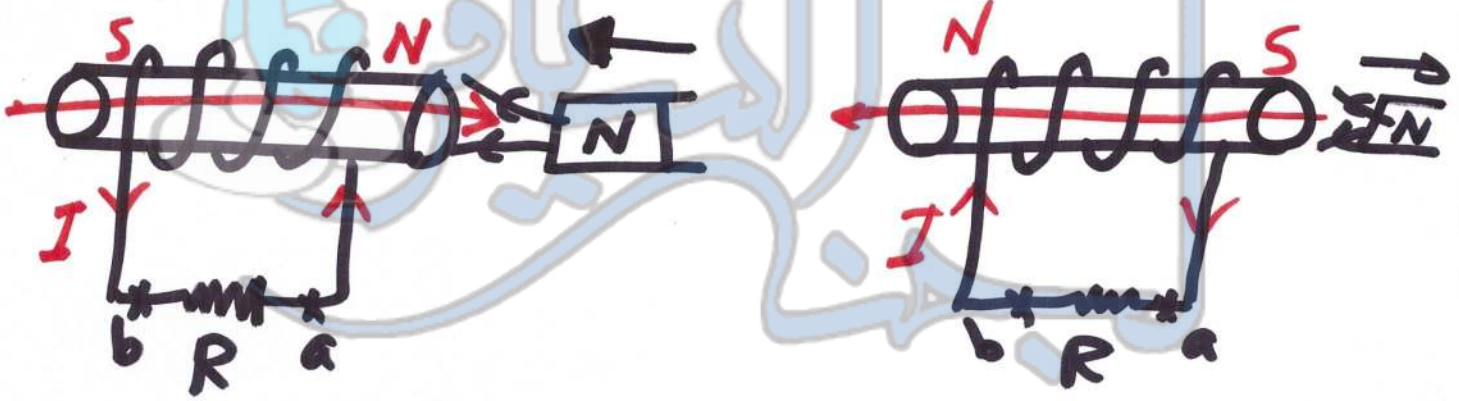
$$F_B = q v B \sin\theta$$

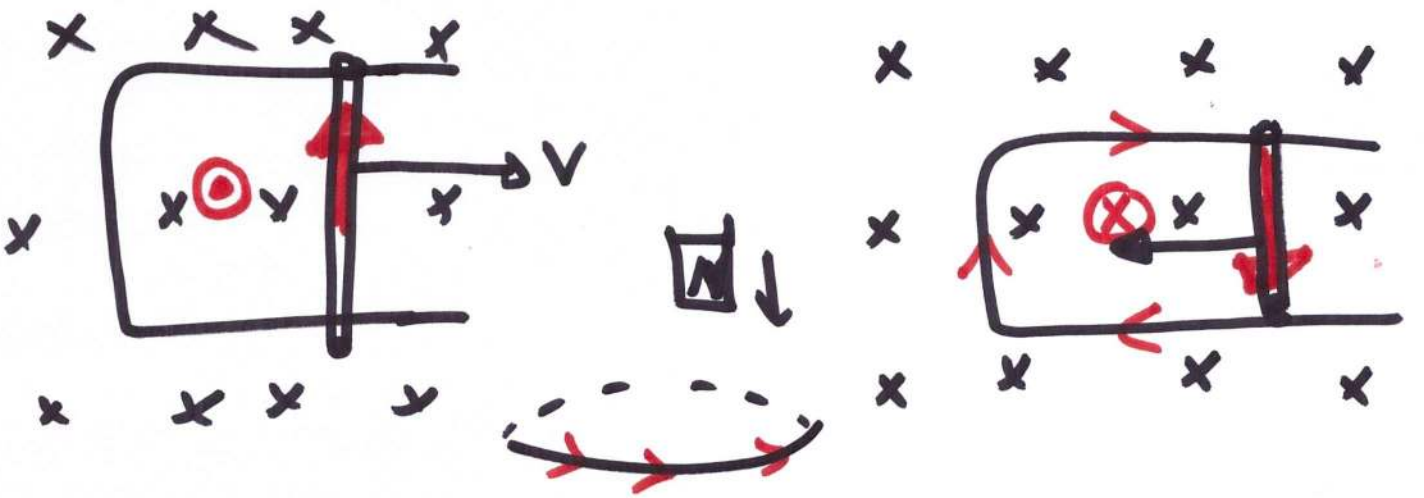
$$F_E = F_B$$

$$qE = qvB$$

$$v = \frac{E}{B}$$

Lenz law :-





§4: Induced emf and Electric fields



B: depent on time

$B \equiv B(t)$

$$\frac{d\Phi}{dt} = \frac{d}{dt}(BA) = A\frac{dB}{dt} \qquad A \begin{cases} A = \pi r^2 \\ a \times b \end{cases}$$

$$\mathcal{E} = -N\frac{d\Phi}{dt} = -NA\frac{dB}{dt}$$

$$q\mathcal{E} = qE(2\pi r)$$

$$\Delta V = E \cdot L \qquad \frac{}{L}$$

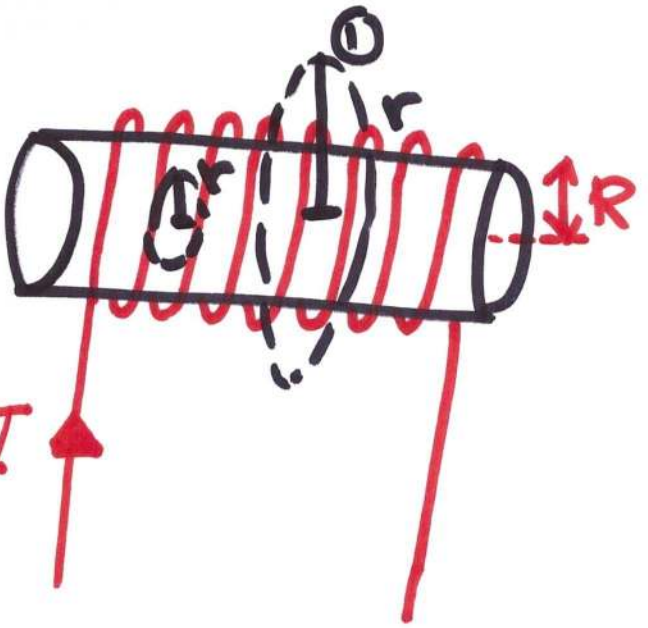
$$E = \frac{\mathcal{E}}{2\pi r} = -\frac{1}{2\pi r} N\pi r^2 \frac{dB}{dt}$$

$$E_{in} = -\frac{N}{2} r \frac{dB}{dt}$$

$$\oint E \cdot ds = -\frac{N}{dt}\frac{d\Phi}{dt}$$

$$E \cdot \Delta L = \mathcal{E}$$

Ex 31.7
948



$\underline{n} = \frac{N}{L}$

$I = I_{max} \cos \omega t$

1) $r > R$

$$\mathcal{E} = -\frac{d\Phi}{dt} = -\frac{d}{dt}(BA) = -\frac{d}{dt}(\mu_0 n I \pi R^2)$$

$$= -\mu_0 n \pi R^2 \frac{dI}{dt}$$

$$= -\mu_0 n \pi R^2 (-I_{max} \omega \sin \omega t)$$

$$\mathcal{E} = \mu_0 n \pi \omega R^2 I_{max} \sin \omega t .$$

$$E(\Delta L) = \mathcal{E}$$

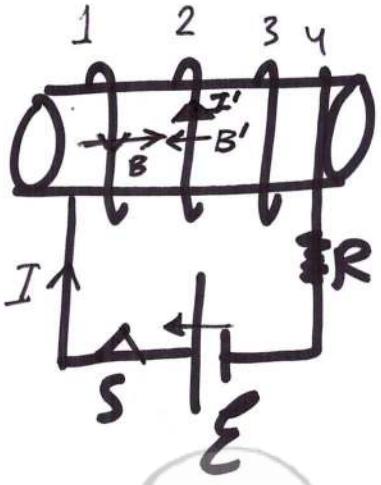
$$E(2\pi r) = \mu_0 n \pi \omega R^2 I_{max} \sin \omega t$$

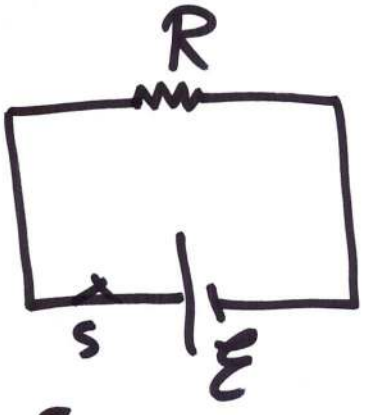
$$E_{out} = \frac{\mu_0 n \omega R^2 I_{max}}{2r} \sin \omega t \quad (r > R)$$

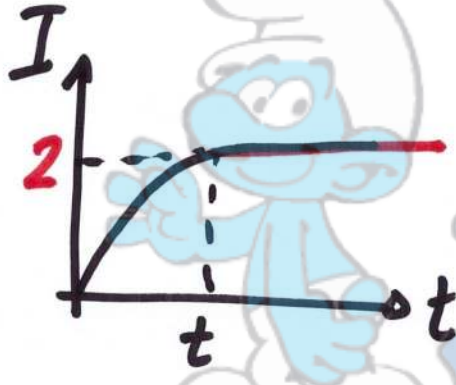
$$E_{ins} = \frac{\mu_0 n \omega I_{max}}{2} r \sin \omega t$$

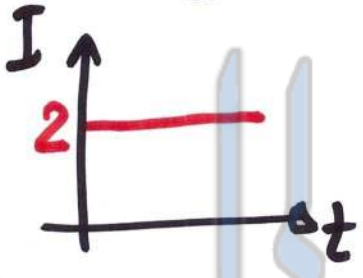
CH:31

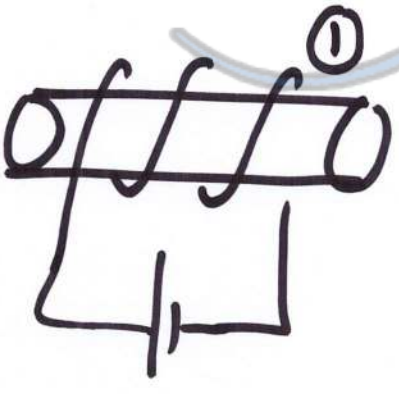
CH:32 Self Induction and Inductance









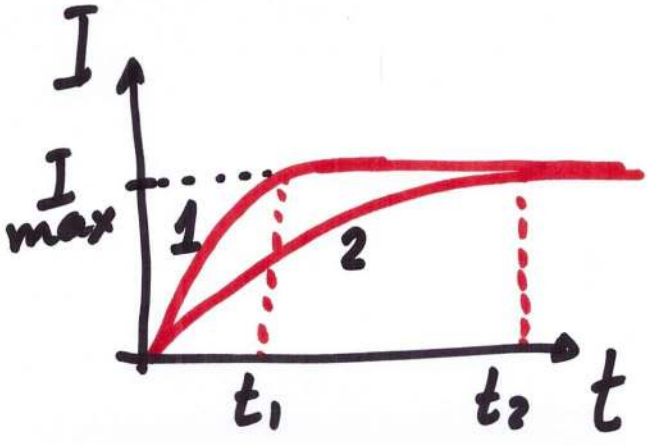




Inductance

⇓

L



$L_2 > L_1$

self Induction:

$$\mathcal{E}'_L = -N\frac{d\phi}{dt} = -N\frac{\Delta\phi}{dt}$$ general

$\phi = BA\cos\theta$

$d\phi = d(BA\cos\theta)$

$\Delta\phi = \Delta(BA\cos\theta)$

$$\mathcal{E}'_L = -L\frac{di}{dt} = -L\frac{\Delta i}{\Delta t}$$

$$i' = \frac{|\mathcal{E}'|}{R}$$

$$L = \frac{N\Delta\phi}{\Delta i} = \frac{N\phi_B}{i}$$ general

$$L = \frac{\mu_0 N^2 A}{\ell} = \mu_0 n^2 \ell A = \mu_0 n^2 Vol$$ solenoid

$L \longrightarrow$ Henry (H)

$C = \frac{q}{V}$

$R = \frac{V}{i}$

$L = \frac{N\phi}{i}$

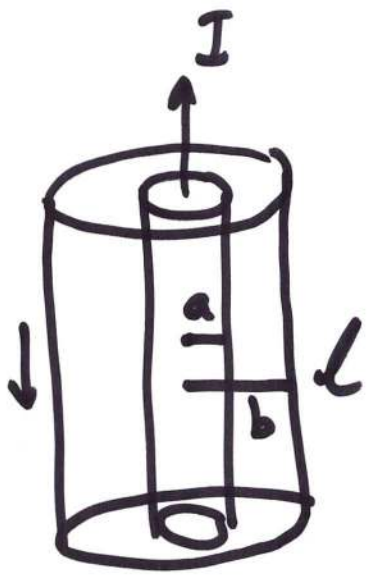
$N = n\ell$

$$U_B = \frac{1}{2}LI^2$$ (magnetic energy stored in the magnetic field inside the inductor)

$$u_B = \frac{U}{Vol} = \frac{B^2}{2\mu_0}$$ energy density or "per unit

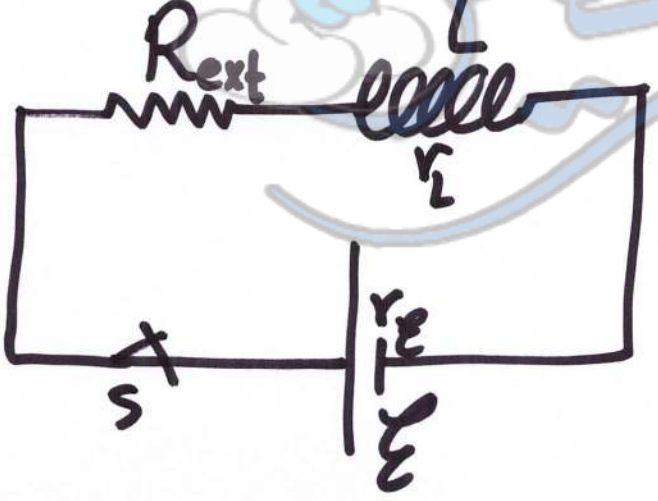
$u_E = \frac{1}{2}\epsilon_0 E^2$

$$L_{\text{coaxial cable}} = \frac{\mu_0 l}{2\pi} \ln\left(\frac{b}{a}\right)$$



$$C = \frac{2\pi\varepsilon l}{\ln\left(\frac{b}{a}\right)}$$

RL - circuit :



$$R = R_{ext} + r_L + r_\varepsilon$$

$$\varepsilon = iR + L\frac{di}{dt}$$

تشغيل الدارة

	$t=0$	$t=\infty$
i	0	i_{max}
$\frac{di}{dt}$	$\left(\frac{di}{dt}\right)_{max}$	0

$$i_{max} = \frac{\varepsilon}{R}$$

time constant τ

$$\left(\frac{di}{dt}\right)_{max} = \frac{\varepsilon}{L}$$

$$\tau = \frac{L}{R}$$

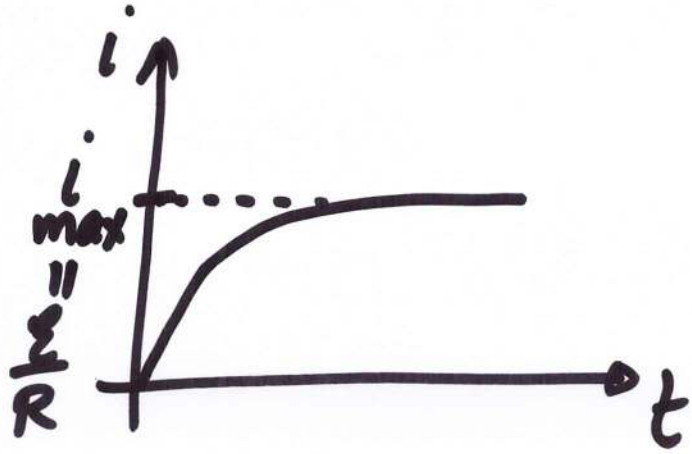
i	di/dt
1) i	1) di/dt
2) $V_R = iR$	2) $V_L = L\frac{di}{dt} + ir_L$
3) $V_\mathcal{E} = \mathcal{E} - ir_\mathcal{E}$	3) $\mathcal{E}' = -L\,di/dt$
4) $P_R = i^2R$	4) $P_{Power\ (L)} = iL\frac{di}{dt}$
5) $P_\mathcal{E} = i\mathcal{E}$	
6) $U_L = \frac{1}{2}Li^2$	

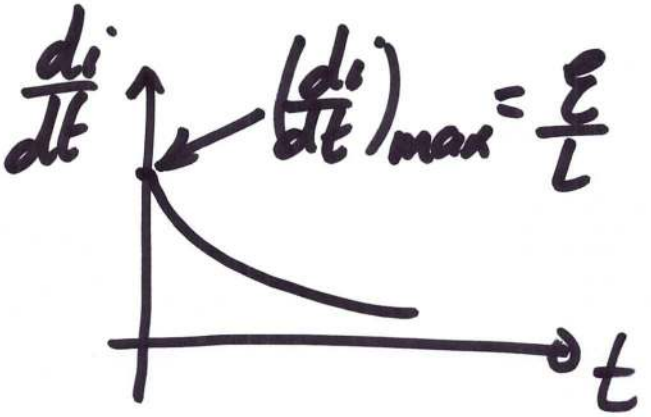
$$\mathcal{E} = iR + L\frac{di}{dt}$$

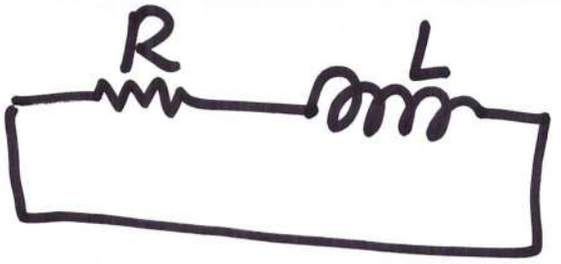
$$i = \frac{1}{2} i_{max}$$

$$i = i_{max}\left(1 - e^{-Rt/L}\right) \qquad \tau = \frac{L}{R}$$

$$\frac{di}{dt} = \left(\frac{di}{dt}\right)_{max} e^{-Rt/L}$$





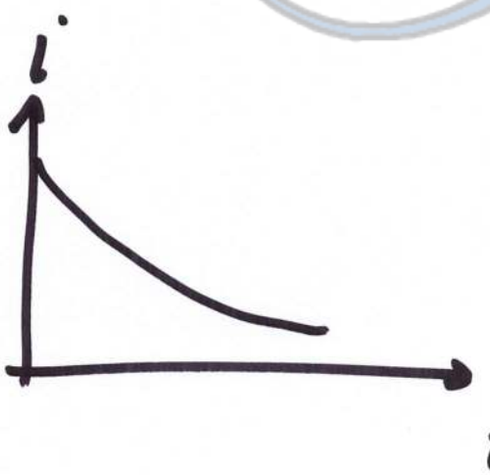


$$0 = iR + L\frac{di}{dt}$$

$$i_{max} = \frac{\mathcal{E}}{R} \quad , \quad \left(\frac{di}{dt}\right)_{max} = \frac{\mathcal{E}}{L} \quad , \quad \tau = \frac{L}{R}$$

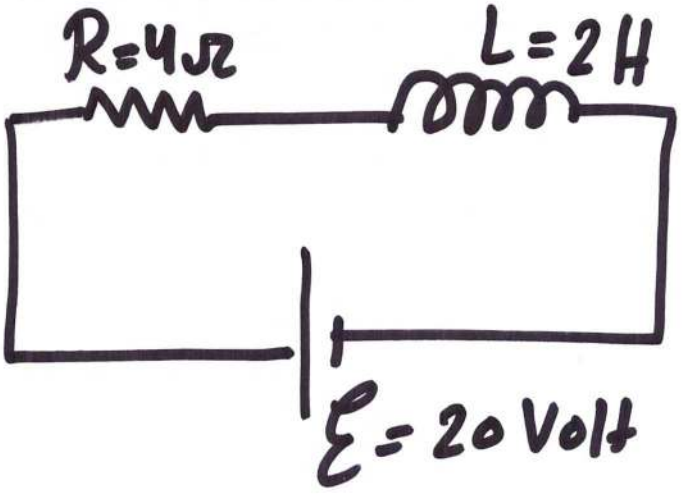
$$i = i_{max}\, e^{-t/\tau}$$

$$\left(\frac{di}{dt}\right) = -\left(\frac{di}{dt}\right)_{max} e^{-t/\tau}$$





Ex:



Find: 1) i_{max} , $\left(\frac{di}{dt}\right)_{max}$, τ .

2) if $i = \frac{1}{5} i_{max}$, Find :

a) $\frac{di}{dt}$ b) Voltage accross the inductor

c) induced emf ($\mathcal{E}'$) d) Power in L .

3) if $\frac{di}{dt}$ is equal to 40% of its max. value, what is :

a) current (i) b) Voltage accross (R)

c) Power in R d) " = (ℰ)

e) " " = ℰ f) energy in the inductor

4) at $t = 2$-sec, find:

a) current b) change rate of current $\left(\frac{di}{dt}\right)$

5) find Current (i) and $\frac{di}{dt}$ after 3-time constant

6) find time needed to reach to $\frac{1}{2}$ max. Current

7) " " " " " $\frac{1}{10}$ " $\frac{di}{dt}$.

Sol: $R = 4\Omega$, $L = 2\,H$, $\mathcal{E} = 20$ Volt

① $\tau = \frac{L}{R} = \frac{2}{4} = 0.5$ sec

$i_{max} = \frac{\mathcal{E}}{R} = \frac{20}{4} = 5\,A$

$\left(\frac{di}{dt}\right)_{max} = \frac{\mathcal{E}}{L} = \frac{20}{2} = 10\,A/s$

2) $i = \frac{1}{5} i_{max} = \frac{1}{5} * 5 = 1A$ $\left(\frac{1}{5} = 20\%\right)$

a) $\mathcal{E} = iR + L\frac{di}{dt}$

$20 = 1*4 + 2\frac{di}{dt}$

$\frac{di}{dt} = 8 A/s$

b) $V_L = L\frac{di}{dt} + \cancel{iR}$

$= 2 * 8$

$= 16$ Volt

c) $\mathcal{E}' = -L\frac{di}{dt} = -16V$

d) $P_L = i L \frac{di}{dt}$

$= 1 * 2 * 8$

$= 16$ watt

3) $\frac{di}{dt} = \frac{40}{100}\left(\frac{di}{dt}\right)_{max} = \frac{40}{100} * 10 = 4 A/s$.

a) $\mathcal{E} = iR + L\frac{di}{dt}$

$20 = i * 4 + 2 * 4$

$i = 3A$

b) $V_R = iR$

$= 3 * 4$

$= 12$ Volt

c) $P_R = i^2 R$

$= 9 * 4 = 36$ Watt

(d) $V_\mathcal{E} = \mathcal{E} - iR$ (a)

$= 20$ Volt

(e) $P_\mathcal{E} = i\mathcal{E} = 3 * 20 = 60$ watt

(f) $U_L = \frac{1}{2} L i^2$

$= \frac{1}{2} * 2 * (3)^2 = 9$ J.

(g) $\underline{t=2}$ $\tau = 0.5$ sec

$i = i_{max} (1 - e^{-t/\tau})$

$= 5(1 - e^{-2/0.5}) = 4.9$ A

$\frac{di}{dt} = \left(\frac{di}{dt}\right)_{max} e^{-t/\tau}$

$= 10\ e^{-2/0.5} = 0.18$ A/s

⑤ $t = 3\tau$

$$i = 5(1 - e^{-3t/\tau}) = 4.75\ A$$

$$\frac{di}{dt} = 10\, e^{-3t/\tau} = 0.5\ A/s.$$

---

⑥ $$i = i_{max}(1 - e^{-t/\tau})$$

$$\frac{1}{2} i_{max} = i_{max}(1 - e^{-t/\tau})$$

$$\frac{1}{2} = 1 - e^{-t/0.5}$$

$$\frac{1}{2} = +e^{-t/0.5} \Rightarrow (\ln) \Rightarrow \frac{-t}{0.5} = \ln 0.5 = -0.693$$

$$t = 0.35\ sec$$

---

⑦ $$\frac{di}{dt} = \left(\frac{di}{dt}\right)_{max} e^{-t/\tau}$$

$$0.1 \left(\frac{di}{dt}\right)_{max} = \left(\frac{di}{dt}\right)_{max} e^{-t/\tau} \Rightarrow 0.1 = e^{-t/\tau} \Rightarrow (\ln)$$

$$\ln 0.1 = \frac{-t}{0.5} \Rightarrow t = 1.15\ sec$$

ch:32 الدرس 1